

Strategic Multinational Location Choice*

Sungwan Hong
University of Pittsburgh

Hyungjin Kim
Korea Development Institute

October 7, 2026

Preliminary draft

Abstract

We study how market-access restrictions reshape multinational networks when firms choose locations strategically and prices adjust endogenously. We develop profit bounds that use the same location choices for market benefits and costs, yielding globally optimal portfolio best responses without complementarity or single-crossing restrictions. We apply the framework to a calibrated model of eight cloud providers in 42 countries. Relative to unrestricted sourcing, EU-wide requirements increase EU core presence, whereas national requirements reduce it by fragmenting the customer base supporting each location. Competitor responses offset about half of the additional worldwide contraction under national rather than EU-wide sourcing while leaving the price-index increase for regulated customers nearly unchanged. Higher offshore delivery costs in China increase Chinese providers' mainland presence but reduce their worldwide networks and operating profits.

Keywords: multinational production, location choice, portfolio equilibrium, variable markups, cloud computing, digital trade

JEL Codes: F23, F12, F13, F21, L86, C61

*We thank Jaedo Choi, Ryo Makioka, and Hongyong Zhang for helpful comments. We are grateful to Rowan Shi for helpful comments and for sharing code. Hong: sungwan.hong@pitt.edu. Kim: hyungjin.kim@kdi.re.kr.

1. INTRODUCTION

A restriction on market access changes the customers a facility can serve and, with them, the value of locations throughout a multinational network. A firm’s response depends on its other facilities and on how competitors reorganize their networks. Investment and customer prices can therefore change beyond the country imposing the restriction. We study how multinational firms’ joint location choices and competitors’ responses transmit market-access restrictions across countries.

One cloud facility can serve customers in several countries, so restrictions on service origins change the markets that facility can reach. The European Commission’s cloud procurement framework, for example, incorporates sovereignty requirements (European Commission, 2026a).¹ We model multinational cloud providers that observe private location costs and choose the countries in which to operate core infrastructure before competing in customer markets. Networks determine delivered costs and accessibility; prices and shares determine operating profits. The share-dependent markup schedule of Atkeson and Burstein (2008, henceforth AB) allows network choices to change both costs and markups.

The first difficulty is finding a multinational firm’s best network, even with rivals’ networks fixed. Evaluating all possible networks quickly becomes impractical, while a search limited to one-country improvements can miss profitable combinations.

The direction of these interactions can also change as the network expands. Consider adding the same core at the same cost. With a small network, service may be too expensive or difficult to access for the addition to attract enough business. Network expansion can make that same addition profitable, through complementarities familiar from sourcing and production models (Antràs et al., 2017, 2024). In our setting, better customer access can raise the return to a cost-reducing location. As the provider captures more of the market, however, fewer customers remain to be won, and the gain can again fall below the location cost. The same location’s net return can thus turn from negative to positive and

¹Australia’s hosting certification scheme also incorporates sovereignty and control (Australian Government, 2021). Digital sovereignty encompasses several claims about authority over data and technology (Pohle and Thiel, 2020); our applications isolate geographic supply restrictions while holding legal ownership and operating arrangements fixed.

back, even with constant markups. Variable markups also make these returns depend on competitors' price responses.

This double reversal violates single crossing in either direction (Proposition 3). Existing combinatorial methods exploit complementarity in strategic chain entry (Jia, 2008), single-crossing differences (Arkoulakis, Eckert and Shi, 2026, henceforth AES), or known signs of pairwise interactions (Castro-Vincenzi et al., 2025). These restrictions order marginal returns, or their signs, across feasible networks; AES also branches when its elimination procedure leaves unresolved choices. We instead bound complete-portfolio profits without inferring the sign of a location's return from its return in another network.

Our *completion bounds* limit the profit attainable by every network in an unexamined group. If this upper bound is below the profit of a feasible network already found, the whole group can be discarded. Our tighter bounds retain common location choices across market benefits and location costs: with only one core left to add, a firm cannot combine one country's advantage in Europe, another's in Asia, and a third country's low cost. These bounds support globally optimal portfolio best responses, including endogenous network size. With 20 optional countries, the search matches exhaustive enumeration's optimal profits with a median 235-fold speedup.

The second difficulty is strategic feedback: optimal portfolios can change as competitors adjust their networks. For strategic portfolio games with asymmetric firms and private costs, Sabal (2026) bounds equilibrium choice distributions using submodularity: marginal variable profits decline as own and rival product sets expand.² We compute numerical equilibria by updating distributions over complete networks and using completion bounds to check gains from portfolio deviations. In the EU application, these checks cover every feasible portfolio for every simulated cost type at the reported distributions. We also derive a sufficient uniqueness condition linking private-cost dispersion to strategic feedback.³

Reducing the profit contribution of one location can also change investment elsewhere in a firm's network. In a two-location illustration, the loss can discourage investment in

²His algorithm also extends to supermodularity and allows multiple equilibria. His estimation bounds require weaker conditions than submodularity.

³Increasing the scale of private-cost shocks lowers a uniform upper bound on choice sensitivity to competitors' networks.

a complementary location or redirect it toward a substitute. Under AB pricing, the direction can differ across rival configurations even when the firm’s initial market share and location-cost distribution are held fixed.

We apply the framework to eight cloud providers choosing core presence across 42 countries, with other suppliers represented in aggregate. The EU experiment restricts service origins for public-sector customers. Under EU-wide eligibility, facilities anywhere in the participating EU countries can serve them; under national eligibility, only facilities in the customer’s own country can do so.⁴ The China experiment varies the cost of supplying mainland customers from offshore facilities, holding mainland operating arrangements fixed. The offshore-cost shock applies to Chinese and foreign providers alike whenever they serve mainland customers from abroad. We jointly calibrate provider-specific location-cost levels and relative competitiveness to core counts and worldwide market shares. Elasticities, spatial cost coefficients, service connections, and private-cost dispersion are assigned inputs.

Relative to unrestricted sourcing, both EU policies reduce worldwide core presence and raise public-sector prices, but they have opposite effects on investment within the EU. EU-wide eligibility adds about 2.1 expected EU core presences, whereas national eligibility removes about 2.6. EU-wide requirements redirect regulated demand toward EU facilities while preserving access to customers across member states. National requirements favor domestic supply but remove that shared customer base. Profits from several customer markets help cover a core’s location cost; losing access to some of those customers can therefore make the core unprofitable to retain. In the calibration, the loss of market access outweighs the incentive to duplicate facilities across domestic markets.

We next compare national with EU-wide sourcing to isolate the additional effect of fragmenting the EU public-sector customer market. A country-by-country cutoff approximation fitted to baseline participation predicts a smaller worldwide contraction than joint portfolio responses and misses the contraction outside the EU when rivals’ networks are held at baseline.

⁴The Commission’s 2026 proposal for a Cloud and AI Development Act includes an EU-wide sovereignty framework (European Commission, 2026c); national schemes include France’s SecNumCloud (Agence nationale de la sécurité des systèmes d’information, 2022). The experiment isolates geographic eligibility rather than the full qualifications in these schemes.

Allowing competitors to adjust then halves the additional worldwide network contraction under national rather than EU-wide sourcing, from 8.59 to 4.32 core presences. Rival withdrawal can increase the profit a core adds to its owner’s network by redirecting customers toward that provider. A core that would otherwise close can then earn enough to cover its location cost. Most of this attenuation occurs outside the EU. Yet the public-sector price difference remains nearly unchanged, at about 28 percent: network adjustment restores little of the cross-border supply lost under national requirements.

Higher costs of supplying mainland customers from offshore facilities increase Chinese providers’ probabilities of operating a mainland core but reduce their worldwide networks and operating profits. Offshore cores serve several markets, with profits from mainland customers helping cover their location costs. Raising the cost of supplying China can therefore make some offshore cores less profitable to retain, even as mainland investment becomes more attractive.

The paper builds on quantitative trade models in which destinations affect later entry costs (Morales et al., 2019; Alfaro-Ureña et al., 2025), sourcing locations jointly determine input costs (Antràs et al., 2017), and production locations jointly determine market access (Tintelnot, 2017; Castro-Vincenzi, 2024; Antràs et al., 2026). Related interactions shape fulfillment and plant networks (Houde et al., 2023; Oberfield et al., 2024). We study joint core-location decisions and responses to competitors’ networks and prices, connecting multinational investment to strategic interaction across markets (Bulow et al., 1985) and product and network choice (Holmes, 2011; Wollmann, 2018; Fan and Yang, 2020; Nishida, 2015).

The applications also complement work on data restrictions and productivity (Ferracane et al., 2020), cloud inputs under privacy law (Demirer et al., 2024), digital-service trade (Sun and Trefler, 2026; Sun, 2026), and cloud competition (Manganelli and Schnurr, 2024).⁵ We show how restrictions on service origins change the customer base supporting shared facilities and, through providers’ joint location choices and competitors’ responses, affect investment and prices beyond the regulated market.

⁵Policy research distinguishes data-transfer restrictions from requirements to store or process data locally (Casalini et al., 2021; Del Giovane et al., 2023).

Sections 2 and 3 present the model and theory. Section 4 describes the data and calibration, and Section 5 presents the policy experiments. The appendix contains proofs and supporting evidence.

2. LOCATION PORTFOLIOS AND ENDOGENOUS PRICING

Providers observe private location costs and choose their core networks simultaneously. They know rivals' cost distributions but not their realizations. The chosen networks and geographic eligibility rules determine which facilities can supply each market, their delivered costs, and their accessibility to customers. Given these networks, the pricing system determines market shares, markups, and operating profits.

2.1 Service availability, costs, and accessibility

Service availability. Firm f chooses a set S_f of countries hosting its core infrastructure. Core regions are groups of data centers supplying cloud services. We group each provider's core regions within a country into one location alternative, so $|S_f|$ counts its core-hosting countries. The choice concerns core presence across countries rather than the scale of facilities within a country. Let i index supply countries and n customer markets.

Existing local facilities remain at their observed locations, but their service availability can change with core choices. They can supply service from countries where the provider has no core. A local facility may rely on a supporting core region, called its parent, which can be in another country (Figure 1). When the parent is known, local service is available if and only if the parent's country is in the chosen core portfolio S_f .⁶

Policy determines which supply locations are eligible to serve each market. Geographic eligibility applies to the service origin, the country in which the core or local service is located. For each provider, multiple operating local facilities in the same country count as one service origin. Parent support determines whether a local facility can operate; service-origin eligibility determines which customers it can serve.

⁶For local facilities with unknown parents, we assume that the support needed to operate them remains available as S_f changes. These facilities are subject to the same geographic eligibility rules. Section 4.1 describes the parent assignments and the local facilities with unknown parents.

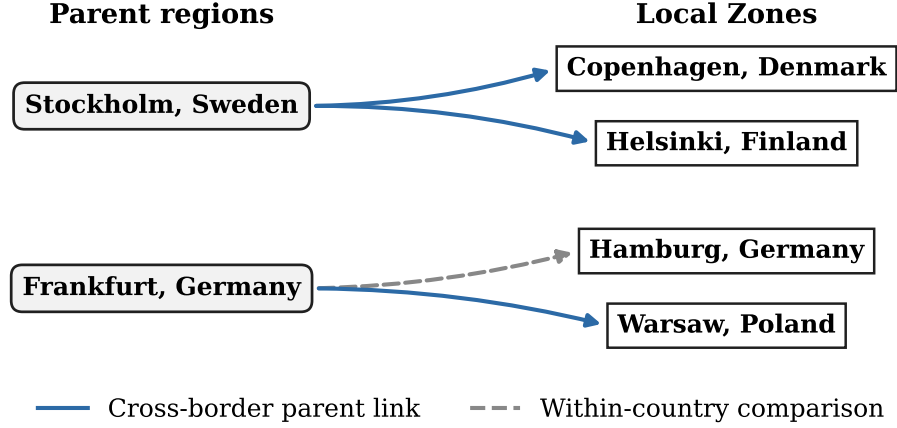


Figure 1: A core region can support local cloud services in another country

Notes: Solid blue lines connect AWS Local Zones to parent regions abroad; the dashed gray line stays within Germany. Lines show parent-region relationships, not necessarily data routes. Sources: the AWS Local Zone roster and architecture guide, archived September 11, 2026.

Figure 1 illustrates the distinction. AWS’s Copenhagen Local Zone operates only if Sweden is in its core portfolio, but its service origin is Denmark. Under the model’s domestic sourcing rule, it can therefore serve Danish customers even though its parent core is abroad.

Core and local facilities can both perform general-purpose virtual-machine (VM) workloads, although their product ranges differ. Within this workload class, a core has supply intensity one, local-only service has lower intensity γ_{loc} , and an origin with no eligible service has intensity zero. Core and local supply in the same country are not added together. Let $u_{fin}(S_f)$ indicate that firm f has a core in country i eligible to serve market n , and let $o_{fin}(S_f)$ indicate service from that country through either an eligible core or an eligible, operating local facility. Thus $o_{fin}(S_f) \geq u_{fin}(S_f)$, and effective supply is

$$t_{fin}(S_f) = \underbrace{u_{fin}(S_f)}_{\text{core supply}} + \underbrace{\gamma_{\text{loc}}[o_{fin}(S_f) - u_{fin}(S_f)]}_{\text{local-only supply}}.$$

Delivered costs. A unit of service requires a fixed bundle of VM workloads. The provider assigns each workload to one eligible facility at the lowest delivered resource cost. Different workloads may use different facilities, and all workload requirements

increase proportionally with service quantity. Unit cost is the weighted average of these minimum costs.⁷

Each eligible origin can lower the cost of serving a customer. Let w_i denote its operating input-cost index, τ_{in} the cost of delivering service from i to market n , and a_f the provider's relative competitiveness. Delivered marginal cost aggregates effective supply as

$$\Theta_{fn}(S_f) = \sum_i t_{fin}(S_f)(w_i \tau_{in})^{-\theta}, \quad c_{fn}(S_f) = a_f^{-1}[\Theta_{fn}(S_f)]^{-1/\theta}. \quad (2.1)$$

To obtain this expression, we assume workload-specific efficiencies follow independent Fréchet distributions across eligible locations. An additional origin offers another opportunity to perform each workload efficiently. For a given contribution to effective supply, the cost reduction is larger when existing supply is limited. Integrating the minimum workload costs gives (2.1). With positive eligible supply, unit cost is finite for every $\theta > 0$; Appendix D.1 gives the derivation. The parameter θ governs efficiency dispersion across origins, whereas σ below governs substitution across providers. Substitution between workloads within a provider and customer substitution across providers are distinct margins (Antràs et al., 2017, 2024). The appendix also derives this supply exponent with CES workload aggregation. Fréchet sourcing underlies the multinational production model of Tintelnot (2017) as well.

Accessibility. Resource costs determine the cost of performing workloads; accessibility affects how customers value the available service locations. An eligible origin can therefore influence both marginal cost and demand. Let q_{in} be an index of accessibility from origin i to market n . The access index captures both the accessibility of individual origins and the breadth of eligible service locations. An additional origin can therefore improve access even when the most accessible origin is unchanged. We aggregate access in A_{fn} and map it into the demand weight ψ_{fn} :

$$A_{fn}(S_f) = \lambda_A^{-1} \log \left(1 + \sum_i o_{fin}(S_f) e^{\lambda_A q_{in}} \right), \quad \psi_{fn}(S_f) = e^{\chi[A_{fn}(S_f) - A_n^{\text{frontier}}] + d_{fn}}. \quad (2.2)$$

⁷A workload includes resources that must be deployed together, such as a virtual machine and its required storage and network functions. These resources are supplied together at the assigned facility. The production technology abstracts from capacity constraints and congestion.

Here A_n^{frontier} is access with all origins available, a common reference level across policy regimes. The parameter λ_A controls aggregation across origins, χ scales the demand advantage from access, and the demand shifter d_{fn} permits a home-market advantage. Both cost and access depend on the complete portfolio because local supply can require a parent core elsewhere. A provider with no eligible supply has zero share in that market.

In the Copenhagen example, with Sweden in the portfolio and Danish service eligible, adding a Danish core raises supply intensity from γ_{loc} to one. It does not add a new origin to accessibility, because Denmark already enters through $o_{fin} = 1$.

2.2 Market shares, variable markups, and pricing

Customer expenditure R_n in each market is fixed across policy regimes. The real quantity of the service composite, R_n/P_n , therefore falls when its price index rises. Operating input prices are also fixed: this is a partial-equilibrium model in which core locations and cloud-service prices adjust.⁸ For each portfolio profile and regime, price-index contributions and revenue shares satisfy

$$y_{fn} = \psi_{fn} p_{fn}^{1-\sigma}, \quad s_{fn} = \frac{y_{fn}}{b_n + \sum_g y_{gn}}, \quad b_n \geq 0. \quad (2.3)$$

The nonnegative term b_n is the aggregate contribution of other suppliers (the residual) to the price index. It summarizes their available services and prices. For each policy regime, this contribution is exogenous and these suppliers do not choose portfolios. Their expenditure share, $b_n/(b_n + \sum_g y_{gn})$, rises when the modeled providers' aggregate contribution falls. Losing eligible services can therefore redirect expenditure toward the residual even when b_n is fixed.

For a portfolio profile S , the corresponding price index is

$$P_n(S) = \left[b_n + \sum_f \psi_{fn}(S_f) p_{fn}(S)^{1-\sigma} \right]^{1/(1-\sigma)}. \quad (2.4)$$

⁸The demand system abstracts from customer-specific switching histories and explicit fees for switching or using multiple clouds. These differ from the geographic service costs studied here. Competition and Markets Authority (2025, Chapter 6) describes technical barriers and egress fees affecting switching and multi-cloud use.

This is the cost of the market's CES service composite under the maintained quality weights. It can rise because suppliers charge more, because a network loses accessibility, or because a previously available service becomes ineligible. A sourcing restriction can therefore make the composite more expensive without higher posted prices for continuing services.⁹

We adopt the AB share-dependent markup schedule, also used by Edmond et al. (2015), as a conduct specification. AB derive this schedule under Cournot competition with nested-CES demand. We retain its share–markup relationship while holding expenditure in each market fixed. Downstream payoffs follow this pricing system:

$$p_{fn} = \mu(s_{fn})c_{fn}, \quad \mu(s) = \frac{\mathcal{E}(s)}{\mathcal{E}(s) - 1}, \quad \frac{1}{\mathcal{E}(s)} = \frac{1-s}{\sigma} + \frac{s}{\eta}, \quad \sigma > \eta > 1. \quad (2.5)$$

A larger supplier faces a lower perceived elasticity and charges a higher markup, bounded between $\sigma/(\sigma - 1)$ and $\eta/(\eta - 1)$. With market expenditure held fixed, η governs the share–markup relationship rather than an expenditure response. In the constant-markup limit $\eta = \sigma$, prices are uniquely given by $\sigma/(\sigma - 1)$ times marginal cost: they still change with the network's costs, while markups remain fixed.

Revenue is $R_n s_{fn}$, and the pricing rule implies an operating margin of $1/\mathcal{E}(s_{fn})$. Operating profit is therefore

$$\begin{aligned} VP_{fn} &= \frac{R_n s_{fn}}{\mathcal{E}(s_{fn})} \\ &= R_n \left[\underbrace{\frac{s_{fn}}{\sigma}}_{\text{constant-markup term}} + \underbrace{\left(\frac{1}{\eta} - \frac{1}{\sigma} \right) s_{fn}^2}_{\text{variable-markup term}} \right], \quad \Pi_f = \sum_n VP_{fn}. \end{aligned} \quad (2.6)$$

Gaining market share raises revenue and the operating margin. The share-squared term captures the extra profit from a higher markup. A network expansion that lowers cost or improves access raises the provider's share. Its markup rises, while rivals that lose share reduce theirs, partly offsetting its competitive gain. These price responses also affect how locations interact in a portfolio.

⁹Feenstra (1994) studies the related role of changing varieties in CES price measurement. Security benefits and network choices by unmodeled suppliers are outside the comparison; these price effects do not measure the full welfare consequences of regulation.

Proposition 1 (Unique prices). *For $\sigma > \eta > 1$ and a fixed portfolio profile, the pricing system (2.3)–(2.5) has a unique solution.*

Unique prices assign a well-defined operating profit to every network profile. Appendix A.6 gives a globally convergent pricing iteration and establishes smooth price responses to positive marginal costs on each fixed active set.

2.3 Private location costs

Given its private shocks Z_f , firm f 's total location cost sums the costs of the countries in its core portfolio S_f . It therefore depends on both the number and identity of selected countries, but not on service output:

$$C_f(S_f, Z_f) = \ell_f^{\text{ref}} \sum_{i \in S_f} \left[\underbrace{\delta_f + c_i^0 + \kappa D_{fi}}_{\text{systematic net cost}} + \underbrace{\sigma_\varepsilon Z_{fi}}_{\text{private shock}} \right], \quad Z_{fi} \stackrel{\text{iid}}{\sim} N(0, 1). \quad (2.7)$$

The firm-specific intercept δ_f governs worldwide portfolio size: a higher per-core cost makes larger portfolios less attractive. Systematic costs vary across countries through a fixed location-cost adjustment c_i^0 and distance from the firm's home market, D_{fi} . The private shocks are normal and enter net location costs in levels, as do additive disturbances to profits or fixed costs in empirical entry models (Bresnahan and Reiss, 1991; Berry, 1992; Jia, 2008; Ciliberto and Tamer, 2009).¹⁰

The fixed, positive provider-specific cost scaling factor ℓ_f^{ref} multiplies both systematic costs and private shocks. Whereas δ_f shifts the firm's baseline per-core cost, ℓ_f^{ref} also scales cost differences across countries and the dispersion of private costs. The cost of an additional unit of distance is $\kappa \ell_f^{\text{ref}}$, and the standard deviation of a per-core cost shock is $\sigma_\varepsilon \ell_f^{\text{ref}}$, both in operating-profit units.

An additional core adds one location cost to the portfolio and can contribute to operating profit in several customer markets. Eligibility restrictions therefore affect the return to a core by changing the markets it can serve. Firm f chooses its core portfolio to maximize

¹⁰Private payoff shocks appear in Seim (2006); Aguirregabiria and Mira (2007); Dickstein and Morales (2018) use normal fixed-export-cost shocks. Lognormal costs appear in market entry (Eaton et al., 2011), plants (Tintelnot, 2017), sourcing (Antràs et al., 2017), capacity (Castro-Vincenzi, 2024), and market entry and product development (Sabal, 2026). Online Appendix S.2.1 extends the results to nonlinear shocks that retain additivity across choices.

expected operating profit net of $C_f(S_f, Z_f)$, taking rivals' portfolio distributions as given. The policy comparisons are between static equilibria.¹¹

3. STRATEGIC LOCATION CHOICE AND EQUILIBRIUM

Section 2 determines operating profits for a given network profile. We now characterize firms' optimal network choices and the distributions consistent with equilibrium. We define the portfolio game, characterize location interactions, and construct bounds for globally optimal portfolio best responses.

3.1 Portfolio game and equilibrium

There are F firms. Firm f has a finite set $\mathcal{A}_f$ of portfolio elements and a nonempty set of feasible portfolios $\mathcal{X}_f \subseteq 2^{\mathcal{A}_f}$. In the cloud application, portfolio elements are core-country choices, with location costs given by (2.7). For every pure portfolio profile $S = (S_1, \dots, S_F)$, the pricing system in Section 2.2 determines prices $p^\dagger(S)$ and operating profit $\Pi_f(S; p^\dagger(S))$.¹² The payoff net of deterministic location costs is

$$r_f(S) = \underbrace{\Pi_f(S; p^\dagger(S))}_{\text{operating profit}} - \underbrace{\sum_{j \in S_f} C_{fj}^0}_{\text{deterministic location costs}}. \quad (3.1)$$

Each element j has a deterministic cost C_{fj}^0 and a positive private-shock loading ℓ_{fj} .

Firm f observes the vector of private cost shocks $\varepsilon_f = (\varepsilon_{fj})_{j \in \mathcal{A}_f}$. Its payoff is

$$u_f(S; \varepsilon_f) = r_f(S) - \sigma_\varepsilon \sum_{j \in S_f} \ell_{fj} \varepsilon_{fj}, \quad \ell_{fj} > 0. \quad (3.2)$$

Throughout, $\sigma_\varepsilon > 0$. The loading ℓ_{fj} converts the standardized private-cost shock into payoff units.¹³

¹¹A lower expected core count describes a different network configuration. Modeling construction delays and the transition between configurations would require a dynamic framework, such as Ericson and Pakes (1995).

¹²The general portfolio game permits any fixed, finite downstream-payoff table. If downstream equilibria are multiple, $p^\dagger(S)$ denotes a selection fixed independently of private costs and beliefs.

¹³For core locations, $C_{fj}^0 = \ell_f^{\text{ref}}(\delta_f + c_j^0 + \kappa D_{fj})$, $\ell_{fj} = \ell_f^{\text{ref}}$, and $\varepsilon_{fj} = Z_{fj}$ give the location-cost formula (2.7).

Assumption 1 (Private cost shocks). Private types are independent across firms, and each firm's type vector has a joint density. Within a firm, the coordinates may be dependent. For every j , the conditional density of ε_{fj} given $\varepsilon_{f,-j}$ is bounded by $\bar{g}_{fj} < \infty$ almost surely. Independent normal or logistic cost shocks satisfy this condition.

Private shocks capture information known to the firm but not its rivals, such as negotiated site terms, conditional on publicly known costs. Portfolios containing the same location share its shock. Additivity restricts private costs while allowing complementarity and substitutability through operating profits. Private profitability information is familiar in entry games (Seim, 2006); Sabal (2026) combines it with additive product costs in a portfolio game.

Holding rivals' distributions and the firm's other cost shocks fixed, compare the best networks containing and excluding a location, reoptimizing the remaining choices in each case. The difference in their payoffs before deducting that location's private cost determines the highest private cost at which the firm includes it. Changes in rivals' behavior move this cutoff; the conditional density bound limits the probability mass between the old and new cutoffs. The cutoff concerns changes in a location's own cost, whereas single crossing concerns its net return as other locations are added. Computing the cutoff still requires reoptimizing the remaining network.

Let $Q_f \in \Delta(X_f)$ be firm f 's portfolio distribution before its private costs are observed. Each cost type chooses a complete network; heterogeneity in those costs generates the distribution of network choices. Location choices within a firm can be dependent even when its location-cost shocks are independent.

A rival that operates both a German and a French core or neither creates different competitive conditions from one that always operates exactly one, even if their country-level entry probabilities coincide. Online Appendix S.4.6 isolates this distinction by changing which rival locations occur together while preserving every country's entry frequency. We therefore retain distributions over complete portfolios and write $Q_{-f} = \bigotimes_{g \neq f} Q_g$. Firm f 's interim value of portfolio S_f , averaging over rivals' choices, is

$$V_f(S_f; Q_{-f}) = \sum_{S_{-f}} r_f(S_f, S_{-f}) Q_{-f}(S_{-f}). \quad (3.3)$$

Independence across firms gives the product Q_{-f} .

The type-specific global best response and the distribution it induces are

$$\text{BR}_f(\varepsilon_f; Q_{-f}) \in \arg \max_{S_f \in \mathcal{X}_f} \left\{ \underbrace{V_f(S_f; Q_{-f})}_{\text{payoff net of deterministic costs}} - \underbrace{\sigma_\varepsilon \sum_{j \in S_f} \ell_{fj} \varepsilon_{fj}}_{\text{realized private costs}} \right\}, \quad (3.4)$$

$$\mathcal{B}_f(Q_{-f})(S_f) = \Pr\{\text{BR}_f(\varepsilon_f; Q_{-f}) = S_f\}. \quad (3.5)$$

Definition 1 (Portfolio Bayesian Nash equilibrium). A portfolio Bayesian Nash equilibrium is a profile $Q^* = (Q_1^*, \dots, Q_F^*)$ satisfying

$$Q_f^* = \mathcal{B}_f(Q_{-f}^*), \quad f = 1, \dots, F.$$

Each type chooses a globally optimal portfolio in $\mathcal{X}_f$, with ties resolved by a fixed rule.

Proposition 2 (Equilibrium existence). *Fix the finite payoff table $r_f(S)$ in (3.1). A portfolio Bayesian Nash equilibrium exists if firms' private cost vectors are independent across firms and each has a joint density.*

Although each network choice is discrete, a small payoff change switches the choices of only a small mass of cost types. Absolute continuity makes ties negligible and the distributional response map continuous, allowing Brouwer's theorem to establish existence (Appendix A.1).

3.2 Location interactions and investment incentives

An additional location can improve a firm's ability to serve several markets, but its return depends on the network already in place. At low supply, expanding that network can increase the return to a further location. At high supply, diminishing gains in market share can reverse this relation, which under variable markups also depends on competitors' repricing. The following one-market calculation isolates these forces before subtracting location costs.

Let $x > 0$ summarize provider f 's supply in one market and let its pre-markup competitiveness be $\xi_f = \psi_f c_f^{1-\sigma} = \omega_f x^{\nu_s}$, where the supply exponent ν_s is the elasticity of competitiveness with respect to supply and $\omega_f > 0$ collects factors held fixed as supply

changes. Rival competitiveness, the residual price-index contribution, and expenditure are fixed, but all modeled providers reprice as x changes. In the core-only benchmark without an accessibility margin, $v_S = (\sigma - 1)/\theta$ under (2.1).

The constant-markup case gives the simplest intuition. At $\eta = \sigma$, operating profit is

$$\Pi_f(x) = \frac{R}{\sigma} \frac{\omega_f x^{v_S}}{H + \omega_f x^{v_S}},$$

where $H > 0$ combines rival competitiveness and residual supply after factoring out the common markup. Own competitiveness enters the denominator as well as the numerator: expansion raises the provider's share of a fixed market. The sign of $\Pi_f''(x)$ is the sign of

$$\underbrace{(v_S - 1)}_{\text{competitiveness gain at low supply}} - \underbrace{2v_S s_f}_{\text{diminishing scope to gain share}}.$$

For $v_S > 1$, profit is convex at low shares and concave at high shares. The unit-cost reduction from a given supply increment diminishes as existing supply grows, but its operating-profit gain can rise as customers switch toward the provider. Once its share is large, fewer customers remain to be won, and the additional profit declines. This complementarity can therefore arise even with additive core supply. In a monopolistically competitive sourcing model with a given aggregate price index, a supply exponent above one can sustain complementarity as the network expands (Antràs et al., 2017). Here, the firm's own expansion changes that index, limiting further gains in market share.

Let $\Pi_f(T)$ denote operating profit when the firm adds locations T to a fixed base network. The net gain from location $j \notin T$ is

$$\Delta_j(T) = \Pi_f(T \cup \{j\}) - \Pi_f(T) - c_j,$$

where c_j is the same realized location cost in every comparison. With both supply and accessibility, competitiveness takes the form $\xi_f = \zeta \Theta^{v_S} W^{v_A}$, where $\zeta > 0$, W is the argument of the logarithm in (2.2), and $v_A = \chi/\lambda_A$. In a core-only network, a location can add $d_\ell > 0$ to supply Θ and $a_\ell > 0$ to accessibility W in different proportions.

Proposition 3 (Sign reversals in location returns). *In a market with $R > 0$, fix the positive pre-markup competitiveness of finitely many active rivals and the residual contribution $b \geq 0$, with at least one active rival or $b > 0$.*

(a) Fixed accessibility. *With $v_S > 1$, positive base supply x_0 and contributions d_j, d_k, d_h can be chosen for the additive supply index $x(T) = x_0 + \sum_{\ell \in T} d_\ell$.*

(b) Supply and access. *For every $v_S > 0$ and $v_A > 0$, positive contributions (d_ℓ, a_ℓ) and bases $\Theta_0 > 0, W_0 > 1$ can be chosen for $\Theta(T) = \Theta_0 + \sum_{\ell \in T} d_\ell$ and $W(T) = W_0 + \sum_{\ell \in T} a_\ell$.*

In each case, these choices admit a single cost $c_j > 0$ such that

$$\Delta_j(\emptyset) < 0, \quad \Delta_j(\{k\}) > 0, \quad \Delta_j(\{k, h\}) < 0.$$

All subsets of $\{j, k, h\}$ are feasible. Both results hold under AB pricing ($\sigma > \eta > 1$), with all providers repriced at each portfolio, and under constant markups ($\eta = \sigma > 1$).

The strict $-, +, -$ pattern violates single crossing in either direction and rules out submodularity and supermodularity. Part (b) permits any positive pair of exponents by choosing location-specific supply and access contributions. A location that mainly lowers service costs becomes more valuable when another improves customer access: the firm can supply more attractive service at lower cost. Further expansion eventually reduces the additional business the first location attracts. Thus reversals can occur even when $v_S + v_A \leq 1$. With additive supply and access in one CES market, a fixed pair whose joint competitiveness gain exceeds the sum of its separate gains is complementary at low initial shares and substitutable above a share threshold. This threshold holds the base network and geographic contributions fixed (Lemma A.5). Appendix A.7 provides the proofs of Proposition 3 and Lemma A.5, together with finite examples and a strict reversal under the calibrated baseline inputs.

Holding accessibility fixed again isolates the role of rivals' repricing. Under CES pricing, rival supply enters profit through a single competitor aggregate. Under AB pricing, each rival's markup responds to its own market share. How competing sales are divided among suppliers therefore matters, even at the same own share. Appendix A.7 derives the curvature expression (A.21).

These interactions also determine how a loss of return at one location affects investment elsewhere. Holding the existing network fixed, the firm can add neither candidate

location, a alone, b alone, or both. An additive profit loss affects only location a . If the locations are substitutes, the loss can shift investment toward b ; if they are complements, it can reduce investment in b as well (Appendix F.1). Table 1 holds initial own share and the cost distribution fixed. Losing profit at a raises the inclusion probability of b with one rival, but lowers it with seven equal rivals. Here, the large rival lowers its markup more as it loses customers, limiting the gain from joint additions. In a broader set of constructed markets, dividing rival sales among smaller providers widens the range of own shares at which the additions are complements (Online Appendix S.4.7).

Table 1: Rival composition and investment in an unexposed location

Rivals	Location interaction	Change in inclusion probability of b (pp)
One	Substitutes	+0.298
Seven equal	Complements	-0.128

Notes: Constructed AB markets with $(\sigma, \eta, \theta) = (5, 2, 3)$, initial own share 0.08, and no residual supply. Accessibility and rivals' networks are fixed. Each of two locations adds 0.1 supply; costs are independently uniform on $[0, 0.003]$. An additive operating-profit loss of 0.001 affects only location a . Choices optimize over all four portfolios. Initial inclusion probabilities of b are 0.623 and 0.723, respectively. Appendix F.1 gives the construction.

With this single supply index, discrete interactions depend on curvature over the whole interval spanned by the additions, not only at the initial network. Across markets, the expenditure-weighted interactions add (Appendix A.7).

Additive core supply with fixed accessibility and $0 < \nu_S \leq 1$ gives decreasing location returns under CES and under repriced AB with $\sigma \geq 2$, $1 < \eta < \sigma$ (Appendix Lemma A.4); AES elimination then applies. Restoring accessibility and local-service support produces complementarities even at $\nu_S = 2/3$ in controlled comparisons holding geographic inputs and rivals fixed (Online Appendix S.4.3). Appendix C.1 compares computation times where decreasing returns are guaranteed.

3.3 Completion bounds for portfolio best responses

We find a globally optimal portfolio at given rival distributions by bounding the best profit attainable by groups of unexamined portfolios. This is the logic of *branch and bound* (Land and Doig, 1960): if a group's profit upper bound is below the payoff of a feasible network already found, we discard the group. Otherwise we split it by fixing another location

decision. The bounds use the service and cost structure without requiring an ordering of marginal returns.

A search node (S_f, U_f) fixes locations in S_f as included, leaves those in U_f undecided, and excludes the rest. Its feasible completions form $\mathcal{X}_f(S_f, U_f)$. For a given type ε_f , write $\tilde{V}_f(T_f; \varepsilon_f, Q_{-f})$ for expected net profit in (3.4). A completion bound $B_f(S_f, U_f)$ satisfies

$$\begin{aligned} \mathcal{X}_f(S_f, U_f) &= \{T_f \in \mathcal{X}_f : S_f \subseteq T_f \subseteq S_f \cup U_f\}, \\ B_f(S_f, U_f) &\geq \underbrace{\max_{T_f \in \mathcal{X}_f(S_f, U_f)} \tilde{V}_f(T_f; \varepsilon_f, Q_{-f})}_{\text{best profit among feasible completions}}. \end{aligned} \quad (3.6)$$

Every surviving complete network is evaluated using the original profit function. Once no remaining subproblem has an upper bound above the best feasible payoff found, the associated portfolio is globally optimal.

We construct three bounds. The first links benefits and costs through the number of locations selected. The second uses a common affine ranking of locations. The third retains common core and service choices within a relaxation of the nonlinear profit problem.

A bound that fixes only the number of locations. If the firm adds k locations, give it the k largest contributions to each market statistic and subtract only the k lowest location costs. Finding these contributions requires sorting lists of locations, not evaluating every k -location network. The selected countries may differ across statistics and costs, so the result can exceed the profit of any actual network. Conditioning on size preserves the trade-off between gains from additional locations and their costs. Comparing across all feasible k then lets the firm choose how large its network should be. Total element cost is $C_{fj} = C_{fj}^0 + \sigma_\varepsilon \ell_{fj} \varepsilon_{fj}$. Appendix B states the result and constructs the pricing envelopes, including parent-dependent local service.

Making benefits and costs come from the same locations. In Figure 2, choosing the best market contributions and lowest cost separately gives a bound of 17, although no available location earns more than 7. Requiring a common location therefore rules out the group when another feasible portfolio earns 8.

One additional core must serve both customer markets

Core location	Europe benefit	Asia benefit	Cost	Net profit
Germany	9	1	6	4
Singapore	2	9	5	6
Poland	4	4	1	7

Separate market choices	One common location choice
Germany for Europe; Singapore for Asia; Poland for cost. $9 + 9 - 1 = 17$ An infeasible combination that overstates profit.	Each country's benefits are combined with its own cost. $\max\{4, 6, 7\} = 7$ A bound below a known profit of 8.

Figure 2: Why common location choices tighten a profit bound

Notes: Hypothetical payoffs in arbitrary units. In this illustrative subproblem, one core must be chosen and operating profit is the sum of the two market benefits. A feasible portfolio outside this group is assumed to earn 8. The separate-market bound cannot exclude the group, whereas the common-choice bound can.

The joint-affine refinement uses a common affine upper bound to rank countries by their total market benefit net of cost. Proposition B.2 in Appendix B.2 gives the construction and proves that its bound is no larger than the cardinality bound built from the same component envelopes. The ranking combines contributions across all markets and rival-network configurations.¹⁴

A convex relaxation with common location choices. We first combine supply and accessibility in a single index without changing the firm's profit. Competitiveness combines effective supply Θ_n and accessibility W_n , the argument of the logarithm in (2.2), with elasticities $\nu_S = (\sigma - 1)/\theta$ and $\nu_A = \chi/\lambda_A$. For positive elasticities, set $\vartheta = \nu_S/(\nu_S + \nu_A)$ and define the supply–access index

$$z_n = \underbrace{\Theta_n^\vartheta}_{\text{supply component}} \cdot \underbrace{W_n^{1-\vartheta}}_{\text{access component}}, \quad \xi_n = \zeta_n z_n^{\nu_S + \nu_A}. \quad (3.7)$$

The factor $\zeta_n > 0$ is independent of location choices. This weighted geometric mean is concave in supply and accessibility and preserves competitiveness exactly (Appendix B.4).

¹⁴Branch-and-bound searches and bounds that add the k largest contributions are standard in assortment optimization, where a seller chooses which products to offer (Feldman and Topaloglu, 2015; Leitner et al., 2024; Zhang et al., 2025).

Holding rivals' portfolio distributions fixed, let $H_n(z)$ be expected operating profit in market n at own competitiveness $\zeta_n z^{\nu_S + \nu_A}$, repricing all modeled providers in every rival state before averaging.

We then enlarge the optimization problem in two ways. Core and service choices may be fractional, and each market's profit curve H_n is replaced by a nondecreasing concave upper bound $\hat{H}_n$. Actual profits can be nonconcave; concavity is imposed only on this upper bound. Every discrete network remains feasible and is credited with at least its actual profit, so the relaxed optimum bounds what it can earn. The firm still chooses a discrete portfolio, evaluated with the original profit function.

Fix a search node (S_f, U_f) and k additional locations. Let x_j indicate a core choice and let v collect service-availability and parent-support variables. Define $\mathcal{R}_f(S_f, U_f, k)$ by linear constraints on (x, v) , each in $[0, 1]$, tracking included and excluded cores, the number of additions, and local-service support. Every feasible completion T must be represented at $x = \mathbf{1}_T$ with its actual supply and accessibility. The same choices determine all markets' benefits and costs.

Throughout this set, $\Theta_n \geq 0$ and $W_n > 0$ are affine functions of (x, v) : linear in the choices with a possible constant term. For each market, let $I_n = [\underline{z}_n, \bar{z}_n] \subseteq [0, \infty)$ contain all attainable supply-access indices. Require $\hat{H}_n$ to be continuous and to satisfy $\hat{H}_n(z) \geq H_n(z)$ throughout I_n .

Proposition 4 (A common-location profit bound). *Under these conditions, the following problem is convex, and its value bounds net profit for every feasible completion with k additional locations:*

$$\begin{aligned}
B_{f,k}^{\text{geo}} = \max_{x,v,z} \quad & \underbrace{\sum_n \hat{H}_n(z_n)}_{\text{relaxed operating profit}} - \underbrace{\sum_{j \in \mathcal{A}_f} C_{fj} x_j}_{\text{costs of the same locations}} \\
\text{subject to} \quad & (x, v) \in \mathcal{R}_f(S_f, U_f, k), \\
& z_n \in I_n, \quad 0 \leq z_n \leq \Theta_n(x, v)^{\vartheta} W_n(x, v)^{1-\vartheta} \quad \text{for every } n
\end{aligned} \tag{3.8}$$

Taking the maximum over feasible k bounds the best profit across all remaining portfolio sizes.

The bound applies to both service structures in Proposition 3, without restricting the sum of the supply and access exponents. Appendix B.4 proves the result, constructs the service constraints, and gives a linear-programming relaxation. At each completion size, we take the minimum of the common-location, cardinality, and joint-affine bounds.

Adding the common-location relaxation therefore weakly tightens the bound used in the search.

We next compare the speed of the search with exhaustive enumeration. The benchmark uses the eight-provider calibration of Section 4, with 20 optional countries and all other core choices fixed. Different cost types face the same operating-profit opportunities but deduct different location costs. For given rivals and policy, the market-profit curves H_n and their concave bounds can therefore be prepared once and reused across types; the reported times include preparation. The search recovers the same optimal profits as exhaustive enumeration with a median 235-fold speedup (Figure 3). Preparation costs can outweigh the savings from search in smaller problems. Relative to the search using only the cardinality and joint-affine bounds, adding the common-location bound yields a median 10.6-fold speedup. Appendix C.1 gives the protocol and timings.

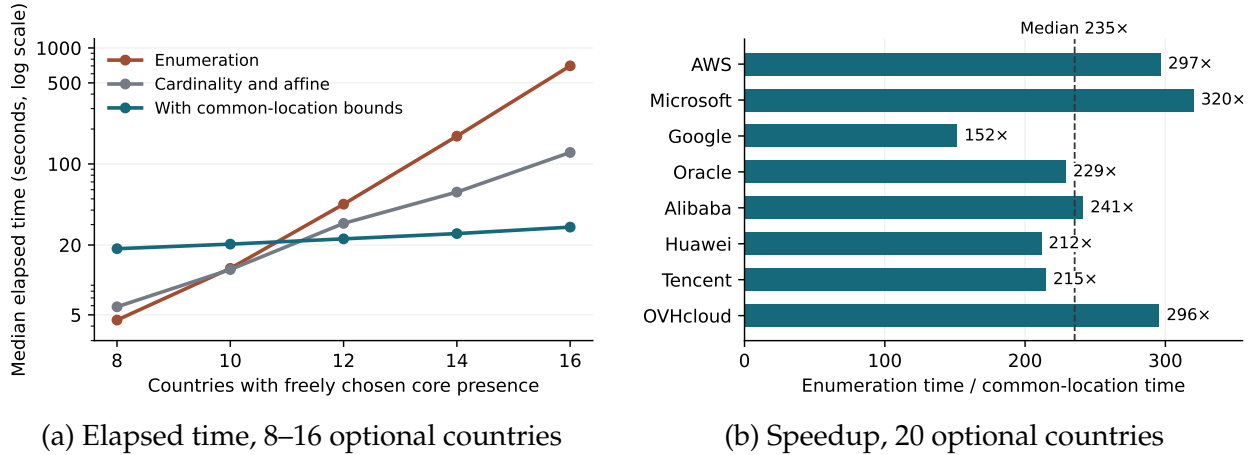


Figure 3: Portfolio search compared with exhaustive enumeration

Notes: The sample comprises all eight providers, one prespecified nested sequence of optional countries, and two private-cost vectors per firm. Other core choices are fixed to the observed network. Times include setup and the solution of both types; each method can reuse its own operating-profit evaluations. All searches finish, and optimal profits agree across methods. Panel (a) reports median elapsed times with two firms running concurrently. Panel (b) reports enumeration/common-location-bound time ratios with eight firms running concurrently; each method uses one computation thread. Each comparison uses a single timing run.

From optimal networks to a portfolio equilibrium. The search solves the best-response problem for a given rival distribution; equilibrium additionally requires the distributions of optimal portfolios to reproduce the distributions taken as given. Averaging each firm's optimal choices over private costs gives $\mathcal{B}_f(Q_{-f})$ in (3.5).

We update distributions on candidate networks and use global searches to check for profitable alternatives. Discovered portfolios enlarge the candidate sets before distributions are recomputed. Section 4.3 describes the procedure and final checks.¹⁵

Equilibrium uniqueness. Increasing the scale of private-cost shocks lowers a uniform upper bound on the aggregate sensitivity of location choices to competitors (Mason and Valentinyi, 2007; Melo, 2022). Appendix A.5 gives a sufficient condition for a unique equilibrium and convergence of simultaneous best-response iteration: dispersion must dominate a bound on strategic feedback. Online Appendix S.1.1 retains the signs of market interactions in an alternative response bound.¹⁶

4. QUANTIFYING MULTINATIONAL CLOUD NETWORKS

Market shares inform providers’ relative competitiveness, while core-country counts inform their net location costs. Because competitiveness also affects the profitability of expanding a network, we calibrate these parameters jointly. The calibration matches world-wide shares and network size; regional composition and host-country characteristics provide separate checks of the implied geographic allocation.

4.1 Data and assigned inputs

Providers and networks. Our 2026 sample covers the eight providers in Table 3. The five covered by Gartner’s 2024 worldwide IaaS data account for 82.1 percent of that \$171.8 billion market (Gartner, 2025); with the assigned shares of Oracle, Tencent, and OVHcloud, the eight providers account for 88 percent in the calibration. The sample contains 169 core presences across 42 countries, giving 336 provider-country choices. Multiple core regions of the same provider in one country count as a single presence (Section 2.1). All 42 countries are potential core locations in the EU experiment; the China experiment adds operating-mode restrictions. The model EU comprises 12 sample countries.¹⁷

¹⁵Appendix B.5 separates global-search, integration, and equilibrium approximation errors.

¹⁶Online Appendix S.3.2 shows that the global normal-cost uniqueness condition fails at the baseline parameters.

¹⁷Austria, Belgium, Denmark, Finland, France, Germany, Ireland, Italy, the Netherlands, Poland, Spain, and Sweden.

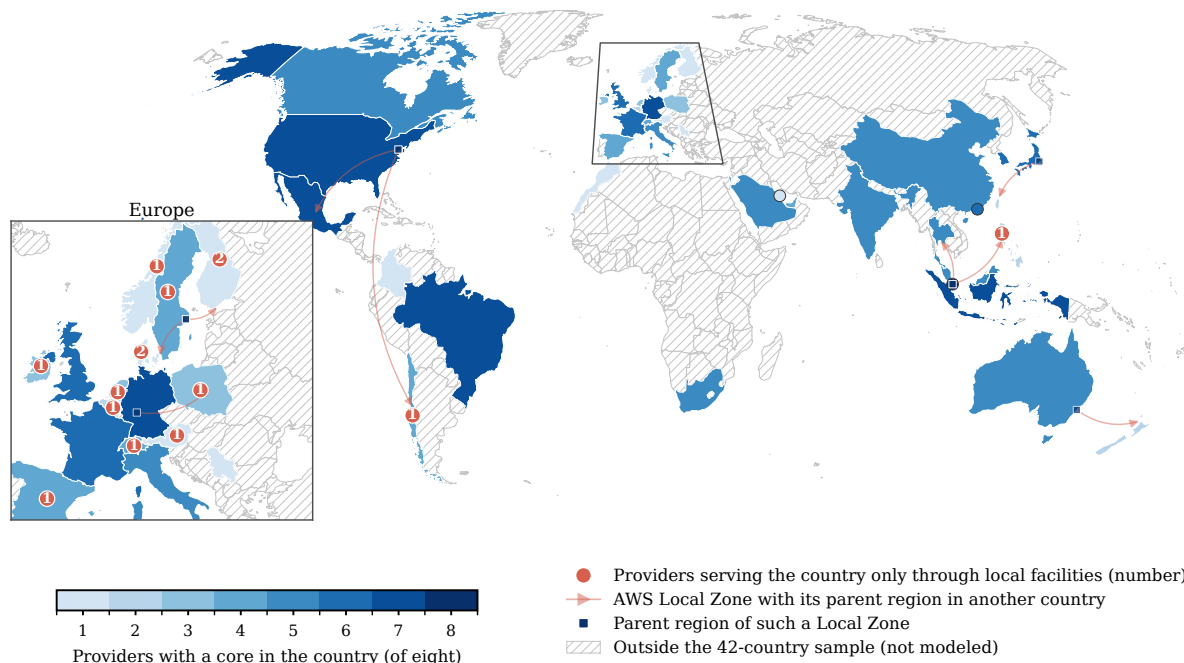


Figure 4: Observed core presence and local service

Notes: Shading gives the number of the eight providers with a core presence in each of the 42 sample countries; Bahrain, Hong Kong, and Singapore are also shown as points. Circles give the number of providers that serve a country only through local facilities. Arrows connect AWS parent regions to Local Zones in other countries; city locations are approximate. The inset enlarges Europe. Hatched countries are outside the sample and are not modeled. AWS's and Microsoft's mainland China regions are operated by local partners (Section 5.2). Sources: Appendix E.1; country boundaries from Natural Earth.

The local-service data contain 60 entries, with documented parents for 34 and unknown parents for 26.¹⁸ Among the 33 AWS entries with a documented parent, nine rely on a parent in another country. Figure 4 maps core presence and local-only service, including these cross-border links; Appendix E.1 lists the sources and parent assignments. Figure 1 illustrates how a core supports local service in another country. Worldwide size and regional concentration differ: OVHcloud has cores in ten sample countries and Alibaba in 17, yet OVHcloud has four EU core presences compared with Alibaba's two.

Market demand. We proxy country cloud expenditure by $(\text{GDP} \times \text{internet-use rate})^{0.7}$, normalized to one in the United States. The proxy gives more weight to large, digitally connected markets while moderating cross-country differences in size. In each model-EU

¹⁸We calibrate investment costs to core-country counts. The facility listings describe local locations and parent links but do not supply separate measures of local entry costs.

country, we assign 18 percent of expenditure ($\omega = 0.18$) to public-sector customers covered by procurement rules and 82 percent to unrestricted customers. The groups face the same supply rules in the baseline but different eligibility rules in the EU policy experiments (Section 5.1).¹⁹

We set $\sigma = 5$ for substitution across providers and $\eta = 2$ for the share–markup schedule. At given prices and portfolios, home preference multiplies a provider’s price-index contribution by four. This applies to US providers serving US customers, Chinese providers serving China, and OVHcloud serving the model EU. Relative competitiveness is calibrated conditional on these demand shifts.

Service costs and accessibility. Input prices and delivery costs determine the cost of serving a market from a given location; accessibility determines how customers value the available service locations. We set $\theta = 3$ for supply aggregation and $(\chi, \lambda_A) = (0.4, 1.5)$ for accessibility. Together with σ , these values give the supply and access exponents $\nu_S = 4/3$ and $\nu_A = 4/15$, with combined exponent 1.6 (Section 3.3).²⁰

We use the AWS Local Zone price premium as a benchmark for the relative delivered cost of local service. The median premium over parent-region prices is 25 percent across the 33 AWS local entries, comparing the same instance type.²¹ At $\theta = 3$, supply intensity γ_{loc} implies a delivered-cost factor $\gamma_{\text{loc}}^{-1/3}$ relative to a core at the same location. Using the price premium as this cost benchmark gives $\gamma_{\text{loc}} = 1.25^{-3} \approx 0.51$; we set $\gamma_{\text{loc}} = 0.5$. This weight also determines the supply gain from adding a core where local service already exists. Appendix E.2 describes the overlap in virtual-machine supply and differences in supported features.

The operating input-cost index w_i combines 2025 industrial electricity prices from Eurostat, the IEA, and the US EIA (Eurostat, 2025; International Energy Agency, 2025; U.S. Energy Information Administration, 2025) with a human-capital-adjusted labor-cost

¹⁹Treating the two groups as separate markets gives 54 customer markets. The regulated expenditure share is a scenario parameter. The Commission’s figures for the scope of SecNumCloud concern shares of public-sector use cases, with a different denominator from total cloud expenditure (European Commission, 2026b, Part 1, p. 43).

²⁰Supply exponents above one also appear in related sourcing and production models. The benchmark estimates in Antràs et al. (2017) imply $(\sigma - 1)/\theta \approx 1.59$. AES use the analogous exponent $(\sigma - 1)/(\epsilon - 1) = 1.5$ as their complements benchmark, chosen symmetrically to their substitutes benchmark of $2/3$.

²¹On-demand Linux prices from the AWS EC2 pricing page, excluding GPU instances, retrieved September 30, 2026.

Table 2: Calibrated parameters and assigned inputs

Parameter or input	Value	Economic role
<i>Panel A. Calibrated parameters</i>		
δ_f	Provider-specific	Average net location cost; fitted to worldwide core counts.
a_f/a_{AWS}	$a_{\text{AWS}} = 1$	Relative competitiveness; fitted to global shares jointly with residual supply.
Residual intensity	$b \simeq 2.045$	Common price-index contribution of other suppliers; fitted to their aggregate share.
<i>Panel B. Assigned parameters and inputs</i>		
(σ, η, θ)	$(5, 2, 3)$	Substitution across providers, the share-markup schedule, and supply aggregation.
Country cost adjustment	$c_i^0 \in \{0, 0.614\}$	Country differences in net location costs; Appendix D.3 gives the country assignments.
Distance cost gradient	$\kappa = 0.5$	Net location-cost gradient in distance from the provider's home market.
ℓ_f^{ref}	$G_f^{\text{ref}}/K_f^{\text{ref}}$	Scale of country-cost differences and private shocks; reference operating profit per core country.
σ_ε	0.5	Dispersion of private location costs before provider-specific scaling.
γ_{loc}	0.5	Supply intensity of local-only service relative to a core; benchmarked to AWS Local Zone prices.
Accessibility	$\lambda_A = 1.5, \chi = 0.4$	Customers' valuation of access to available service locations.
Home preference	$d_{fn} = \log 4$ at home; 0 elsewhere	Domestic demand advantage; the home market for OVHcloud is the model EU.

Notes: Appendix Table D.1 reports the provider-specific calibrated intercepts and competitiveness parameters. Parameters and assigned inputs are held fixed across policy regimes. Values are rounded for display.

proxy from the Penn World Table (Feenstra et al., 2015). Delivery costs rise with geographic separation. Within the EU, measured round-trip-time rankings assign higher accessibility indices to foreign origins with shorter round-trip times; domestic indices and pairs outside the EU retain their geographic specification. Appendix D.2 gives the delivery-cost values, input formulas, and sources.

Table 2 distinguishes the calibrated parameters from the assigned inputs.

Location costs. We set the distance gradient and private-cost dispersion at $\kappa = \sigma_\varepsilon = 0.5$, with distance measured in 10,000 kilometers. A fixed country adjustment c_i^0 is 0.614 in ten sample countries and zero elsewhere (Appendix D.3). Before calibration, we fix the provider-specific cost scaling factors at $\ell_f^{\text{ref}} = G_f^{\text{ref}}/K_f^{\text{ref}}$, reference operating profit G_f^{ref} per reference core country. These factors scale country-cost differences and private shocks, while the calibrated intercept δ_f shifts each provider’s average net location cost. Appendix D.3 defines the reference networks and reports the factors.

4.2 Calibration and baseline fit

Provider-specific intercepts δ_f are calibrated to target worldwide core counts, while relative competitiveness and aggregate residual supply target global market shares. We normalize $a_{\text{AWS}} = 1$ and fit a common residual price-index contribution $b_n = b$ across markets. Residual suppliers’ realized shares vary with the modeled providers’ contributions. Table 3 lists the targets.²²

Expected core counts sum country-inclusion probabilities; worldwide shares average market shares across portfolio profiles and weight markets by expenditure. We adjust these parameters jointly, recomputing baseline portfolio distributions and prices after each update to account for the feedback between competitiveness, network choices, and profits.

Model shares are within 0.55 percentage points of their targets, and expected worldwide core presence is 171.35, compared with 169 observed (Table 3). The largest share error is for Tencent; other errors are at most 0.22 points. The core-count RMSE is 1.01 countries, with Tencent accounting for the largest discrepancy: 15.80 cores against 13 observed.

Regional core counts provide an untargeted check of geographic allocation. The model predicts 52.88 EU core presences against 40 observed, a 32.2 percent overprediction. The calibration uses common spatial-cost coefficients without an EU-specific cost intercept. Across providers, the model compresses differences in the EU share of core locations, overpredicting that share for AWS, Oracle, and the Chinese providers and underpredicting it for OVHcloud. Appendix D.4 reports these and other composition checks.

²²The global share data cover a broader IaaS market than the representative workload class; Appendix E.2 discusses this difference.

Table 3: Worldwide market shares and core-country counts

Provider	World share (%)		World core count	
	Target	Model	Target	Model
AWS	37.7	37.57	28	27.96
Microsoft	23.9	23.76	33	32.93
Google	9.0	8.88	29	28.80
Oracle	4.0	4.01	27	27.14
Alibaba	7.2	6.98	17	16.69
Huawei	4.1	4.22	12	12.29
Tencent	2.0	2.55	13	15.80
OVHcloud	0.1	0.10	10	9.73
Eight providers	88.0	88.07	169	171.35
Residual	12.0	11.93	—	—

Notes: Core-country counts refer to 2026, while public worldwide market-share data refer to 2024. Model shares and expected core counts are evaluated at the reported baseline portfolio distribution. Shares average over 4,096 portfolio profiles; counts are computed directly from firms’ distributions. Worldwide counts and share ratios are calibration targets. Parameters are held fixed when candidate sets and numerical integration are refined after calibration. Appendix F reports numerical accuracy checks.

Core presence and service coverage also differ because local facilities can supply a country without a domestic core. Appendix Figure D.1 compares both with the data: the model underpredicts cores in Germany and overpredicts them in Finland, Austria, and Denmark. Appendix Table F.7 reports the fixed-network price effect under observed geography.

4.3 Equilibrium computation and numerical accuracy

We first compute portfolio distributions on a restricted set of complete networks for each firm, then use global searches to check for more profitable networks outside these sets, following Section 3.3.

Each firm has 1,536 simulated cost types and a candidate set of complete networks common across regimes within an application. The EU sets contain 10,622–13,323 networks per firm. We integrate rivals’ portfolios with a fixed quadrature of 4,096 points and solve AB prices in every portfolio profile. Appendix F.2 describes the integration procedure.

At each iteration, each cost type selects its best candidate network against the current rival distributions. To stabilize the iteration, we average each firm’s current portfolio

distribution with its best-response distribution on the candidate set, updating all firms simultaneously. Appendix F.2 gives the update and stopping rules.

The branch-and-bound search then maximizes profit over all feasible portfolios at the computed rival distributions. When a search finds a network that improves on the best candidate, we add it to all regimes' candidate sets within the application and recompute distributions. The final EU searches evaluate each regime at its computed equilibrium rival distributions, covering every simulated type in all eight firms and three regimes. All 36,864 searches meet the stated global optimality tolerance, and none finds an improving network outside the candidate sets. For every firm and regime, mean regret relative to all feasible portfolios is at most $1.3 \times 10^{-5} G_f^{\text{ref}}$.²³ Table notes report differences from an independent 4,096-point validation quadrature.

In the EU application, responses to unchanged rivals and one-firm responses start from each type's best candidate network and improve it by adding or dropping one country at a time, yielding local optima. The China responses to unchanged rivals select the best candidate network without this additional search. Appendix F.3 defines the EU comparisons, and Online Appendix S.4 compares alternative response procedures.

5. INTERNATIONAL NETWORK ADJUSTMENT

The EU experiment restricts which service origins can supply public-sector customers; the China experiment changes the cost of supplying mainland customers from offshore facilities. Both alter the return to facilities serving several customer markets. In both applications, competitor responses substantially moderate worldwide network contraction while leaving price effects in directly affected markets nearly unchanged.²⁴

We use three network comparisons. With *fixed networks*, every provider retains its baseline portfolio distribution. Under *responses to unchanged rivals*, all eight providers can adjust, but each expects rivals to retain their baseline distributions. We combine the resulting

²³The regret upper bound is the search's bound on attainable profit minus the type's expected profit under its computed strategy, plus a pricing-error allowance. Closing a search identifies the best attainable payoff; regret can remain when the computed strategy places weight on less profitable portfolios. The reported bounds condition on simulated cost types and fixed rival quadrature. Appendix F.2 reports type-level maxima.

²⁴Online Appendix Tables S.8.1 and S.8.3 separate effects on US providers, US customers, and cores located in the United States.

network choices and solve prices jointly. Under *equilibrium responses*, each provider instead anticipates the policy-specific rival distributions generated by the others' choices. Prices are solved jointly in all three comparisons. Appendix F.3 defines them formally.

5.1 EU public-sector cloud procurement: EU-wide eligibility and national fragmentation

Public-sector cloud procurement restricts who can supply a service and where it can be supplied from. The European Commission's *Cloud Sovereignty Framework*, applied in a 2026 procurement for EU institutions, assesses legal, operational, and technological control (European Commission, 2026a). Member states also operate separate qualification schemes, such as France's SecNumCloud (Agence nationale de la sécurité des systèmes d'information, 2022). Our experiment isolates the geographic criterion by comparing eligibility throughout the EU with national fragmentation.

The rules apply to 18 percent of cloud expenditure in each model-EU country ($\omega = 0.18$). These public-sector customers may purchase services supplied from anywhere in the model EU under EU-wide eligibility. Under National, every model-EU country restricts its public-sector customers to domestic service origins.²⁵ The remaining customers face no geographic sourcing restriction. The two groups have separate prices and market shares but are served through the same provider networks. The unrestricted baseline imposes neither rule.

Market access and the location of investment. EU-wide sourcing requirements increase core presence within the EU, but national requirements reduce it (Table 4). The unrestricted baseline has 171.35 expected core presences worldwide, including 52.88 within the EU. EU-wide eligibility increases the EU total to 54.95, a 3.9 percent rise, whereas National eligibility reduces it to 50.27, a 4.9 percent decline. Both policies reduce worldwide core presence, by 2.9 and 5.4 percent, respectively. The public-sector price index, normalized to 100 under unrestricted sourcing, rises to 127.73 under EU-wide and 164.04 under National eligibility.

²⁵Eligibility depends on the service origin rather than the provider's nationality. A domestic local facility can serve regulated customers under National even if its parent core is abroad, provided that parent is in the provider's chosen portfolio.

Table 4: Core presence and public-sector prices by sourcing regime

Outcome	Unrestricted	EU-wide	National
World core presences	171.35	166.45	162.13
EU core presences	52.88	54.95	50.27
Non-EU core presences	118.46	111.50	111.86
EU public-sector price index	100.00	127.73	164.04

Notes: All entries use equilibrium responses. Core counts cover the eight modeled providers in the 42-country sample; residual suppliers do not choose locations. The price index includes residual supply and refers to covered public-sector customers in the 12 model-EU countries. It exponentiates the expenditure-weighted expected log index, normalizing unrestricted sourcing to 100 (Appendix F.3). The rules apply to 18 percent of expenditure in each EU country.

EU-wide eligibility directs regulated demand toward EU facilities while preserving access to customers across member states. National eligibility removes that shared customer base while favoring domestic supply. A subsidy lowers domestic location costs; the sourcing rule changes the customers a network can serve. National sourcing therefore weakens the return to facilities serving several EU markets while encouraging domestic replication.

A core incurs its location cost regardless of how many markets it serves. Losing access to customers elsewhere in the EU can therefore make a core unprofitable to retain. In the calibration, this loss of shared market access dominates the replication incentive. Appendix F.1 formalizes these opposing investment incentives. In the paired pure best-response comparison, National yields only 0.01 additional EU cores against 2.62 removals relative to unrestricted sourcing (Appendix Table F.1).

Against baseline rivals, two illustrative cost draws yield contraction as National coverage expands within public-sector spending: AWS drops Spain and Oracle drops the Netherlands, with all other locations unchanged. In each case, the lower public-sector return to the core makes the location-cost saving exceed the profit forgone across all customer markets (Appendix Table F.2).

We next compare National with EU-wide sourcing to isolate the additional effect of fragmenting the EU public-sector customer market. Table 5 reports this contrast under the

three network comparisons. Both responses to unchanged rivals use the same unrestricted baseline distributions Q^O .²⁶

Table 5: National fragmentation: core networks, prices, and competitor responses

Outcome	Fixed networks	Responses to unchanged rivals	Equilibrium responses
<i>Core networks</i>			
World core presences	0.00	-8.59	-4.32
EU core presences	0.00	-5.83	-4.68
Non-EU core presences	0.00	-2.76	0.36
<i>Price indices (%)</i>			
Regulated EU	27.52	28.28	28.43
Private-sector EU	0.00	1.00	0.60
Non-EU	0.00	0.72	0.31

Notes: Core entries are differences in expected provider-country counts (National minus EU-wide). Price entries are percentage changes relative to EU-wide, computed from expenditure-weighted expected log price indices. The network comparisons are defined in Section 5. Prices are solved jointly for each portfolio profile before aggregation. Residual supply remains locally eligible with a fixed price-index contribution. All columns use the main 4,096-point rival quadrature; recomputing them on the independent 4,096-point validation quadrature changes entries by at most 0.03. Appendix Table F.6 reports the associated eligibility and residual-share measures.

Competitor responses and customer prices. For the National–EU-wide contrast, the worldwide core-count reduction is 8.59 against rivals’ unrestricted-baseline distributions and 4.32 under equilibrium responses (Table 5). Rival feedback therefore attenuates this additional contraction by about 50 percent.

A rival’s withdrawal can make a core worth retaining by increasing the additional business it brings to the provider’s network. What matters is how much the core adds to profit: the provider may also gain business without it. Appendix F.1 gives conditions under which the incremental return rises as rivals withdraw. Most of the attenuation arises outside the EU, where the National–EU-wide difference changes from a loss of 2.76 core presences to a gain of 0.36. Non-EU core presence remains below the unrestricted baseline under both policies (Table 4). Oracle’s non-EU change, for example, moves from -0.58 to $+0.08$.

²⁶Price effects are based on expenditure-weighted expected log price indices; Appendix F.3 gives the aggregation formula.

For one Oracle cost draw, rival adjustment changes whether the Philippine core is included under either sourcing rule. Against baseline rivals, both policy responses exclude it; against equilibrium rivals under each policy, both include it. These decisions compare the best networks with and without the Philippine core, reoptimizing all 41 other locations. Appendix F.1 reports the inclusion thresholds.

The public-sector price index is about 28 percent higher under National than under EU-wide eligibility in all three columns. Competitor responses therefore leave this price effect nearly unchanged. Most of it arises from the loss of eligible supply with networks held fixed. The price index reflects this loss as well as prices at the remaining providers. Network adjustment restores little eligible service in the calibration. A core's country alone does not determine its contribution: domestic local facilities can be supported by an overseas parent core. Across the two response comparisons, the loss of eligible modeled providers is similar (2.79 versus 2.68), and residual suppliers gain about 57 percentage points of market share in both (Appendix Table F.6). Under equilibrium responses, their share rises from about one-third under EU-wide eligibility to nine-tenths under National.

Geographic adjustment and international price spillovers. Under equilibrium responses, every model-EU country loses expected core presences relative to EU-wide eligibility. The EU total falls by 8.5 percent (Figure 5, panel (a)). Appendix Table F.5 reports the corresponding EU-wide–baseline comparison.

Portfolio adjustment transmits the policy to customers whose eligibility rules have not changed. Their prices are unchanged at fixed networks, because the model has no cross-market capacity or input-price channel. With equilibrium responses, private-sector EU and non-EU price indices are 0.60 and 0.31 percent higher under National than under EU-wide eligibility. Both increases are smaller than under responses to unchanged rivals (Table 5). Figure 5, panel (b), shows how these spillovers vary across countries.

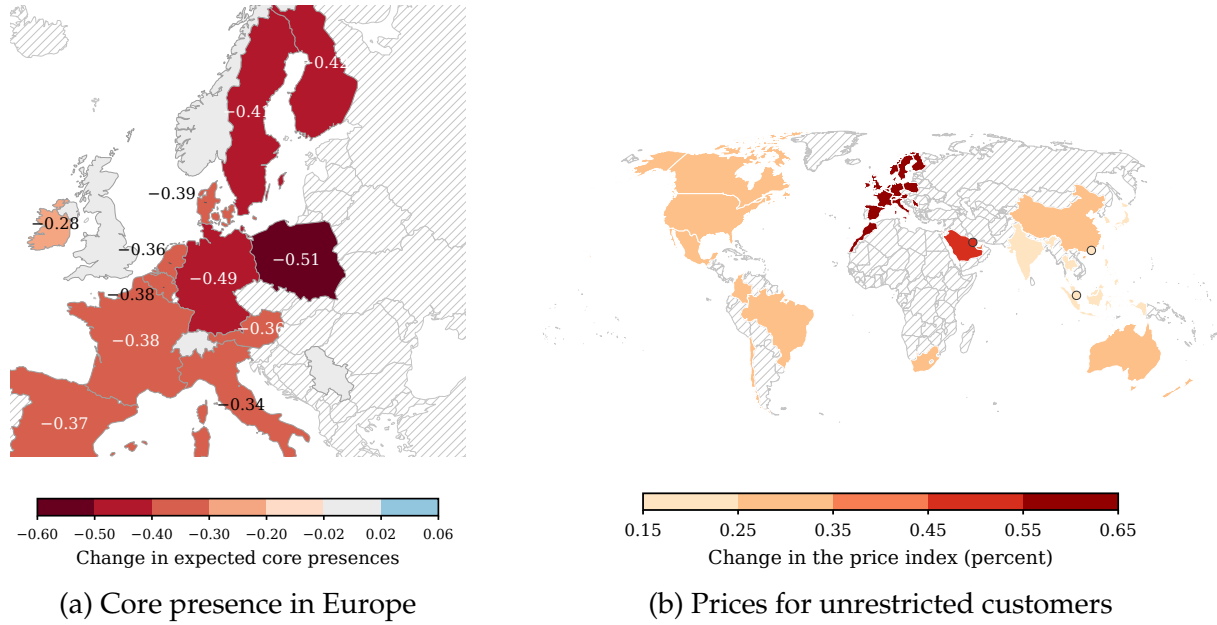


Figure 5: National versus EU-wide eligibility: core presence and price spillovers

Notes: National relative to EU-wide eligibility under equilibrium responses and AB pricing. Panel (a) reports country-level changes in expected core presence, summed across providers; grey denotes absolute changes below 0.02. Panel (b) reports price-index changes (percent) for private-sector customers in the model EU and all customers elsewhere. Hatched countries are outside the sample. Boundaries: Natural Earth.

National eligibility thus leaves non-EU prices higher than EU-wide eligibility despite having slightly more non-EU cores. EU cores contribute to both the cost and accessibility of service outside Europe. Delivery costs and accessibility differ across origins and destinations, so the additional non-EU locations do not fully replace the contributions lost through EU contraction. Prices depend on these market-specific service contributions, not just the number of locations.

Provider networks and operating profits. All eight providers have fewer EU cores under National than under EU-wide; outside the EU, changes are small and differ in sign across providers (Figure 6, panels (a) and (b)). OVHcloud accounts for about 30 percent of the worldwide count reduction.

Every modeled provider loses worldwide operating profit despite the rise in customer prices (Figure 6, panel (c)). National eligibility redirects regulated expenditure toward eligible residual suppliers. The modeled providers lose sales, which accounts for most of their operating-profit loss (Appendix F.5).

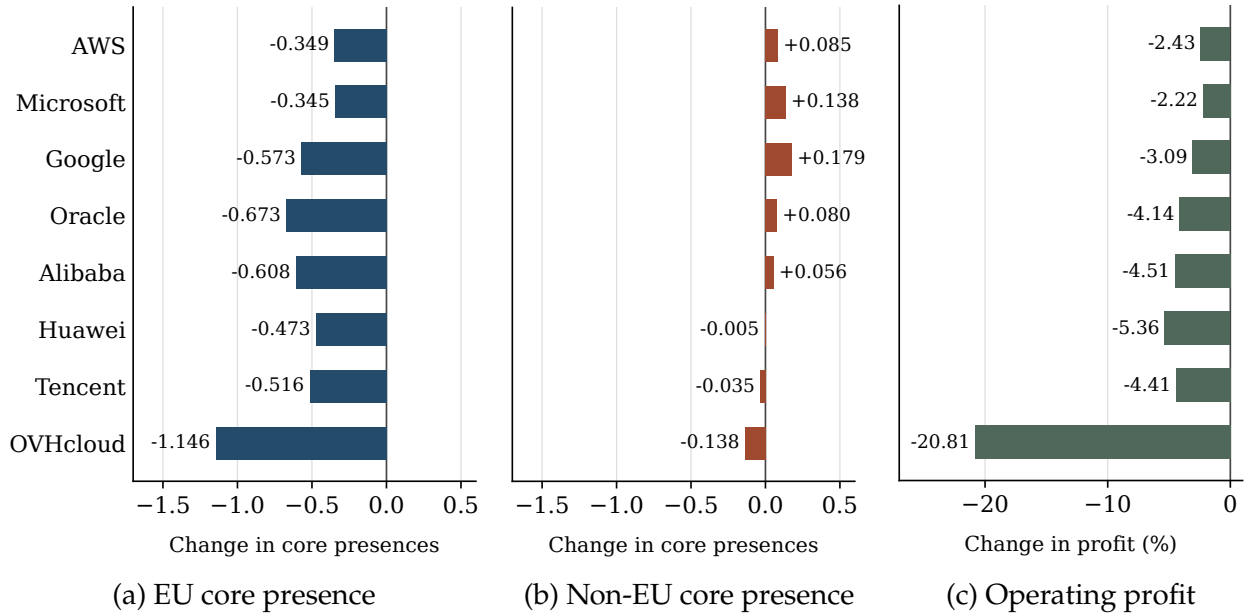


Figure 6: Provider networks and operating profits under national fragmentation

Notes: National relative to EU-wide eligibility under equilibrium responses and AB pricing. Panels (a) and (b) show changes in expected core-country presences; EU and non-EU changes add to each firm's worldwide change. Panel (c) shows the percentage change in expected worldwide operating profit, before location costs.

OVHcloud loses 20.81 percent of its worldwide operating profit, much more in proportional terms than the other providers. Under EU-wide eligibility, regulated EU markets generate about 15 percent of its worldwide profit, compared with roughly 2–3 percent for the US and Chinese provider groups. Providers lose similarly large proportions of their profits in these markets, so OVHcloud's greater exposure magnifies the worldwide effect. About two-thirds of its loss arises in regulated markets; the rest arises through portfolio adjustment in unregulated markets (Appendix F.5).

Joint portfolio choice and independent cutoffs. Independent country cutoffs arise in separable export models (Chaney, 2008) and in the separable sourcing case of Antràs et al. (2017). We compare the reported portfolio responses with a baseline-fitted country-by-country cutoff approximation (Table 6). Each country is selected when its private location cost falls below its cutoff. The cutoffs are fitted to reproduce baseline participation probabilities and updated under each policy, holding rivals at their baseline distributions.

Despite matching baseline participation, the independent cutoff approximation misses the adjustment of non-EU cores induced by changes in the EU network. Its non-EU cutoffs

do not change when sourcing eligibility changes only within the EU. Joint portfolio choice allows changes in the EU network to alter the return to cores elsewhere, because locations jointly determine delivered costs and accessibility across markets.

Conditioning inclusion on the other current choices retains those interactions, but repeated one-country improvements can still miss joint changes. For one Tencent cost type under EU-wide eligibility, removing Mexico, the Netherlands, and Singapore together from its response to unchanged baseline rivals raises net profit even though every one- and two-country change lowers it. This response selects the best candidate portfolio and improves it through profitable one-country additions or removals.

Rivals' joint networks also matter: preserving their country-level entry frequencies while changing which locations occur together reverses portfolio rankings in a controlled comparison. Appendix C.3 defines the cutoff approximation and reports these comparisons.

Table 6: EU policy responses: joint portfolios and independent cutoffs

Response procedure	Change in expected core presences		
	World	EU	Non-EU
Reported portfolio responses	-8.59	-5.83	-2.76
Baseline-fitted independent cutoffs	-5.17	-5.17	0.00

Notes: National minus EU-wide eligibility, holding rivals' baseline portfolio distributions fixed.

Alternative specifications and response procedures. Geographic fragmentation reduces core presence and raises the price index for regulated customers under constant markups as well (Table 7). This comparison changes the markup schedule while retaining the supply exponent $\nu_S = 4/3$ and recalibrating to the same targets. Under CES pricing, competitor responses offset about two-fifths of worldwide contraction, compared with one-half under AB, while the price increase for these customers remains close to 28 percent. The specifications differ in where networks adjust: non-EU core presence falls slightly under CES instead of rising. Proportional operating-profit losses are similar, although baseline profits for the largest providers are higher under AB (Online Appendix Figure S.6.1).

Table 7: National fragmentation under AB and constant-markup CES pricing

Outcome	AB pricing		CES pricing	
	Unchanged rivals	Equilibrium	Unchanged rivals	Equilibrium
<i>Core networks</i>				
World core presences	-8.59	-4.32	-8.02	-4.68
EU core presences	-5.83	-4.68	-5.47	-4.54
Non-EU core presences	-2.76	0.36	-2.55	-0.14
Attenuation of world contraction (%)	49.71		41.68	
Regulated EU price index (%)	28.28	28.43	27.63	27.74

Notes: Core entries are differences in expected provider-country counts (National minus EU-wide). Price entries are percentage changes relative to EU-wide, computed from expenditure-weighted expected log price indices. The network comparisons are defined in Section 5. CES sets $\eta = \sigma = 5$ and is recalibrated to the same targets; Appendix C.2 describes the candidate-set expansion. Attenuation is one minus the ratio of the equilibrium worldwide core-count reduction to the reduction against unchanged rivals. All entries use the main 4,096-point rival quadrature.

The supply alternatives remaining after regulation determine the size of the price increase. In the central specification, residual supply remains eligible with a fixed price-index contribution. For the National–EU-wide comparison, repricing fixed networks under alternative residual-supply assumptions yields public-sector price-index increases of about 18–48 percent (Appendix Table F.7). These comparisons show how eligible residual supply buffers the loss of modeled supply. The baseline network configuration also matters: with observed cores and the same local topology and parameters, the fixed-network price effect is 22.02 percent, compared with 27.52 percent at the calibrated baseline.

Network adjustment is more sensitive than prices to how responses to unchanged rivals are computed. Add-or-drop searches starting from baseline portfolios imply attenuation of about 38 percent, compared with 50 percent for the reported responses; selecting the most profitable portfolios found across the tested procedures gives 48 percent. The price-index effect for regulated customers remains between 28.23 and 28.28 percent (Online Appendix S.4.1).

5.2 China: offshore delivery costs and the multinational network

China’s Great Firewall restricts access to foreign online services (Sun, 2026), while restrictions on cross-border data flows affect digital-service trade (Sun and Trefler, 2026). Our

experiment varies the cost of offshore cloud supply while retaining the existing operating arrangements.

The model allows Chinese providers to serve mainland customers from both domestic and offshore facilities.²⁷ Higher offshore delivery costs therefore affect both their foreign competitors and their own networks.

Alibaba, Huawei, and Tencent can choose both mainland and offshore cores. The five non-Chinese providers cannot add a directly controlled mainland core in this scenario, but can continue offshore service to China. AWS and Microsoft also retain service through fixed partner-operated facilities, which are excluded from selectable core counts.²⁸ The baseline distribution is recomputed under these arrangements. It has 168.11 expected core presences worldwide, including 2.28 selectable mainland presences across the three Chinese providers.

The experiment multiplies the delivery cost $\tau_{i,\text{CHN}}$ from each offshore origin i to mainland customers by e^d , lowering d from zero to -0.10 or raising it to 0.25 (a 9.52 percent decrease or a 28.40 percent increase) relative to this segmented baseline; Online Appendix S.5.1 gives the specification. The change applies to offshore service by Chinese and foreign providers alike. Domestic supply costs and delivery costs to other markets remain unchanged.

Mainland investment and the international network. Tightening increases mainland core presence for all three Chinese providers, yet their worldwide networks contract (Figure 7). A mainland core avoids the higher delivery cost. Offshore cores, however, serve several markets, with Chinese sales helping cover their location costs. Losing some of those profits can make offshore locations less attractive to retain even as mainland investment becomes more attractive.

²⁷Tencent Cloud offers traffic packages for mainland users accessing services deployed in Hong Kong and Singapore. Alibaba Cloud also describes mainland access to its Hong Kong and other offshore server regions, while recommending domestic regions for mainland-facing workloads (Tencent Cloud, 2026; Alibaba Cloud, 2026). These examples establish the availability of offshore service, not its share of mainland demand.

²⁸Local operators supply these branded services through separate contracts and accounts (Online Appendix Table S.7.2). The restrictions describe the maintained operating arrangements, not a general legal prohibition of foreign participation. Partner-operated facilities have no private core-cost draws; the central accounting assigns their modeled branded operating profit to the provider. Partnership formation and profit-sharing changes are outside the experiment.

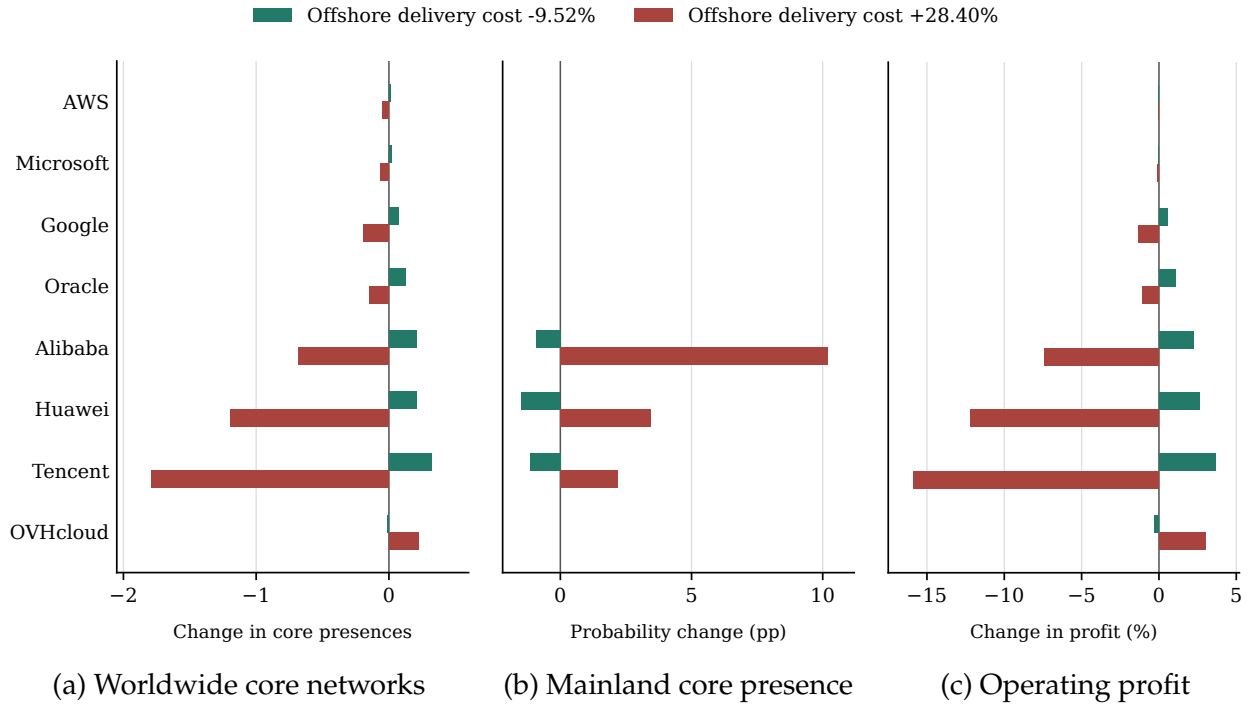


Figure 7: China frictions: provider networks and operating profits

Notes: Equilibrium responses relative to the segmented baseline. Panel (a) shows changes in worldwide core presences; panel (b) shows changes in the probability of a mainland core for the three Chinese providers, the only providers that can choose one; panel (c) shows changes in worldwide operating profit before location costs. AWS's and Microsoft's partner-operated mainland facilities are not counted as cores.

The three Chinese providers account for about 94 percent of the net worldwide contraction. Across all eight firms, mainland core presence rises from 2.28 to 2.44 (6.9 percent), while offshore presence falls by 4.06. Additional mainland cores offset only a small part of the offshore contraction. The three Chinese providers' worldwide operating profits fall by about 7–16 percent (Figure 7, panel (c)).²⁹ Under easing, their mainland-core probabilities decline while worldwide core presence rises from 168.11 to 169.06 (0.6 percent) and operating profits increase.

Competitor responses and customer prices. As in the EU experiment, rival withdrawal can raise the incremental return to retaining a core and moderate contraction. Under tightening, worldwide core presence falls by 7.36 against unchanged rivals (4.4 percent

²⁹Profits net of location costs also fall (Online Appendix Figure S.5.1).

Table 8: China: prices and core networks under easing and tightening

Network response	China price index (%)	Non-China price index (%)	World cores (presences)	China cores (presences)
<i>Easing: offshore delivery cost -9.52%</i>				
Fixed networks	-7.84	0.00	0.00	0.00
Responses to unchanged rivals	-8.11	-0.14	+1.43	-0.03
Equilibrium responses	-8.05	-0.11	+0.96	-0.04
<i>Tightening: offshore delivery cost +28.40%</i>				
Fixed networks	+19.37	0.00	0.00	0.00
Responses to unchanged rivals	+20.72	+0.70	-7.36	+0.12
Equilibrium responses	+20.25	+0.42	-3.90	+0.16

Notes: All scenarios use common candidate sets and the same baseline reference, which retains existing segmentation. The three response comparisons are defined in Section 5. Prices are solved jointly for every profile; reported changes use expenditure-weighted expected log indices. Core counts are expected provider-country presences, excluding fixed partner-operated facilities. Residual supply is unexposed to the shock ($\phi = 0$); its share changes are reported in Online Appendix Table S.5.1.

of the baseline total), compared with 3.90 under equilibrium responses (2.3 percent). Rival adjustment therefore attenuates the contraction by about 47 percent (Table 8).³⁰ All eight providers retain more worldwide cores under equilibrium responses than against unchanged rivals, so both Chinese and foreign providers contribute to the attenuation.

Retaining facilities does little to offset the higher cost of serving China from abroad: the price index rises by 19.37 percent at fixed networks and 20.25 percent under equilibrium responses. Outside China, supply conditions are unchanged at fixed networks, so prices rise only through network adjustment. The smaller equilibrium contraction reduces this increase from 0.70 to 0.42 percent.

Supply origins and the incidence across providers. Suppliers whose service is unaffected by the offshore restriction gain market share as the modeled providers become more costly. In the central calculation, residual supply remains unexposed and its China share rises from 9.78 to 20.39 percent under tightening. The three Chinese providers' combined share falls by 4.56 percentage points.³¹

³⁰The China responses to unchanged rivals are best responses in the common candidate sets, without the add-or-drop search used for the EU application. Computed in the same way, the EU attenuation is 44 percent (Online Appendix Table S.4.1).

³¹Online Appendix S.5.1 compares the baseline China shares with published market statistics and discusses differences in product coverage.

This incidence depends on the origins of residual supply. Exposing its entire contribution to the same cost increase raises the China price effect from 20.25 to 25.63 percent and changes the Chinese providers' share effect to approximately +0.09 percentage points (Online Appendix Table S.5.2). This comparison reprices the central policy portfolios, holding locations fixed. Thus the origins of competing supply affect which providers gain or lose market share.

6. CONCLUSION

Geographic restrictions change the customer base of facilities throughout a multinational network, including facilities outside the regulated market. Relative to unrestricted sourcing, both EU policies reduce worldwide core presence and raise public-sector prices. EU-wide eligibility increases aggregate EU core presence, whereas national eligibility reduces it. Competitor responses offset about half of the additional worldwide contraction under national rather than EU-wide eligibility while leaving the public-sector price difference nearly unchanged. Compared with EU-wide eligibility, national eligibility also lowers every modeled provider's operating profit, with the largest proportional loss for the EU-based OVHcloud. Higher costs of serving China from abroad raise Chinese providers' mainland presence while shrinking their worldwide networks and operating profits.

The framework makes complete-portfolio optimization tractable even when a location's net return switches from negative to positive and back as the network expands. Bounds with common location choices across markets and costs permit endogenous portfolio size. Global response checks bound the profit gains from any feasible deviation at the reported rival distributions, without imposing complementarity or single crossing.

REFERENCES

- Agence nationale de la sécurité des systèmes d'information**, "Prestataires de services d'informatique en nuage (SecNumCloud): référentiel d'exigences, version 3.2," Technical Report, ANSSI 2022. 8 March 2022.
- Aguirregabiria, Victor and Pedro Mira**, "Sequential Estimation of Dynamic Discrete Games," *Econometrica*, 2007, 75 (1), 1–53.
- Alfaro-Ureña, Alonso, Juanma Castro-Vincenzi, Sebastián Fanelli, and Eduardo Morales**, "Firm Export Dynamics in Interdependent Markets," *NBER Working Paper 31610*, 2025.
- Alibaba Cloud**, "Simple Application Server: Regions and Network Connectivity," 2026. Updated May 29, 2026; accessed September 30, 2026.
- Antràs, Pol, Evgenii Fadeev, Teresa C. Fort, and Felix Tintelnot**, "Export-Platform FDI: Cannibalization or Complementarity?," 2024. Manuscript, January 22.
- , —, —, and —, "Exporting, Global Sourcing, and Multinational Activity: Theory and Evidence from the United States," *Review of Economics and Statistics*, 2026, 108 (3), 553–571.
- , **Teresa C. Fort, and Felix Tintelnot**, "The Margins of Global Sourcing: Theory and Evidence from US firms," *American Economic Review*, 2017, 107 (9), 2514–2564.
- Arkolakis, Costas, Fabian Eckert, and Rowan Shi**, "Combinatorial Discrete Choice: Theory and Application to Multinational Production," Working Paper 31877, National Bureau of Economic Research 2026. Revised March 29, 2026.
- Atkeson, Andrew and Ariel Burstein**, "Pricing-to-Market, Trade Costs, and International Relative Prices," *American Economic Review*, 2008, 98 (5), 1998–2031.
- Australian Government**, "Hosting Certification Framework," Australian Government framework overview 2021. Framework released March 2021; framework overview accessed September 19, 2026.

- Berry, Steven T.**, “Estimation of a Model of Entry in the Airline Industry,” *Econometrica*, 1992, 60 (4), 889–917.
- Boyd, Stephen and Lieven Vandenberghe**, *Convex Optimization*, Cambridge University Press, 2004.
- Bresnahan, Timothy F. and Peter C. Reiss**, “Entry and Competition in Concentrated Markets,” *Journal of Political Economy*, 1991, 99 (5), 977–1009.
- Bulow, Jeremy I., John D. Geanakoplos, and Paul D. Klemperer**, “Multimarket Oligopoly: Strategic Substitutes and Complements,” *Journal of Political Economy*, 1985, 93 (3), 488–511.
- Casalini, Francesca, Javier López González, and Taku Nemoto**, “Mapping Commonalities in Regulatory Approaches to Cross-Border Data Transfers,” OECD Trade Policy Papers 248, OECD Publishing 2021.
- Castro-Vincenzi, Juanma**, “Climate Hazards and Resilience in the Global Car Industry,” *Working Paper*, 2024.
- , **Eugenia Menaguale, Eduardo Morales, and Alejandro Sabal**, “Market Entry and Plant Location in Multiproduct Firms,” 2025. Working paper.
- Chaney, Thomas**, “Distorted Gravity: The Intensive and Extensive Margins of International Trade,” *American Economic Review*, 2008, 98 (4), 1707–1721.
- Ciliberto, Federico and Elie Tamer**, “Market Structure and Multiple Equilibria in Airline Markets,” *Econometrica*, 2009, 77 (6), 1791–1828.
- Competition and Markets Authority**, “Cloud Infrastructure Services: Final Decision Report,” Technical Report, Competition and Markets Authority July 2025.
- Del Giovane, Chiara, Janos Ferencz, and Javier López González**, “The Nature, Evolution and Potential Implications of Data Localisation Measures,” OECD Trade Policy Papers 278, OECD Publishing 2023.
- Demirer, Mert, Diego Jiménez-Hernández, Dean Li, and Sida Peng**, “Data, Privacy Laws and Firm Production: Evidence from the GDPR,” 2024. NBER Working Paper No. 32146, revised February 2026.

- Dickstein, Michael J. and Eduardo Morales**, “What Do Exporters Know?,” *Quarterly Journal of Economics*, 2018, 133 (4), 1753–1801.
- Eaton, Jonathan, Samuel Kortum, and Francis Kramarz**, “An Anatomy of International Trade: Evidence From French Firms,” *Econometrica*, 2011, 79 (5), 1453–1498.
- Edmond, Chris, Virgiliu Midrigan, and Daniel Yi Xu**, “Competition, Markups, and the Gains from International Trade,” *American Economic Review*, 2015, 105 (10), 3183–3221.
- Ericson, Richard and Ariel Pakes**, “Markov-Perfect Industry Dynamics: A Framework for Empirical Work,” *Review of Economic Studies*, 1995, 62 (1), 53–82.
- European Commission**, “Commission Advances Cloud Sovereignty through Strategic Procurement,” European Commission announcement 2026a. April 17.
- , “Impact Assessment Report Accompanying the Proposal for a Cloud and AI Development Act, SWD(2026) 502, Parts 1 and 2,” Technical Report, European Commission 2026b. Brussels, 3 June 2026.
- , “Proposal for a Regulation of the European Parliament and of the Council Establishing a Framework of Measures for Strengthening Europe’s Cloud and AI Ecosystem (Cloud and AI Development Act), COM(2026) 502 final,” 2026c. Brussels, 3 June 2026.
- Eurostat**, “Electricity Prices for Non-Household Consumers, Bi-annual Data (nrg_pc_205),” Eurostat dissemination database 2025.
- Fan, Ying and Chenyu Yang**, “Competition, Product Proliferation, and Welfare: A Study of the U.S. Smartphone Market,” *American Economic Journal: Microeconomics*, 2020, 12 (2), 99–134.
- Feenstra, Robert C.**, “New Product Varieties and the Measurement of International Prices,” *American Economic Review*, 1994, 84 (1), 157–177.
- , **Robert Inklaar, and Marcel P. Timmer**, “The Next Generation of the Penn World Table,” *American Economic Review*, 2015, 105 (10), 3150–3182.
- Feldman, Jacob and Huseyin Topaloglu**, “Bounding Optimal Expected Revenues for Assortment Optimization under Mixtures of Multinomial Logits,” *Production and Operations Management*, 2015, 24 (10), 1598–1620.

- Ferracane, Martina Francesca, Janez Kren, and Erik van der Marel**, “Do Data Policy Restrictions Impact the Productivity Performance of Firms and Industries?,” *Review of International Economics*, 2020, 28 (3), 676–722.
- Gartner**, “Gartner Says Worldwide IaaS Public Cloud Services Market Grew 22.5% in 2024,” Worldwide IaaS market-share release 2025. August 6.
- Holmes, Thomas J.**, “The Diffusion of Wal-Mart and Economies of Density,” *Econometrica*, 2011, 79 (1), 253–302.
- Horn, Roger A. and Charles R. Johnson**, *Matrix Analysis*, second ed., Cambridge: Cambridge University Press, 2013.
- Houde, Jean-François, Peter Newberry, and Katja Seim**, “Nexus Tax Laws and Economies of Density in E-Commerce: A Study of Amazon’s Fulfillment Center Network,” *Econometrica*, 2023, 91 (1), 147–190.
- International Energy Agency**, “End-Use Prices Data Explorer,” IEA end-use prices data 2025.
- Jia, Panle**, “What Happens When Wal-Mart Comes to Town: An Empirical Analysis of the Discount Retailing Industry,” *Econometrica*, 2008, 76 (6), 1263–1316.
- Land, A. H. and A. G. Doig**, “An Automatic Method of Solving Discrete Programming Problems,” *Econometrica*, 1960, 28 (3), 497–520.
- Leitner, Markus, Andrea Lodi, Roberto Roberti, and Claudio Sole**, “An Exact Method for (Constrained) Assortment Optimization Problems with Product Costs,” *INFORMS Journal on Computing*, 2024, 36 (2), 479–494.
- Manganelli, Antonio and Daniel Schnurr**, “Competition and Regulation of Cloud Computing Services: Economic Analysis and Review of EU Policies,” CERRE report 2024. CERRE Report, Centre on Regulation in Europe.
- Mason, Robin and Ákos Valentinyi**, “The Existence and Uniqueness of Monotone Pure Strategy Equilibrium in Bayesian Games,” Discussion Papers in Economics and Econometrics 710, University of Southampton 2007.

- Melo, Emerson**, “On the Uniqueness of Quantal Response Equilibria and Its Application to Network Games,” *Economic Theory*, 2022, 74, 681–725.
- Morales, Eduardo, Gloria Sheu, and Andrés Zahler**, “Extended Gravity,” *Review of Economic Studies*, 2019, 86 (6), 2668–2712.
- Nishida, Mitsukuni**, “Estimating a Model of Strategic Network Choice: The Convenience-Store Industry in Okinawa,” *Marketing Science*, 2015, 34 (1), 20–38.
- Oberfield, Ezra, Esteban Rossi-Hansberg, Pierre-Daniel Sarte, and Nicholas Trachter**, “Plants in Space,” *Journal of Political Economy*, 2024, 132 (3), 867–909.
- Pohle, Julia and Thorsten Thiel**, “Digital Sovereignty,” *Internet Policy Review*, 2020, 9 (4), 1–19.
- Sabal, Alejandro**, “Product Entry in the Global Automobile Industry,” 2026. Working paper, University of Chicago, June 26.
- Seim, Katja**, “An Empirical Model of Firm Entry with Endogenous Product-Type Choices,” *The RAND Journal of Economics*, 2006, 37 (3), 619–640.
- Sun, Ruiqi**, “Blocking the Giants: Theory and Evidence from the Great Firewall,” 2026. CESifo Working Paper No. 12913.
- **and Daniel Trefler**, “The Impact of AI and Cross-Border Data Regulation on International Trade in Digital Service,” *Journal of International Economics*, 2026, 164, 104328.
- Tencent Cloud**, “Quality Traffic Packages: Overview,” 2026. Updated September 2, 2026; accessed September 30, 2026.
- Tintelnot, Felix**, “Global Production with Export Platforms,” *Quarterly Journal of Economics*, 2017, 132 (1), 157–209.
- U.S. Energy Information Administration**, “Electric Power Monthly, Table 5.3: Average Price of Electricity to Ultimate Customers,” EIA Electric Power Monthly 2025.
- Wollmann, Thomas G.**, “Trucks without Bailouts: Equilibrium Product Characteristics for Commercial Vehicles,” *American Economic Review*, 2018, 108 (6), 1364–1406.

Zhang, Le, Shadi Sharif Azadeh, and Hai Jiang, “Exact and Heuristic Algorithms for Cardinality-Constrained Assortment Optimization Problem under the Cross-Nested Logit Model,” *European Journal of Operational Research*, 2025, 324 (1), 183–199.

APPENDIX

The appendix contains proofs, computational and portfolio-response comparisons, calibration details, and EU mechanisms and numerical checks. The accompanying online appendix gives cost-distribution extensions, additional uniqueness results, response-method and pricing comparisons, detailed measurement sources, and supplementary policy accounting.

A. EQUILIBRIUM AND PRICING PROOFS

This appendix establishes portfolio equilibrium existence and sufficient uniqueness conditions, and derives the conditional pricing and location-return results.

A.1 Existence and inclusion cutoffs

The fixed payoff table in Proposition 2 is obtained by selecting a downstream outcome independently of private-cost types and beliefs, with finite payoffs for every portfolio profile. Existence requires absolute continuity and independence across firms; the conditional density caps in Assumption 1 are used for the response bounds below.

Proof of Proposition 2. The state space $Q = \prod_f \Delta(\mathcal{X}_f)$ is nonempty, compact, and convex, and each portfolio's interim value is continuous in Q . Distinct portfolios have distinct private-cost loading vectors, so their indifference set is an affine hyperplane of probability zero. If $Q^k \rightarrow Q$, the best-response indicator therefore converges for almost every type. Although a firm's choice is discrete, the probability of switching tends to zero as the payoff change vanishes. Dominated convergence gives continuity of $\mathcal{B} : Q \rightarrow Q$, and Brouwer's theorem gives a fixed point. This argument applies to any continuous interim-value tables; it does not require multilinearity. $\square$

Inclusion after reoptimizing the remaining network. Fix Q_{-f} and the other costs $\varepsilon_{f,-j}$, and put $v_f(S) = V_f(S; Q_{-f})$. For a location j whose inclusion varies across $\mathcal{X}_f$, define

$$V_{fj}^{\text{in}} = \max_{\substack{S \in \mathcal{X}_f \\ j \in S}} \left\{ v_f(S) - \sigma_\varepsilon \sum_{k \in S \setminus \{j\}} \ell_{fk} \varepsilon_{fk} \right\}, \quad V_{fj}^{\text{out}} = \max_{\substack{S \in \mathcal{X}_f \\ j \notin S}} \left\{ v_f(S) - \sigma_\varepsilon \sum_{k \in S} \ell_{fk} \varepsilon_{fk} \right\}.$$

Lemma A.1 (Inclusion cutoff after reoptimization). *With probability one, the optimal portfolio contains j if and only if*

$$\varepsilon_{fj} < \bar{\varepsilon}_{fj}(\varepsilon_{f,-j}; v_f) = \frac{V_{fj}^{\text{in}} - V_{fj}^{\text{out}}}{\sigma_\varepsilon \ell_{fj}}. \quad (\text{A.1})$$

The two maximizing networks can contain different sets of other locations. The cutoff is therefore a reservation cost after reoptimization, not a value assigned to j independently of the remaining costs. Computing it requires the best complete networks containing and excluding j . This cutoff representation is used in the probability-flow proof below.

For the proof, condition on $\varepsilon_{f,-j}$. All include- j portfolios have slope $-\sigma_\varepsilon \ell_{fj}$ in ε_{fj} , and all exclude- j portfolios have slope zero. Their optimized values therefore differ by $V_{fj}^{\text{in}} - V_{fj}^{\text{out}} - \sigma_\varepsilon \ell_{fj} \varepsilon_{fj}$, which proves the cutoff formula with probability one. Coordinates constant across the feasible portfolios are not decision margins.

A.2 Portfolio-response bounds

Let $P_f(v)$ be the distribution of optimal portfolios when deterministic payoffs are v , and let $\mathcal{J}_f$ contain the coordinates varying across $\mathcal{X}_f$. Put $\gamma_{fj} = \bar{g}_{fj}/\ell_{fj}$. A payoff perturbation h changes a type's choice only through changes $h(S) - h(T)$ in relative portfolio payoffs. The cover below assigns a nonnegative weight c_j to each location so that, for every pair of portfolios, the weights on the locations where they differ sum to at least the change in their payoff difference. The density caps convert these weights into a bound on the change in choice probabilities, and the minimum selects the weights giving the smallest bound. The weights are an analytical device, not location costs. Define

$$C_f(h) = \min_{c \geq 0} \left\{ \sum_{j \in \mathcal{J}_f} \gamma_{fj} c_j : |h(S) - h(T)| \leq \sum_{j \in S \Delta T} c_j \quad (S, T \in \mathcal{X}_f) \right\}. \quad (\text{A.2})$$

Lemma A.2 (Probability flow across portfolio boundaries). *Under Assumption 1, C_f is a seminorm and*

$$d_{\text{TV}}(P_f(v+h), P_f(v)) \leq C_f(h)/\sigma_\varepsilon. \quad (\text{A.3})$$

It is no larger than the include-exclude bound $\|h\|_{\partial f} = \sum_j \gamma_{fj} \max_{S \ni j, T \not\ni j} |h(S) - h(T)|$.

For rival portfolios A, B and other rivals' portfolios U , set $\Delta r_{A,B,U}^{fg}(S) = r_f(S, A, U) - r_f(S, B, U)$ and

$$K_{fg}^\# = \max_{A,B,U} C_f(\Delta r_{A,B,U}^{fg}), \quad K_{ff}^\# = 0. \quad (\text{A.4})$$

The comparison proved below gives

$$d_{\text{TV}}(\mathcal{B}_f(Q_{-f}), \mathcal{B}_f(Q'_{-f})) \leq \sigma_\varepsilon^{-1} \sum_{g \neq f} K_{fg}^\# d_{\text{TV}}(Q_g, Q'_g). \quad (\text{A.5})$$

Total variation need not equal the fraction of types switching under a common draw of private costs.

A.3 Proof of the probability-flow lemma

Suppress the firm index. Remove constant coordinates: their costs cancel from every comparison, and conditional density caps on the remaining coordinates survive marginalization by conditional expectation. Write $z_S \in \{0, 1\}^m$ for the distinct remaining incidence vectors and $L = \text{diag}(\ell_j)$. A singleton family has zero sensitivity.

The cover exists because $c_j = \text{osc}(h)$ is feasible. Truncating any component at that value preserves feasibility, so the minimum is attained. Scaling covers and adding them prove homogeneity and the triangle inequality. The include-exclude maximum for coordinate j covers every pair that differs at j , proving the stated comparison.

Take a feasible cover c and interpolate $v_t = v + th$, $0 \leq t \leq 1$. Couple choices using the same cost vector throughout. If N counts changes in the maximizing portfolio along this segment, then

$$d_{TV}(P(v + h), P(v)) \leq \Pr\{\text{endpoint choices differ}\} \leq \mathbb{E}N.$$

There are finitely many switches almost surely: the choice is the upper envelope of finitely many affine functions of t . A switch between S, T requires $d_{ST} = h(S) - h(T) \neq 0$. Allocate its unit weight across changed coordinates as $\omega_{j,ST} = c_j / \sum_{k \in S \Delta T} c_k$. The cover implies $\omega_{j,ST} |d_{ST}| \leq c_j$. These weights allocate one whole-portfolio switch across the locations that change, avoiding multiple counts of the same switch. The c_j cover relative payoff changes; they are not individual locations' marginal returns.

First let the density p be smooth with $p(e) \leq \bar{g}_j p_{-j}(e_{-j})$. For each switching pair and $j \in S \Delta T$, the tie equation maps t to e_j with absolute derivative $|d_{ST}| / (\sigma_\epsilon \ell_j)$. At fixed (t, e_{-j}) , the best include- j and exclude- j portfolios are unique almost everywhere and meet at the single cutoff $\bar{e}_j(t, e_{-j})$ from Lemma A.1. Hence summing the weighted crossings and changing variables gives

$$\mathbb{E}N \leq \frac{1}{\sigma_\epsilon} \sum_j \frac{c_j}{\ell_j} \int_0^1 \int p(\bar{e}_j(t, e_{-j}), e_{-j}) de_{-j} dt \leq \frac{1}{\sigma_\epsilon} \sum_j \frac{c_j \bar{g}_j}{\ell_j}.$$

Within-group ties are nontrivial hyperplanes in the other costs. Triple top-choice ties have zero crossing measure because three distinct binary incidence vectors cannot be collinear. Thus neither creates an additional switching term. For an arbitrary absolutely continuous density, convolve with a nonnegative product kernel φ_ϵ whose coordinate factors integrate to one. Each conditional cap is preserved because

$$p \leq \bar{g}_j p_{-j} \implies p * \varphi_\epsilon \leq \bar{g}_j (p_{-j} * \varphi_{\epsilon, -j}) = \bar{g}_j (p * \varphi_\epsilon)_{-j}.$$

Convergence in L^1 gives total-variation convergence of the distributions induced by each fixed endpoint choice map, passing the inequality to the original choices. Minimize over covers to obtain the result. Bounds valid throughout the payoff segment can replace the global coordinate caps.

Boundary weights for smooth densities. For later use, let $\Gamma_{ST}(t)$ be a shared boundary face (facet) between optimal-choice regions and define $w_{ST}(t) = \int_{\Gamma_{ST}(t)} p(e) / \|L(z_S - z_T)\|_2 d\mathcal{H}^{m-1}(e)$. Projection onto e_{-j} has Jacobian $\ell_j / \|L(z_S - z_T)\|_2$. The same single-cutoff argument therefore gives

$$\sum_{\{S,T\}:j \in S \Delta T} w_{ST}(t) = \ell_j^{-1} \int p(\bar{\varepsilon}_j(t, e_{-j}), e_{-j}) de_{-j} \leq \bar{g}_j / \ell_j. \quad (\text{A.6})$$

This is the boundary-mass bound used for Gaussian costs in Online Appendix S.1.1.

The same weights give a signed response for a portfolio outcome $K(S)$. For a differentiable payoff path v_t , a fixed smooth cost density, and integrable boundary flows, at a regular t ,

$$\frac{d\mathbb{E}K}{dt} = \frac{1}{\sigma_\varepsilon} \sum_{\{S,T\}} w_{ST}(t) [K(S) - K(T)] [\dot{v}_t(S) - \dot{v}_t(T)]. \quad (\text{A.7})$$

Here K has no direct dependence on t or realized costs, and each unordered pair of adjacent optimal-choice regions is counted once. Probability flows toward the portfolio with the larger payoff increase: the flow into S is $w_{ST}(t) [\dot{v}_t(S) - \dot{v}_t(T)] / \sigma_\varepsilon$, with the opposite flow into T . Weighting these flows by K proves the identity; triple ties have zero boundary measure as above.

A.4 Independent normal costs

When all subsets are feasible, put $D_j(h) = \max_{S:j \notin S} |h(S \cup \{j\}) - h(S)|$. Adjacent pairs require $c_j \geq D_j(h)$; toggling each differing coordinate once shows that these values cover every pair. Therefore

$$C_f(h) = \sum_{j \in \mathcal{J}_f} \gamma_{fj} D_j(h). \quad (\text{A.8})$$

For independent standard normal shocks and any set of feasible portfolios, write $L_f = \text{diag}(\ell_{fj})$ and $m_a(S) = \sum_{j \in S \cap \mathcal{J}_f} a_j$. Split the payoff change as $h = m_a + (h - m_a)$. The modular part m_a adds the same return a_j to location j regardless of which other locations the firm chooses. In the choice problem it is equivalent to a cost shift:

$$m_a(S) - \sigma_\varepsilon \sum_{j \in S \cap \mathcal{J}_f} \ell_{fj} \varepsilon_{fj} = -\sigma_\varepsilon \sum_{j \in S \cap \mathcal{J}_f} \ell_{fj} \left(\varepsilon_{fj} - \frac{a_j}{\sigma_\varepsilon \ell_{fj}} \right).$$

This cost translation yields the following bound:

$$C_f^N(h) = \inf_a \left\{ \frac{\|L_f^{-1} a\|_2}{\sqrt{2\pi}} + C_f(h - m_a) \right\}, \quad \gamma_{fj} = \frac{1}{\ell_{fj} \sqrt{2\pi}}. \quad (\text{A.9})$$

This seminorm satisfies $C_f^N \leq C_f$ and the probability bound (A.3) with C_f^N in place of C_f . To prove it, first compare v with $v + m_a$. The identity translates the normal costs by $-L_f^{-1}a/\sigma_\varepsilon$ without changing the choice rule. Applying a fixed choice rule cannot increase total variation, and the distance between $N(0, I)$ and $N(d, I)$ is $2\Phi(\|d\|_2/2) - 1 \leq \|d\|_2/\sqrt{2\pi}$. Next compare $v + m_a$ with $v + h$ using the probability-flow lemma for the residual $h - m_a$. The triangle inequality and minimization over a give the stated bound. Scaling and combining decompositions establish the seminorm property; $a = 0$ establishes domination. Replacing C_f by C_f^N in (A.4) defines $K^N \leq K^\#$.

When all subsets are feasible, let $\underline{d}_j, \bar{d}_j$ be the minimum and maximum marginal perturbations, and put $b_j = (\bar{d}_j + \underline{d}_j)/2$, $r_j = (\bar{d}_j - \underline{d}_j)/2$. Then

$$C_f^N(h) = \frac{1}{\sqrt{2\pi}} \left[\|L_f^{-1}b\|_2 + \sum_j \frac{r_j}{\ell_{fj}} \right]. \quad (\text{A.10})$$

The midpoint b_j isolates a common location-return shift; the half-width r_j bounds how much that return varies with the rest of the portfolio. To see this, $D_j(h - m_a) = r_j + |b_j - a_j|$ and $\|L_f^{-1}b\|_2 \leq \|L_f^{-1}a\|_2 + \|L_f^{-1}(b - a)\|_1$, with equality in the minimizing bound at $a = b$. Enclosing the marginal extrema by wider intervals gives a valid upper bound. Equation (A.10) solves the minimization in (A.9).

A.5 Equilibrium comparison and uniqueness

Private-cost dispersion can limit strategic feedback (Mason and Valentinyi, 2007; Melo, 2022): a payoff change alters choices only for types sufficiently close to a boundary between optimal portfolios. Overlapping portfolios share location costs, so these boundaries must be considered jointly.

For one optional location with deterministic net benefit a , inclusion has probability $F_\varepsilon(a/(\sigma_\varepsilon \ell))$. If the shock density is bounded by $\bar{g}$, changing the benefit by Δa changes inclusion probability by at most $\bar{g}|\Delta a|/(\sigma_\varepsilon \ell)$. The portfolio counterpart assigns changes in relative payoffs to the locations on which competing portfolios differ. Appendix A.2 constructs the resulting response-bound matrix $K^\#$ using a pair-cover argument: its entry $K_{fg}^\#/\sigma_\varepsilon$ bounds the change in firm f 's response per unit change in rival g 's distribution, measured in total variation. Because it covers all feasible portfolios and rival distributions, $K^\#$ is a uniform bound derived from the model, not a response evaluated at one equilibrium.

The condition below limits feedback around cycles of strategic responses. With two firms, let $M_{fg} = K_{fg}^\#/\sigma_\varepsilon$, and let d_f be the total-variation distance between firm f 's portfolio distributions in two equilibria. Then $d_1 \leq M_{12}d_2$ and $d_2 \leq M_{21}d_1$, so $d_1 \leq M_{12}M_{21}d_1$. When $M_{12}M_{21} < 1$, both distances are zero. Here $\rho(K^\#) = (K_{12}^\#K_{21}^\#)^{1/2}$, so $M_{12}M_{21} < 1$ is condition (A.11) in the theorem below; the spectral radius extends it to more firms.

Theorem A.1 (Uniqueness under dispersed private costs). *Under the assumptions in Section 3.1, if*

$$\underbrace{\sigma_\varepsilon}_{\text{private-cost dispersion}} > \underbrace{\rho(K^\sharp)}_{\text{bound on strategic feedback}}, \quad (\text{A.11})$$

the portfolio equilibrium is unique. The iteration $Q^{t+1} = \mathcal{B}(Q^t)$ converges to it from every $Q^0 \in \mathcal{Q}$.

Independent normal costs permit a refinement $K^N \leq K^\sharp$ by absorbing part of a payoff change into a shift of the cost vector; K^N can replace $K^\sharp$ in (A.11). The comparison argument below proves the theorem and gives the corresponding error bound for an approximate equilibrium.

Let $d_f(Q, Q') = d_{\text{TV}}(Q_f, Q'_f)$. Suppose a nonnegative matrix M satisfies $d(\mathcal{B}(Q), \mathcal{B}(Q')) \leq Md(Q, Q')$ componentwise for every $Q, Q' \in \mathcal{U} \subseteq \mathcal{Q}$, where $\mathcal{U}$ contains every equilibrium. For $\tilde{Q} \in \mathcal{U}$, define its exact-response residual by $e_f = d_{\text{TV}}(\tilde{Q}_f, \mathcal{B}_f(\tilde{Q}_{-f}))$.

Proposition A.1 (Equilibrium comparison). *Under these conditions, if $\rho(M) < 1$, the following hold.*

(i) **Uniqueness.** *The portfolio equilibrium Q^* is unique.*

(ii) **Error bound.** *For every $\tilde{Q} \in \mathcal{U}$,*

$$d(\tilde{Q}, Q^*) \leq (I - M)^{-1}e. \quad (\text{A.12})$$

(iii) **Convergence.** *The iteration $Q^{t+1} = \mathcal{B}(Q^t)$ converges from every initial point in $\mathcal{U}$ if $\mathcal{B}(\mathcal{U}) \subseteq \mathcal{U}$, and from every point in $\mathcal{Q}$ if $\mathcal{B}^L(\mathcal{Q}) \subseteq \mathcal{U}$ for some finite L .*

Proof. Any difference between two equilibria must be reproduced by firms' responses to one another. Two fixed points have distance vector $d \leq Md \leq M^k d \rightarrow 0$. For an approximate point, $d \leq e + Md$; iterating and using $(I - M)^{-1} = \sum_{k \geq 0} M^k \geq 0$ gives the residual bound. Each term $M^k e$ allows another round of propagation of the initial residual. Iterates in $\mathcal{U}$ obey $d^{t+1} \leq Md^t$. Invariance keeps them there; finite-step containment does so from time L onward because $\mathcal{B}^t(Q) = \mathcal{B}^L(\mathcal{B}^{t-L}(Q))$. Existence is established above. $\square$

To prove (A.5), change one rival's portfolio distribution at a time and sum the resulting payoff changes. The positive and negative parts of $Q'_g - Q_g$ each have mass $\delta_g = d_{\text{TV}}(Q_g, Q'_g)$. Coupling their normalized parts by $\pi_g(A, B)$ expresses the associated payoff change as

$$h_g(S) = \delta_g \sum_{A, B, U} \pi_g(A, B) \omega_g(U) \Delta r_{A, B, U}^{fg}(S),$$

where ω_g is the product distribution of the other rivals: those already changed use their marginals from Q' , and the rest use those from Q . Homogeneity and subadditivity give $C_f(h_g) \leq \delta_g K_{fg}^\sharp$. Summing and applying the probability-flow bound proves the comparison; the same argument gives K^N under normal costs. Take $\mathcal{U} = \mathcal{Q}$ and $M = K^\sharp / \sigma_\varepsilon$ in the proposition to prove Theorem A.1.

In particular, $a = (I - M)^{-1}\mathbf{1} \gg 0$ makes the response map a contraction in $\max_f d_f/a_f$, since $Ma = a - \mathbf{1} < a$. The spectral radius is the infimum of the comparison constants over positive firm weights (Horn and Johnson, 2013, Chapter 8). A finite-candidate-set or sampled-type residual needs a separate approximation allowance before it can be used as the exact residual in (A.12).

A small AB game in Online Appendix S.3.1 shows that the failure of an ordering condition need not imply multiple equilibria. It satisfies the refined uniqueness condition even though a location's net return changes sign in both directions along nested portfolios.

Total variation compares the probabilities of complete portfolios, so it treats portfolios with similar market effects as entirely different. An alternative condition summarized in Online Appendix S.1.1 retains market effects and signed responses, with an explicit allowance for omitted interactions; it establishes uniqueness without guaranteeing convergence of response iteration. Online Appendix S.3.2 evaluates the global normal-cost condition at the baseline parameters.

A.6 Conditional AB pricing and derivatives

Fix $\sigma > \eta > 1$, an active set, and a market, suppressing its index. Let $y_f = \psi_f p_f^{1-\sigma}$ and $D_f = b + \sum_{g \neq f} y_g$, where $b \geq 0$ is fixed residual price-index contribution. For a given $D_f > 0$, hold $\xi_f = \psi_f c_f^{1-\sigma} > 0$ fixed. A scalar pricing response solves

$$y = \xi_f \mu \left(\frac{y}{y + D_f} \right)^{1-\sigma}.$$

A higher trial value of y raises the firm's share and markup, lowering the right-hand side while the left-hand side rises. The markup bounds keep the right-hand side positive and bounded, so the curves meet exactly once. To differentiate this response, put $s_f = y/(y + D_f)$ and

$$\beta(s) = (\sigma - 1) \frac{d \log \mu(s)}{ds} = \frac{(\sigma - 1)(\sigma - \eta)\mu(s)}{\sigma \eta} > 0, \quad \beta_f = \beta(s_f).$$

The log equation has derivative $1 + s_f(1 - s_f)\beta_f > 0$ with respect to $\log y$, so the unique response $y_f = \mathcal{Y}_f(D_f)$ is smooth.

Differentiating $\log y_f = \log \xi_f + (1 - \sigma) \log \mu(s_f)$ gives $d \log y_f = -\beta_f ds_f$. Combining this with $ds_f = s_f(1 - s_f)(d \log y_f - d \log D_f)$ gives $d \log y_f = q_f(d \log D_f - d \log y_f)$, where $q_f = \beta_f s_f(1 - s_f)$. Hence

$$\tilde{\kappa}_f = \frac{\partial \log y_f}{\partial \log D_f} = \frac{q_f}{1 + q_f} < 1. \quad (\text{A.13})$$

An increase in rival contribution lowers the firm's share and induces a lower markup. The resulting recovery in its own contribution limits the markup adjustment. Thus its contribution responds less than proportionally to D_f , holding its own competitiveness fixed. Here s_f is evaluated at the solved

scalar response $\mathcal{Y}_f(D_f)$. Define $x_f = \log y_f$ and $\Phi_f(x) = \log \mathcal{Y}_f(b + \sum_{g \neq f} e^{x_g})$. Its Jacobian has

$$\Lambda_{fg} = \tilde{\kappa}_f \frac{y_g}{D_f} \quad (g \neq f), \quad \Lambda_{ff} = 0. \quad (\text{A.14})$$

In this Jacobian, $y_g = e^{x_g}$ for rivals, while $\tilde{\kappa}_f$ uses the solved scalar response, not the share implied by the input x_f . Cross-market entries are zero. Since $\sum_{g \neq f} y_g/D_f \leq 1$, $\beta_f \leq \bar{\beta} = (\sigma-1)(\sigma-\eta)/[\sigma(\eta-1)]$, and $s_f(1-s_f) \leq 1/4$,

$$\|\Lambda(x)\|_\infty \leq \bar{\lambda}_{AB} = \frac{\bar{\beta}/4}{1 + \bar{\beta}/4} < 1. \quad (\text{A.15})$$

In markets with at least two active firms, the scalar-response update $x \mapsto \Phi(x)$ is therefore a global sup-norm contraction. A fixed residual contribution preserves the bound because it reduces the total weight $\sum_{g \neq f} y_g/D_f$ of responsive rivals. The map is defined on the Euclidean space of log contributions x ; the mean-value inequality and Banach's theorem give a unique fixed point. If $b = 0$ and only one firm is active, $s_f = 1$ and $p_f = \mu(1)c_f$ directly. With $b > 0$ and one modeled provider, its scalar response faces the constant $D_f = b$. An empty active set has no modeled-provider pricing variables; if residual supply is also absent, we leave the aggregate price index undefined. These cases prove existence and uniqueness for every portfolio profile in Proposition 1.

On a fixed active set, $I - \Lambda$ is invertible and the implicit-function theorem gives smooth dependence on positive costs and accessibility. Moreover $\|(I - \Lambda)^{-1}\|_\infty \leq (1 - \bar{\lambda}_{AB})^{-1}$. At $(\sigma, \eta) = (5, 2)$ the uniform modulus is 0.375.

Lemma A.3 (Equilibrium share and profit derivatives). *On a nonempty fixed active set I , write $b \geq 0$ for the residual contribution to the price-index sum and set $a_0 = 1 - 1/\sigma$, $b_0 = 1/\eta - 1/\sigma$. For log pre-markup competitiveness $t_i = \log \xi_i$, the equilibrium system is*

$$Y s_i = \xi_i (a_0 - b_0 s_i)^{\sigma-1}, \quad s_0 = b/Y, \quad s_0 + \sum_{i \in I} s_i = 1. \quad (\text{A.16})$$

Define

$$r(s) = \frac{s}{1 + \beta(s)s} = \frac{s(a_0 - b_0 s)}{a_0 + (\sigma - 2)b_0 s}, \quad \mathcal{W} = s_0 + \sum_i r(s_i), \quad \omega_i = r(s_i)/\mathcal{W}.$$

The exact share derivatives are

$$J_{ik} = r(s_i)(\mathbf{1}\{i = k\} - \omega_k), \quad J_{0k} = -s_0 \omega_k, \quad J_{0k} + \sum_i J_{ik} = 0. \quad (\text{A.17})$$

With $\underline{\kappa}_{AB} = (a_0 - b_0)/[a_0 + (\sigma - 2)b_0] > 0$,

$$\underline{\kappa}_{AB} s_i (1 - s_i) \leq J_{ii} \leq s_i (1 - s_i). \quad (\text{A.18})$$

Hold rivals' pre-markup competitiveness and b fixed. For $R \geq 0$ and $G(t_i) = R[s_i/\sigma + b_0 s_i^2]$, if $[s^L, s^U]$ encloses the own share throughout a fixed-active-set competitiveness interval, then

$$G'(t_i) \geq \underline{\kappa}_{AB} R \min\{u(s^L), u(s^U)\}, \quad u(s) = (1/\sigma + 2b_0 s)s(1-s). \quad (\text{A.19})$$

Proof. Logarithmic differentiation of (A.16) gives $ds_i = r(s_i)(dt_i - d \log Y)$ and $ds_0 = -s_0 d \log Y$. The own-share response combines the competitiveness gain with the common price-index adjustment; $r(s_i) \leq s_i$ captures the dampening effect of the firm's markup response. Adding these equations yields $d \log Y = \sum_i \omega_i dt_i$, proving (A.17). The denominator $\mathcal{W}$ is positive on a nonempty active set, including a monopoly without residual supply.

The ratio $r(s)/s = (a_0 - b_0 s)/[a_0 + (\sigma - 2)b_0 s]$, continuously extended at zero, has derivative $-(\sigma - 1)a_0 b_0/[a_0 + (\sigma - 2)b_0 s]^2 < 0$. Its denominator remains positive when $\sigma < 2$, because $a_0 + (\sigma - 2)b_0 = (a_0 - b_0) + (\sigma - 1)b_0 > 0$. Thus $\underline{\kappa}_{AB} s \leq r(s) \leq s$. Put $B_i = s_0 + \sum_{g \neq i} r(s_g)$. Then $\underline{\kappa}_{AB}(1 - s_i) \leq B_i \leq 1 - s_i$ and

$$J_{ii} = \frac{r(s_i)B_i}{r(s_i) + B_i}. \quad (\text{A.20})$$

Monotonicity of $xy/(x+y)$ in nonnegative x, y gives (A.18); both bounds are zero for a monopoly without residual supply. Finally, $G' = R(1/\sigma + 2b_0 s_i)J_{ii} \geq \underline{\kappa}_{AB} R u(s_i)$. The derivative $u'(s) = 1/\sigma + (4b_0 - 2/\sigma)s - 6b_0 s^2$ has one negative and one positive root, is positive at zero and negative at one. Hence u increases then decreases on $[0, 1]$; its minimum on an interval is attained at an endpoint. This proves (A.19), including the zero bounds when $R = 0$ or an endpoint is zero or one. $\square$

Residual price-index contribution is fixed in this calculation, but its share varies. The no-residual formulas follow by setting $s_0 = 0$. Derivatives on a fixed active set do not justify integrating through an entry or exit boundary. With positive residual supply, own profit extends continuously to zero as its competitiveness vanishes.

A.7 Location interactions after repricing

Use the notation of Lemma A.3, and assume some competing supply, so $0 < s_i < 1$. The own-share response $D_i = \partial s_i / \partial \log \xi_i$ and its elasticity $E_i = \partial \log D_i / \partial \log \xi_i$ incorporate rivals' repricing. Put $a_0 = 1 - 1/\sigma$, $b_0 = 1/\eta - 1/\sigma$, and $h(s) = s/\sigma + b_0 s^2$, so operating profit is $Rh(s_i)$. If $\xi_i = \omega_i x^{v_s}$, its curvature is

$$\Pi_i''(x) = \frac{Rv_s h'(s_i) D_i}{x^2} \underbrace{\left[v_s \left\{ \frac{h''(s_i)}{h'(s_i)} D_i + E_i \right\} - 1 \right]}_{\Xi_i}. \quad (\text{A.21})$$

The factor outside brackets is positive. To derive the responses, put $r_i = r(s_i)$, $\omega_i = r_i/\mathcal{W}$, $D_i = r_i(1 - \omega_i)$, and

$$T_{-i} = \frac{s_0 + \sum_{k \neq i} r_k r'(s_k)}{\mathcal{W} - r_i}.$$

This quantity averages the pricing responses of the competing suppliers; the fixed residual contributes weight s_0 and derivative one. In contrast, its market share is endogenous. Writing $t_i = \log \xi_i$, direct logarithmic differentiation of $D_i = r_i(\mathcal{W} - r_i)/\mathcal{W}$ gives

$$E_i \equiv \frac{\partial \log D_i}{\partial t_i} = (1 - \omega_i)^2 r'(s_i) - \omega_i^2 T_{-i}. \quad (\text{A.22})$$

Indeed, $\partial r_i / \partial t_i = r'(s_i) D_i$ and $\partial(\mathcal{W} - r_i) / \partial t_i = -\omega_i(\mathcal{W} - r_i) T_{-i}$; substituting into the log derivative proves the expression without any sign assumption on r' . If $\xi_i = \omega_i x^{v_s}$, then $\Pi_i'(x) = Rv_s h'(s_i) D_i / x$. Another derivative gives (A.21). At $\eta = \sigma$, $r(s) = s$, $\mathcal{W} = 1$, $E_i = 1 - 2s_i$, and $h'' = 0$, giving the stated CES expression.

Table A.1: Local profit curvature differs at the same market share

Other strategic suppliers	Residual share	Local curvature
One rival	0	Concave
Seven equal rivals	0	Convex
One rival	0.20	Concave
Seven equal rivals	0.20	Convex

Notes: Constructed AB markets with $(\sigma, \eta, \theta) = (5, 2, 3)$ and own share $s_f = 0.08$. Modeled competitors divide the share left after own and residual supply. For each row, rival primitives are chosen to support the specified shares and held fixed when differentiating profit with respect to own supply. The curvature factor Ξ_i in (A.21) takes the values reported below.

The rows of Table A.1 correspond to admissible pricing equilibria: choose $Y = 1$, $b = s_0$, and $\xi_k = s_k / (a_0 - b_0 s_k)^{\sigma-1}$. For a positive continuous supply mapping, the four rows give $\Xi_i = -0.131$, 0.083 , -0.007 , and 0.095 , respectively, under the table's parameters. Local interactions between two supply increments have the sign of Π_i'' for sufficiently small increments. For arbitrary finite

increments, the fundamental theorem of calculus instead gives

$$\Pi_i(x + d_j + d_k) - \Pi_i(x + d_j) - \Pi_i(x + d_k) + \Pi_i(x) = \int_0^{d_j} \int_0^{d_k} \Pi_i''(x + u + v) dv du.$$

Its sign therefore depends on the curvature over the whole range spanned by both additions, not only at the initial network. A nonsmooth eligibility or parent-activation mapping requires direct portfolio comparisons; the local derivative alone does not cover it.

Proof of Proposition 3. Part (a). Let $g_d(x) = \Pi_i(x + d) - \Pi_i(x)$ be the operating-profit gain from the same location, contributing d , when existing supply is x . We compare this gain with one fixed location cost at three levels of existing supply. Operating profit $\Pi_i(x)$ is strictly increasing for $x > 0$, since $h' > 0$ and $D_i > 0$, and is bounded above by R/η . Bounded markups and positive competing supply imply $s_i(x) = O(x^{\nu_s})$ as $x \downarrow 0$, hence $\Pi_i(x) = O(x^{\nu_s})$. Choose any $x_M > 0$. As $d \downarrow 0$,

$$\frac{g_d(d)}{d} \longrightarrow 0, \quad \frac{g_d(x_M)}{d} \longrightarrow \Pi_i'(x_M) > 0.$$

Choose a small $d < x_M$ such that $g_d(d) < g_d(x_M)/2$, and set the cost of location j to $c_j = g_d(x_M)/2 > 0$. Hold both d and c_j fixed. Because Π_i is increasing and bounded, $g_d(x) \rightarrow 0$ as $x \rightarrow \infty$. Choose $x_R > x_M$ with $g_d(x_R) < c_j$. A base supply $x_0 = d$ and contributions $d_j = d$, $d_k = x_M - d$, and $d_h = x_R - x_M$ give the three strict signs. The other locations' additive costs cancel from these comparisons. The proof does not require a global S-shape: the small-supply limit and bounded profit suffice. It also applies at $\eta = \sigma$.

Part (b). Write $p = \nu_S > 0$, $q = \nu_A > 0$, and $\xi = \zeta \Theta^p W^q$, where Θ and W denote supply and accessibility. We first establish a positive interaction between these two inputs at sufficiently low own competitiveness. Under constant markups, put $M = R/\sigma$ and $s = \xi/(H + \xi)$, where $H > 0$ is fixed. Direct differentiation gives

$$\Pi_{\Theta W} = \frac{Mpq}{\Theta W} s(1-s)(1-2s) > 0 \quad \text{if } s < \frac{1}{2}.$$

For AB pricing, let $\Phi(\xi)$ be own operating profit after all providers reprice. It is smooth and strictly increasing for $\xi > 0$. Its behavior at zero follows directly from the share system, including the case of a single rival without residual supply. At zero own competitiveness, let $Y_0 > 0$ be the rivals' equilibrium price-index contribution plus b . For Y near Y_0 , each share solves $Y s_g = \xi_g(a_0 - b_0 s_g)^{\sigma-1}$. The derivative of its left-minus-right side with respect to s_g is

$$Y + \xi_g(\sigma - 1)b_0(a_0 - b_0 s_g)^{\sigma-2} > 0.$$

Thus $s_g(Y, \xi_g)$ is smooth, including at own $(s_i, \xi) = (0, 0)$. The clearing equation $b/Y + \sum_g s_g(Y, \xi_g) = 1$ has strictly negative derivative with respect to Y : each positive share decreases

in Y , and some competing supply is positive. The implicit function theorem therefore gives a smooth extension at $\xi = 0$, with

$$s_i(\xi) = \frac{a_0^{\sigma-1}}{Y_0} \xi + O(\xi^2), \quad \Phi(\xi) = K\xi + O(\xi^2), \quad K = \frac{Ra_0^{\sigma-1}}{\sigma Y_0} > 0.$$

In particular, $\Phi'(\xi) = K + O(\xi)$ and $\Phi''(\xi) = O(1)$. For $G(t) = \Phi(e^t)$, this yields

$$G''(\log \xi) = \xi \Phi'(\xi) + \xi^2 \Phi''(\xi) = K\xi + O(\xi^2) > 0, \quad \Pi_{\Theta W} = \frac{pq}{\Theta W} G''(\log \xi) > 0$$

at sufficiently small positive ξ .

Fix any $W_0 > 1$ and choose $\Theta_0 > 0$ small enough to obtain this strictly positive mixed derivative. First let j add $(d, 0)$ and k add $(0, a)$, with $d, a > 0$ sufficiently small that the derivative stays positive throughout the rectangle they span. Their finite interaction is

$$\Gamma_{jk} = \int_0^d \int_0^a \Pi_{\Theta W}(\Theta_0 + u, W_0 + v) dv du > 0.$$

Continuity allows both zero components to be replaced by small strictly positive contributions while preserving $\Gamma_{jk} > 0$. Let $g_0 > 0$ and $g_1 = g_0 + \Gamma_{jk}$ be the resulting gains from j before and after adding k . Choose one cost $c_j \in (g_0, g_1)$.

Finally, fix any finite positive access contribution for h and increase its supply contribution. With or without j , own competitiveness tends to infinity. With bounded markups and rivals' competitiveness fixed, the firm's market share tends to one: own price-index contribution diverges while competing contributions stay bounded. Both profits therefore approach R/η under AB, or R/σ under CES. The additional profit from j tends to zero. A sufficiently large *finite* supply contribution for h makes this gain smaller than c_j , proving the three signs.

These contributions are feasible primitives of the core-only service model. A supply contribution $d_\ell > 0$ is implemented by delivered input cost $(w_\ell \tau_{\ell n}) = d_\ell^{-1/\theta}$, and access contribution $a_\ell > 0$ by quality $q_{\ell n} = \log(a_\ell)/\lambda_A$. One fixed base origin similarly implements Θ_0 and $W_0 - 1$. The positive provider scale can implement any fixed ζ after the accessibility normalization is set. All eight candidate portfolios are feasible; no parent activation or additional interaction term is needed. The construction chooses location contributions, rather than asserting reversals for every given geography. $\square$

A sufficient condition for decreasing location returns.

Lemma A.4 (Diminishing returns to locations). *Hold rivals' portfolio distribution, market expenditures, and residual price-index contributions fixed. In every market and rival state, suppose own competitiveness is $\xi_f(S) = \omega x(S)^{v_S}$, where $\omega > 0$, $x(S) = x_0 + \sum_{j \in S} v_j$, $x_0 \geq 0$, $v_j \geq 0$, and $0 < v_S \leq 1$. Accessibility is fixed, and some competing supply is positive. All own subsets are feasible. Expected operating profit less*

additive realized location costs is submodular under either constant markups ($\eta = \sigma > 1$) or AB pricing with $\sigma \geq 2$ and $1 < \eta < \sigma$, with all providers repriced at every portfolio. It therefore satisfies the single-crossing condition from above used by Arkolakis et al. (2026).

Proof. Under constant markups, operating profit in a market is $\Pi_f(x) = (R/\sigma)\omega x^{v_s}/(H + \omega x^{v_s})$, with $H > 0$ the fixed contribution of competing supply after factoring out the common markup. Its curvature has sign $(v_s - 1) - 2v_s s_f \leq 0$.

For AB pricing, use the share responses in Appendix A.7. Put $\rho = (\sigma - \eta)/[\eta(\sigma - 1)] \in (0, 1)$ and write

$$r(s) = \frac{s(1 - \rho s)}{1 + (\sigma - 2)\rho s}, \quad a(s) = \frac{2(\sigma - 1)\rho r(s)}{1 + 2(\sigma - 1)\rho s}, \quad \kappa_0 = \max\{0, -r'(1)\}.$$

Direct differentiation gives

$$r''(s) = \frac{-2\rho(\sigma - 1)}{[1 + (\sigma - 2)\rho s]^3} < 0, \quad r'(1) + \frac{1}{\sigma - 1} = \frac{(1 - \rho)[\sigma + (\sigma - 2)\rho]}{(\sigma - 1)[1 + (\sigma - 2)\rho]^2} > 0.$$

Since $\sigma \geq 2$, $\kappa_0 < 1$ and $r'(s) \geq -\kappa_0$ on $[0, 1]$. The rival-response average T_{-f} is also at least $-\kappa_0$; its residual component has derivative one. Writing $v = 1 - \omega_f$, the log-competitiveness curvature factor in (A.21) obeys

$$\frac{h''(s_f)}{h'(s_f)} D_f + E_f = a(s_f)v + r'(s_f)v^2 - (1 - v)^2 T_{-f} \leq a(s_f)v + r'(s_f)v^2 + \kappa_0(1 - v)^2.$$

The last expression is convex in $v \in [0, 1]$ because $r'(s_f) + \kappa_0 \geq 0$, so its maximum is $\max\{a(s_f) + r'(s_f), \kappa_0\} < 1$. Indeed,

$$1 - a(s) - r'(s) = \frac{\sigma(\sigma - 1)\rho^2 s^2 [3 + 2(\sigma - 2)\rho s]}{[1 + (\sigma - 2)\rho s]^2 [1 + 2(\sigma - 1)\rho s]} > 0 \quad (0 < s < 1).$$

Thus $\Pi_f''(x) < 0$ for positive supply and expenditure when $0 < v_s \leq 1$. Both pricing cases extend continuously to $\Pi_f(0) = 0$.

The finite-increment identity in Appendix A.7 then gives nonpositive cross differences for every pair of locations in each market and rival state. Nonnegative expenditure and probability weights preserve these inequalities, and additive costs have zero cross differences. Hence, for $A \subseteq B$ excluding j , the expected net gains satisfy $\Delta_j(A) \geq \Delta_j(B)$; in particular, $\Delta_j(B) \geq 0$ implies $\Delta_j(A) \geq 0$. $\square$

The expectation argument uses decreasing marginal returns, a stronger property than single crossing. Statewise single crossing alone does not generally survive averaging. In the full service model, supply and accessibility can change in different proportions, and parent-core choices can alter local-service availability. Proposition 3(b) constructs two reversals with both service margins

even when $v_S + v_A < 1$. Lemma A.4 therefore relies on the additive single supply index and fixed accessibility, not just $v_S \leq 1$.

Finite interactions at a given geography. In one CES market, fix expenditure $R > 0$, competing supply $H > 0$, and a base network with $\Theta_0 > 0$ and $W_0 > 1$. Let $\xi = \zeta \Theta^p W^q$, with $\zeta, p, q > 0$, and let locations j, k each add strictly positive amounts to the two additive indices. All four combinations are feasible. Write $\xi_0, \xi_j, \xi_k, \xi_{jk}$ for their competitiveness levels and define

$$A = \xi_j / \xi_0, \quad B = \xi_k / \xi_0, \quad C = \xi_{jk} / \xi_0, \quad s_0 = \frac{\xi_0}{H + \xi_0}.$$

Additivity implies $1 < A, B < C < AB$. Operating profit is $\Pi(\xi) = (R/\sigma)\xi/(H + \xi)$, and the finite interaction is $\Gamma = \Pi(\xi_{jk}) - \Pi(\xi_j) - \Pi(\xi_k) + \Pi(\xi_0)$. Additive location costs cancel from this difference.

Lemma A.5 (Finite location interactions under constant markups). *Hold the base network and location contributions fixed and vary the firm's competitiveness scale ζ , keeping rival supply fixed. If $C \leq A + B - 1$, then $\Gamma < 0$ at every $s_0 \in (0, 1)$. If $C > A + B - 1$, there is a unique $s^* \in (0, 1)$ such that $\Gamma > 0$ below s^* , equals zero at s^* , and is negative above it.*

Proof. Define

$$g = C - A - B + 1, \quad d = AB - C, \quad e = AB(1 + C) - C(A + B).$$

For either additive index, its base times its level with both additions is strictly below the product of its levels with each addition separately. Raising these inequalities to $p, q > 0$ gives $C < AB$. Thus $d > 0$ and $e = d + C(A - 1)(B - 1) > 0$. With $u = \xi_0/H$, direct subtraction gives

$$\frac{\Gamma}{R/\sigma} = \frac{u(g - 2du - eu^2)}{(1 + u)(1 + Au)(1 + Bu)(1 + Cu)}. \quad (\text{A.23})$$

The numerator's bracket is strictly decreasing for $u > 0$. It is negative throughout if $g \leq 0$. For $g > 0$, its unique positive root is $u^* = g/(d + \sqrt{d^2 + eg})$, and the share threshold is $s^* = u^*/(1 + u^*)$. $\square$

The condition $g > 0$ means that the joint competitiveness gain exceeds the sum of the separate gains. Market-share saturation eventually outweighs this complementarity. The comparison holds A, B, C fixed; adding other locations changes these ratios, so the threshold does not order returns along an expanding network.

A finite construction with $v_S + v_A < 1$. Set $\sigma = \eta = 5$, $\theta = 6$, $\chi = 0.4$, and $\lambda_A = 1.5$. Then $p = 2/3$, $q = 4/15$, and $p + q = 14/15 < 1$. With $R = 5$, $H = 1$, $\Theta_0 = 1$, $W_0 = 1.1$, and $\zeta = (2/23)/1.1^{4/15}$, initial own share is 0.08. The supply/access contributions are $(d_j, a_j) = (1, 0.1)$, $(d_k, a_k) = (0.1, 1)$, and $(d_h, a_h) = (10, 0.1)$. Table A.2 evaluates the finite additions at the single cost $c_j = 0.0455$.

Table A.2: A location's return reverses with supply and accessibility

Existing candidates T	Profit gain from j	Cost c_j	Net gain $\Delta_j(T)$
$\emptyset$	0.0437871	0.0455000	-0.0017129
$\{k\}$	0.0472500	0.0455000	+0.0017500
$\{k, h\}$	0.0157824	0.0455000	-0.0297176

Notes: Constructed CES market with the parameters above. Each row compares T and $T \cup \{j\}$, including the same fixed base network. Other locations' additive costs cancel. Values are direct finite-portfolio comparisons, not local derivative approximations.

Holding accessibility at W_0 with the same initial share and supply contributions instead gives decreasing gross gains of 0.0412923, 0.0399988, and 0.0122588. In the two-input example, location k primarily improves access and raises the value of j 's supply contribution. This remains possible even though $\Theta^p W^q$, and hence CES profit, is jointly concave when $p + q \leq 1$. Joint concavity does not require a nonpositive mixed derivative. Locations that change the two inputs in different proportions can therefore be complements; concavity in aggregate inputs alone does not imply submodularity in location choices.

A sign reversal under the calibrated inputs. The ordering restriction can also fail at the calibrated parameters. Table A.3 reports AWS's net gain from adding a core in the United Arab Emirates to three nested networks, holding the baseline rival distributions fixed. The location cost comes from one of the 512 cost draws used in the initial computational exercise. Each comparison uses the full service topology and the main 4,096-point rival quadrature, with all providers repriced under AB. The same addition is unprofitable at the smallest and largest networks but profitable at the intermediate network.

Table A.3: A strict sign reversal under the calibrated baseline

Existing AWS core countries	Net gain divided by G_f^{ref}
None	-0.01500
Australia, Austria, Belgium, Bahrain	+0.00294
All 41 other countries	-0.00294

Notes: Each network excludes the United Arab Emirates before the addition. Local-service locations and parent-support rules remain as in the model. Independent repricing reproduces all three signs; its maximum profit difference from the compiled pricing routine is 8.81×10^{-11} in unnormalized units. These are feasible comparisons used to test the ordering restriction, rather than policy responses.

The negative-to-positive reversal violates single crossing from above, and the positive-to-negative reversal violates single crossing from below. Both directions of global single crossing therefore fail on this problem's full portfolio lattice.

B. PORTFOLIO COMPLETION BOUNDS

The bounds progressively link the benefits and costs of the remaining location choices. Conditioning on network size uses the same number of additions in every market. Joint-affine bounds link their market benefits to their costs, and common-location bounds require one selection to supply all markets. The last subsection describes how branch search uses these bounds and accounts for numerical error.

B.1 Network size and pricing envelopes

For the subproblem (S_f, U_f) defined in Section 3.3, the bound below credits the best k contributions to each statistic and subtracts the costs of the cheapest k additions.

Write the objective (3.4) equivalently as $\sum_{\varsigma} q_{\varsigma} \pi_{f\varsigma}(T_f) - \sum_{j \in T_f} C_{fj}$, separating operating profit from total costs of portfolio elements: ς indexes a finite rival-portfolio state, $q_{\varsigma} \geq 0$ is the probability the rival portfolio distribution assigns to state ς (a point mass on the realized rival profile in a complete-information benchmark), and $C_{fj} = C_{fj}^0 + \sigma_{\varepsilon} \ell_{fj} \varepsilon_{fj}$ combines deterministic and realized private costs. Actual supply is not additive in the chosen core set: local service can require a parent core, and core and local supply in one country do not add. The additive statistics below need not reproduce this service mapping. They may overstate available service, provided that they dominate it at every portfolio; this yields a computable upper bound even when local service depends on parent cores. Suppose operating profit $\pi_{f\varsigma}(T_f)$ from the completed portfolio in state ς is bounded by a coordinatewise increasing function of nonnegative additive statistics, indexed by $m = 1, \dots, L$:

$$z_{\varsigma m}(T_f) = z_{\varsigma m}^0 + \sum_{j \in T_f} a_{\varsigma m j}, \quad a_{\varsigma m j} \geq 0, \quad \pi_{f\varsigma}(T_f) \leq \bar{\pi}_{f\varsigma}(z_{\varsigma 1}(T_f), \dots, z_{\varsigma L}(T_f)).$$

Monotonicity is required of the bounding function, not of the actual marginal profits from adding a location. The bound may rise with available supply even when the additional profit from a particular location rises and falls as the network expands.

Let $a_{\varsigma m, (h)}^\downarrow(U_f)$ be the h th largest contribution to statistic (ς, m) among undecided elements, and let $C_{f, (h)}^\uparrow(U_f)$ be the h th smallest element cost. Define

$$\begin{aligned}\bar{z}_{\varsigma m}(S_f, U_f, k) &= \underbrace{z_{\varsigma m}(S_f)}_{\text{base and selected locations}} + \underbrace{\sum_{h=1}^k a_{\varsigma m, (h)}^\downarrow(U_f)}_{\text{best } k \text{ contributions}}, \\ \underline{C}_f(S_f, U_f, k) &= \underbrace{\sum_{j \in S_f} C_{fj}}_{\text{selected locations}} + \underbrace{\sum_{h=1}^k C_{f, (h)}^\uparrow(U_f)}_{\text{cheapest } k \text{ additions}}.\end{aligned}$$

Proposition B.1 (Profit bound conditional on network size). *Under the conditions above,*

$$B_f^{\text{card}}(S_f, U_f) = \max_{0 \leq k \leq |U_f|} \left\{ \underbrace{\sum_{\varsigma} q_{\varsigma} \bar{\pi}_{f\varsigma}(\bar{z}_{\varsigma}(S_f, U_f, k))}_{\text{operating-profit bound with } k \text{ additions}} - \underbrace{\underline{C}_f(S_f, U_f, k)}_{\text{total cost lower bound}} \right\} \quad (\text{B.1})$$

is an upper bound on net profit for every feasible completion represented by the subproblem. The result applies to any finite set of feasible portfolios; including cardinalities with no feasible completion only weakens the bound.

A common number k of additions prevents crediting all remaining locations while paying for only a few. The maximum over k remains valid when some realized element costs are negative.

Proof of Proposition B.1. For any completion $T = S \cup H$, $H \subseteq U$, $|H| = k$, each additive statistic satisfies

$$z_{\varsigma m}(T) = z_{\varsigma m}(S) + \sum_{j \in H} a_{\varsigma m j} \leq z_{\varsigma m}(S) + \text{Top}_k(a_{\varsigma m \cdot}).$$

Coordinatewise monotonicity of the operating-profit upper bound therefore bounds its operating profit. Its cost is at least $C(S) + \text{Bot}_k(C_U)$, including when costs are negative. Average over rival states with their nonnegative probabilities, subtract this cost lower bound, and maximize over attainable k to obtain (B.1). Relaxing additional feasibility restrictions preserves an upper bound. $\square$

Under CES, nonnegative service and accessibility contributions and nonnegative accessibility loadings make own competitiveness increasing in the statistics. The share is increasing in competitiveness and variable profit is $R_n s_{fn} / \sigma$. Under AB, variable profit is $R_n h(s_{fn})$, where $h(s) = s / \sigma + (1/\eta - 1/\sigma)s^2$ is increasing for $\sigma > \eta > 1$. A valid increasing share envelope therefore suffices.

A markup-based share envelope. Fix a rival state ς . Let $\bar{\psi}_{fn}$ and $\underline{c}_{fn}$ bound own accessibility above and delivered cost below for every completion. For parent-dependent service, these bounds

must cover every supported service location. Put

$$\bar{y}_{fn} = \bar{\psi}_{fn}(\mu - c_{fn})^{1-\sigma}, \quad \underline{y}_{gn} = \psi_{gn}(\mu + c_{gn})^{1-\sigma}, \quad \mu_- = \frac{\sigma}{\sigma - 1}, \quad \mu_+ = \frac{\eta}{\eta - 1}.$$

The markup interval gives

$$\bar{s}_{fn\varsigma} = \frac{\bar{y}_{fn}}{b_n + \bar{y}_{fn} + \sum_{g \neq f} \underline{y}_{gn}}. \quad (\text{B.2})$$

Inactive rivals are omitted and own share is zero if no completion supplies the market. The residual contribution b_n stays fixed. Complete AB portfolios are evaluated at joint equilibrium prices; the markup envelope is used only to bound completions.

Cardinality conditioning weakly improves the maximal-portfolio envelope when both use the same operating-profit bounds: every top- k sum is at most the sum over U , and $\text{Bot}_k(C_U) \geq \sum_{j \in U} \min\{C_j, 0\}$. This does not imply dominance over bounds constructed from different envelopes. Independently valid caps can be combined by taking their minimum.

B.2 A common affine ranking of benefits and costs

A relaxed portfolio can combine benefits that no set of locations delivers together. The following bound deducts a guaranteed profit loss for benefits that a completion does not deliver and ranks locations by these deductions and their actual costs. Fix (S, U) and a completion size k . Index market-rival-state pairs by a , with weights $\omega_a \geq 0$. Let $G_a(t)$ be repriced market profit at own log competitiveness t , holding the rival state fixed, and let $c_j = C_{fj}$. All conditions below cover the relaxed family $\mathcal{H}_k = \{H \subseteq U : |H| = k\}$.

Proposition B.2 (Joint-affine completion bound). *Suppose positive endpoints $\bar{\xi}_a$ and affine coefficients satisfy*

$$\log \frac{\xi_a(H)}{\bar{\xi}_a} \leq d_a + \sum_{j \in H} A_{aj} \leq 0, \quad H \in \mathcal{H}_k. \quad (\text{B.3})$$

Suppose $\bar{G}_a \geq G_a(\log \bar{\xi}_a)$ and $G'_a(t) \geq \mu_a \geq 0$ throughout every interval from $\log \xi_a(H)$ to $\log \bar{\xi}_a$. Define $\bar{G} = \sum_a \omega_a \bar{G}_a$, $D = \sum_a \omega_a \mu_a d_a$, and $\beta_j = \sum_a \omega_a \mu_a A_{aj}$. Then

$$B^{\text{joint}}(S, U, k) = \bar{G} - C(S) + D + \text{Top}_k(\beta - c) \quad (\text{B.4})$$

bounds every feasible completion and, using the same endpoint caps,

$$B^{\text{joint}}(S, U, k) \leq B^{\text{card}}(S, U, k) := \bar{G} - C(S) - \text{Bot}_k(c). \quad (\text{B.5})$$

For unrestricted size, maximize over admissible k .

Proof. Fix a completion H . Its shortfall from the endpoint log competitiveness $\log \bar{\xi}_a$ is at least $g_a(H) = -d_a - \sum_{j \in H} A_{aj} \geq 0$. The slope lower bound converts this shortfall into a guaranteed loss of operating profit:

$$\begin{aligned} G_a(\log \bar{\xi}_a) - G_a(\log \xi_a(H)) &\geq \mu_a [\log \bar{\xi}_a - \log \xi_a(H)] \\ &\geq \mu_a g_a(H). \end{aligned} \tag{B.6}$$

Write $C(H) = \sum_{j \in H} c_j$ and $L(H) = \sum_a \omega_a \mu_a g_a(H) = -D - \sum_{j \in H} \beta_j$. Thus $L(H)$ is a lower bound on the forgone operating profit, not the loss itself. Since $\bar{G}_a \geq G_a(\log \bar{\xi}_a)$, expected net profit is at most $\bar{G} - C(S) - [C(H) + L(H)]$. Both the cost and the shortfall must come from the same locations. Minimizing their sum gives

$$B^{\text{joint}} = \bar{G} - C(S) - \min_{H \in \mathcal{H}_k} \{C(H) + L(H)\} = \bar{G} - C(S) + D + \text{Top}_k(\beta - c).$$

The gain over the cardinality bound is therefore

$$B^{\text{card}} - B^{\text{joint}} = \min_{H \in \mathcal{H}_k} \{C(H) - \text{Bot}_k(c) + L(H)\}.$$

Each term inside braces is nonnegative, proving dominance without a restriction on cost signs. $\square$

The improvement is strict exactly when no cheapest completion has zero shortfall $L(H)$. Strictness at one k need not survive maximization over sizes.

B.3 Service envelopes for the affine bound

Consider additive service and accessibility statistics

$$\xi_a(H) = \zeta_a \Theta_a(H)^{v_S} W_a(H)^{v_A}, \quad \Theta_a(H) = \Theta_a^0 + \sum_{j \in H} v_{aj}, \quad W_a(H) = W_a^0 + \sum_{j \in H} u_{aj}, \tag{B.7}$$

with nonnegative contributions and exponents, and positive statistics whenever their exponent is positive. Bases include selected and mandatory facilities; the accessibility base also includes the log-sum constant. In the cloud model $v_S = (\sigma - 1)/\theta$ and $v_A = \chi/\lambda_A$. Fixed normalizations enter $\zeta_a > 0$. Set $\bar{\Theta}_a = \Theta_a^0 + \text{Top}_k(v_{a\cdot})$, $\bar{W}_a = W_a^0 + \text{Top}_k(u_{a\cdot})$, and $\bar{\xi}_a = \zeta_a \bar{\Theta}_a^{v_S} \bar{W}_a^{v_A}$. The inequality $\log z \leq z - 1$ supplies (B.3) with

$$d_a = v_S \frac{\Theta_a^0 - \bar{\Theta}_a}{\bar{\Theta}_a} + v_A \frac{W_a^0 - \bar{W}_a}{\bar{W}_a}, \quad A_{aj} = \frac{v_S v_{aj}}{\bar{\Theta}_a} + \frac{v_A u_{aj}}{\bar{W}_a}. \tag{B.8}$$

Zero exponents remove the corresponding term before evaluating ratios. Inactive completions are handled separately without taking log competitiveness.

Holding rivals' pre-markup competitiveness and the residual contribution b fixed, Lemma A.3 supplies $\mu = \underline{\kappa}_{AB} R \min\{u(s^L), u(s^U)\}$ for a share interval covering the entire competitiveness range

from each completion to its relaxed upper endpoint. Let $\xi^L \leq \xi_f \leq \bar{\xi}$, $m_{\min} = (1 - 1/\eta)^{\sigma-1}$, and $m_{\max} = (1 - 1/\sigma)^{\sigma-1}$. The latter two values are the minimum and maximum of the markup term $\mu(s)^{1-\sigma}$, respectively, so their order reverses the markup order. One such interval is

$$\begin{aligned} s^L &= \frac{\xi^L m_{\min}}{\xi^L m_{\min} + \sum_{g \neq f} \xi_g m_{\max} + b}, \\ s^U &= \frac{\bar{\xi} m_{\max}}{\bar{\xi} m_{\max} + \sum_{g \neq f} \xi_g m_{\min} + b}. \end{aligned} \tag{B.9}$$

Bottom- k sums give a lower competitiveness bound. When a denominator is zero, use the share interval $[0, 1]$ and set $\mu = 0$. These are derivative bounds, not concavity assumptions.

Parent-dependent service. The actual mapping need not be additive. It suffices to construct additive statistics $\Theta_a^+(H) \geq \Theta_a(H)$ and $W_a^+(H) \geq W_a(H)$, including all parent-supported local service and permitting duplicate supply. Write $\xi_a^+(H) = \zeta_a[\Theta_a^+(H)]^{v_s}[W_a^+(H)]^{v_A}$ and construct $\bar{\Theta}_a, \bar{W}_a, d_a, A_{aj}$ from these additive statistics as above. For positive competitiveness levels, monotonicity and $\log z \leq z - 1$ give

$$\log \frac{\xi_a(H)}{\bar{\xi}_a} \leq \log \frac{\xi_a^+(H)}{\bar{\xi}_a} \leq d_a + \sum_{j \in H} A_{aj} \leq 0.$$

The policy searches first bound actual profit by profit at the relaxed competitiveness level, and then apply the slope correction there. Since repriced profit is nondecreasing,

$$G_a(\log \xi_a(H)) \leq G_a(\log \xi_a^+(H)) \leq \bar{G}_a + \mu_a \left(d_a + \sum_{j \in H} A_{aj} \right).$$

The slope bound need only cover relaxed competitiveness through the upper endpoint. Bottom- k sums of the additive contributions provide a lower bound for relaxed competitiveness, not necessarily for actual supply. Summing and ranking the same location costs gives the stated joint-affine bound. Zero relaxed supply is handled separately; setting $\mu_a = 0$ retains the endpoint cap whenever no valid positive slope correction is available. Applying Proposition B.2 directly to actual competitiveness levels is also valid if its slope bound covers their entire intervals to the upper endpoints.

The endpoint cap must cover profit at the competitiveness value implied by the relaxation, even when that value is unattainable. A cap covering only feasible portfolios cannot safely receive the slope correction. Appendix B.5 discusses numerical allowances.

B.4 A bound retaining common core and service choices

The supply–access index. Fix a firm, the policy rules, and a rival distribution, and suppress the firm index. Write $\Theta_n = \Theta_{fn}$ and $W_n = 1 + \sum_i o_{fin} e^{\lambda_A q_{in}}$. Equations (2.1)–(2.2) give

$$\xi_n = \psi_{fn} c_{fn}^{1-\sigma} = \zeta_n \Theta_n^{\nu_S} W_n^{\nu_A}, \quad \zeta_n = a_f^{\sigma-1} \exp(d_{fn} - \chi A_n^{\text{frontier}}),$$

where $\nu_S = (\sigma - 1)/\theta$ and $\nu_A = \chi/\lambda_A$. For positive exponents, setting $\vartheta = \nu_S/(\nu_S + \nu_A)$ and $z_n = \Theta_n^\vartheta W_n^{1-\vartheta}$ gives the exact reduction $\xi_n = \zeta_n z_n^{\nu_S + \nu_A}$ in (3.7). In the calibration $\nu_S = 4/3$, $\nu_A = 4/15$, and $z_n = \Theta_n^{5/6} W_n^{1/6}$, including zero supply. With positive residual supply, Lemma A.3 and continuity at zero imply that repriced own profit is nondecreasing in ξ_n . Hence

$$H_n(z) = \sum_{\zeta} q_{\zeta} \pi_{fn}(\zeta_n z^{\nu_S + \nu_A}; S_{-f}^{\zeta})$$

is nondecreasing. The expression averages state-specific pricing equilibria, not rival supply statistics. It need not be concave.

Physical feasibility. Write x_i for core presence and o_i for coverage from either a core or a supported local facility at origin i . For a local facility h requiring all cores in a nonempty parent set P_h , introduce t_h and impose

$$0 \leq x_i, o_i, t_h \leq 1, \quad t_h \leq x_j \ (j \in P_h), \quad t_h \geq \sum_{j \in P_h} x_j - |P_h| + 1.$$

If $\mathcal{L}_i$ lists the alternative local facilities at origin i and $\bar{o}_i \in \{0, 1\}$ indicates fixed local availability, impose $o_i \geq \bar{o}_i$, $o_i \geq x_i$, $o_i \geq t_h$ for $h \in \mathcal{L}_i$, and $o_i \leq \bar{o}_i + x_i + \sum_{h \in \mathcal{L}_i} t_h$. Together with $o_i \leq 1$, this includes fixed local supply without requiring an optional parent core. These constraints reproduce the core/local mapping at binary core choices; their continuous relaxation contains all physical completions. Let e_{in} equal one if supply from country i is eligible to serve market n , and zero otherwise. Then

$$\begin{aligned} \Theta_n(x, o) &= \Theta_n^0 + \sum_i e_{in} (w_i \tau_{in})^{-\theta} [(1 - \gamma_{\text{loc}}) x_i + \gamma_{\text{loc}} o_i], \\ W_n(x, o) &= W_n^0 + \sum_i e_{in} e^{\lambda_A q_{in}} o_i. \end{aligned}$$

Ordinarily $\Theta_n^0 = 0$ and $W_n^0 = 1$. China's fixed partner supply enters these constants only for eligible destinations, with its actual supply weight and overlapping optional supply removed. Forbidden direct cores have $x_i = 0$. Fix $x_i = 1$ on S_f , $x_i = 0$ outside $S_f \cup U_f$, and $\sum_{i \in U_f} x_i = k$ to obtain $\mathcal{R}_f(S_f, U_f, k)$. Other feasibility restrictions may be added provided every admissible completion remains represented.

Proof of Proposition 4. Every feasible completion has a representation in $\mathcal{R}_f$. Set z_n to its actual weighted geometric mean, which lies in I_n . This point is feasible in (3.8), and because $\widehat{H}_n \geq H_n$ on I_n , its objective is at least the completion's expected profit net of all costs, including selected locations and any negative costs. This proves the bound. Convexity serves a separate purpose: it makes the relaxed bound tractable. The weighted geometric mean is concave on the nonnegative orthant, and Θ_n and W_n are affine, so the constraint on z_n defines a convex set. Each $\widehat{H}_n$ is concave and the cost term is linear, so the objective is concave; the remaining constraints are linear (Boyd and Vandenberghe, 2004, Section 3.1.7). Maximizing over completion sizes therefore bounds unrestricted portfolio choice. $\square$

Constructing and evaluating the upper bound. On a grid $z_0 < \dots < z_L$ covering I_n , let $u_i \geq H_n(z_i)$ be nondecreasing upper values, replacing them by cumulative maxima if needed. Here H_n includes segment expenditure. Monotonicity gives $H_n(z) \leq u_{i+1}$ on $[z_i, z_{i+1}]$. The upper concave hull of (z_i, u_{i+1}) for $i < L$ and (z_L, u_L) is therefore a nondecreasing piecewise-linear concave majorant: both endpoints of each grid interval dominate its step bound. This controls values between grid points without assuming concavity of actual profit. Finer grids reduce discretization error but need not remove the gap from replacing a nonconcave function by a concave upper envelope.

With $\vartheta = v_S/(v_S + v_A)$, concavity of the weighted geometric mean gives, for every $\rho > 0$,

$$\Theta^\vartheta W^{1-\vartheta} \leq \vartheta \rho^{\vartheta-1} \Theta + (1 - \vartheta) \rho^\vartheta W.$$

Replace its hypograph, the set of points on or below its graph, by any finite collection of these affine upper constraints. Take $\widehat{H}_n$ to be the upper concave hull constructed above, which is the minimum of finitely many affine functions, and replace $\widehat{H}_n(z_n)$ in the objective by a variable h_n that must lie below each of these functions at z_n . The result is an LP relaxation containing every feasible completion. For another concave $\widehat{H}_n$, any finite set of affine functions lying above it on I_n gives a valid but looser relaxation. The LP may be looser than (3.8). If $\overline{B}_{f,k}^{\text{LP}}$ is a valid upper bound on its maximum and $B_{f,k}^{\text{old}}$ is an existing valid cap, then

$$B_f^{\text{used}} = \max_k \min\{B_{f,k}^{\text{old}}, \overline{B}_{f,k}^{\text{LP}}\} \leq \max_k B_{f,k}^{\text{old}}$$

is a valid completion bound. At each size, the search uses the smaller of the valid LP and existing bounds, weakly improving on the existing bound alone. Discarding nodes numerically requires an upper bound from a dual solution with controlled residuals; a feasible primal objective alone is insufficient.

Finally, if adding a given core never reduces operating profit and preserves feasibility for every portfolio omitting it, every unrestricted optimum contains that core when its realized cost is strictly

negative: adding it strictly increases net profit. The maintained cloud mapping has this monotonicity. This dominance rule applies before searching across all sizes; it need not hold within a separately fixed-size problem. Neither this rule nor the geometric relaxation requires an ordering of marginal location returns.

B.5 Branch search and numerical error

For a fixed type and rival distribution, a node (S, U) represents the feasible portfolios T satisfying $S \subseteq T \subseteq S \cup U$. Start with the value of a fully repriced feasible portfolio. Discard a node if infeasible or if its valid upper bound does not exceed the best value found so far; evaluate a complete node; otherwise split on inclusion of an undecided country. The children partition the completion set without ordering marginal returns. Best-bound selection changes processing order, not validity. Let L be the best attained value and O the open nodes. If nodes are discarded when $B(N) \leq L + \epsilon$, retain the largest bound U_{disc} of those nodes. At interruption,

$$U^* = \max\{L, \max_{N \in O} B(N), U_{\text{disc}}\}, \quad \max \emptyset = -\infty,$$

satisfies

$$L \leq \max_T v_f(T) \leq U^*.$$

The search terminates when the optimality gap is at most ϵ , that is, $U^* - L \leq \epsilon$. The EU application uses $\epsilon = 10^{-8}$ in model profit units, before division by G_f^{ref} . The corresponding normalized gap is $10^{-8}/G_f^{\text{ref}}$; this search criterion differs from the firm-average regret threshold of $10^{-4}G_f^{\text{ref}}$ used to stop distributional iteration. If nodes are discarded only when $B(N) \leq L$, then $U_{\text{disc}} \leq L$ and that term is redundant. The finite choice set ensures eventual termination, while worst-case search can remain exponential in the number of locations. The computational comparisons measure the gains from the bounds on the reported problems.

The searches underlying the policy tables apply bounds within each admissible size and retain the largest remaining bound across sizes. A service-availability envelope includes all supply supportable by a completion; an additive relaxation attributes local service to possible parents, permitting duplicates and relaxing joint parent requirements. Componentwise top- k sums bound that relaxation. The smaller competitiveness envelope is repriced against the supplied rival states and combined with the corresponding cost lower bound and joint-affine cap. The EU search permits every subset of 42 core countries. China excludes direct mainland core choice for the five non-Chinese providers; fixed AWS and Microsoft partner-operated services enter exact payoffs and all upper envelopes identically, without a selectable core or private core cost. The EU searches at the reported distributions, which cover every cost type, also use the linear-programming relaxation of the common-location bound (Appendix B.4). Each node is bounded first by that relaxation. If it does not exclude the node, the smaller of it and the cardinality and joint-affine caps is used; if the

linear program fails numerically, the caps alone are used. A child’s bound never exceeds its parent’s. Appendix C.1 compares computation times with and without the common-location bound; the reported EU policy searches combine all three bounds as described above.

Pricing and bound error. If the active log-contribution vector $x = (\log y_i)_i$ has undamped pricing residual r_x , the conditional contraction bounds its sup-norm error by $r_x/(1-\bar{\lambda}_{AB})$. Residuals in shares must first be converted to these units. Profit allowances must enlarge node bounds and lower the best values found; floating-point residuals alone do not bound rounding error. Shares, slopes, affine coefficients, and sums must be enclosed, including cancellation and signed costs. Every bound entering the minimum must be valid. Complete portfolios are repriced using the original payoff, rather than the upper envelope.

Integration and type approximation. Bounds built at finite rival quadrature apply to that payoff; the full expectation needs an additional integration allowance. At a specified cost type and rival distribution, a full-choice-set search bounds all feasible portfolio deviations. It does not control integration over other cost types or best-response error after rivals change. Discovered portfolios enter the common candidate sets before policy distributions are recomputed and searched again; independent cost draws test coverage beyond previously searched types.

Appendix F reports the application checks. Differences across randomized quasi-Monte Carlo point sets (scrambles) measure numerical stability, not sampling confidence intervals.

Distributional residuals. A uniform contraction would additionally convert a response residual to a distributional distance through Proposition A.1. Without it, regret, total variation, and agreement across starts remain distinct diagnostics; small profit errors need not imply small policy-price errors.

C. COMPUTATIONAL AND RESPONSE COMPARISONS

C.1 Computational benchmarks

Comparison with exhaustive enumeration. The experiment holds rivals at one portfolio distribution, taken from an earlier computation of the EU baseline with 1,024 rival integration points, and uses all eight providers and two independent Gaussian cost vectors per provider. A prespecified random permutation gives nested sets of 8, 10, 12, 14, 16, and 20 optional core countries, common to all firms. Decisions outside each set are fixed to the observed portfolio; portfolio size within the set is endogenous. All 42 customer countries and the core–local service mapping remain. The country sequence and cost vectors are selected before inspecting computation times or profits.

Three methods solve each problem. Enumeration prices every feasible portfolio using the same compiled pricing routine as the search methods. One search uses the cardinality and joint-affine bounds. The strengthened search also uses the common-location LP bound, cardinality-dependent supply intervals, and reusable constraint matrices. It evaluates the LP before calling the cardinality and affine bounds when further tightening is needed. The searches behind the reported EU results use the same order (Appendix B.5). Cost-dependent presolve (fixing choices before branching) is disabled in all three methods to hold the number of optional locations fixed. Initial portfolios are the same observed and cost-ordered candidates, repriced separately by each method; enumeration solutions are never passed to the searches.

Table C.1: Computation time by number of optional locations

Optional locations	Median time (seconds)			Median time ratio	
	Enumeration	Cardinality and affine	With common locations	Enumeration/ common locations	Cardinality–affine/ common locations
8	4.5	5.9	18.6	0.24	0.32
10	12.6	12.3	20.4	0.62	0.58
12	45.0	30.7	22.6	1.99	1.28
14	173.6	57.2	25.1	6.93	2.14
16	700.8	125.6	28.5	24.47	3.92
20	11,956.8	631.6	50.9	235.33	10.57

Notes: Each row covers all eight firms and two cost vectors per firm. Times include preparation and solving both types. Ratios are medians of firm-specific time ratios, not ratios of the time medians. Two firms run concurrently at 8–16 locations and eight at 20 locations; methods within a firm run sequentially in rotating order, with one computation thread. These are single-run elapsed times on the same Apple M4 Pro computer, subject to host-load variation. “With common locations” adds the common-location LP bound to the cardinality and joint-affine bounds. Both search methods are developed in this paper.

Each method runs in a fresh process and may reuse its own operating profits across the two cost vectors. Timing starts before input loading, initial candidate evaluation, and profit-curve construction. Independent recalculation of prices for the selected portfolios follows the timed portion for every method. The three methods agree on optimized profits in all 96 type problems, and every search meets the common profit-gap tolerance. At 20 optional locations, each firm’s enumeration evaluates all 1,048,576 portfolios; none of the reported times is extrapolated from an unfinished run.

Comparison with single-crossing elimination. This separate benchmark restricts the service structure to additive supply with fixed accessibility, using CES with $\sigma = 5$ and AB with $(\sigma, \eta) = (5, 2)$. Geographic weights and rival competitiveness remain fixed; own competitiveness is matched at the observed network in each market. Profits average over eight fixed rival-network states. This design differs from the full-service, 1,024-draw enumeration experiment above.

We compare the AES elimination and branching rules with portfolio-profit bounds only in the additive specifications with $\nu_S \in \{1, 2/3\}$, where Lemma A.4 establishes applicability. The two pricing rules, eight providers, four cost types, and 20 or 42 candidate countries give 256 distinct problems. Each method solves every problem three times, for 1,536 runs in total. Methods use separate value caches and the same initial feasible portfolios, without an exhaustive-enumeration solution supplied as a starting point. All runs meet the stated optimality tolerance within the ten-second search limit, and both methods return matching optimal objective values.

Table C.2: Time to solve controlled 42-country portfolio problems

Pricing	Supply exponent ν_S	AES rules	Profit bounds
CES	1	0.00121	0.335
CES	2/3	0.00176	0.298
AB	1	0.0143	0.465
AB	2/3	0.0248	0.420

Notes: Median seconds across 32 distinct problems per row, each repeated three times. Initial-portfolio evaluation is timed for both methods; each bound run is charged the full preparation time for its profit curves. The search tolerance is $2 \times 10^{-7} G_f^{\text{ref}}$. Both methods are implemented in Python: the AES rules use our implementation, not the authors' Julia package, and profit bounds use a tangent-LP relaxation adapted to additive supply. Rival distributions are fixed.

AES is faster on most problems (Table C.2), with the same ordering of median times in the 20-country comparisons. Profit bounds are faster in one 42-country Alibaba problem under AB with $\nu_S = 2/3$: the roughly fivefold advantage persists across all three repetitions, including curve preparation.

C.2 Counterfactuals under constant-markup CES pricing

This subsection repeats the EU comparison of Section 5 with constant-markup CES pricing. Setting $\eta = \sigma = 5$ fixes every markup at $\sigma/(\sigma - 1)$, so every operating margin is 20 percent, whereas AB margins in the baseline range from 20 to 31.5 percent. The decomposition in (2.6) holds at a given share; changing the pricing rule also changes equilibrium shares.

Recalibration and expansion of candidate sets. We recalibrate the eight cost intercepts δ_f , the seven relative competitiveness parameters, and the residual intensity to the count and share targets of Table 3, using the updating rules of the AB calibration and requiring a core-count RMSE of at most 0.25 and a maximum share error of at most 0.15 percentage points. Apart from η , the assigned parameters and inputs in Table 2 are unchanged, as are the cost scaling factors ℓ_f^{ref} and the 1,536 own-cost types; the candidate sets are extended as described below. The recalibrated CES baseline has a core-count RMSE of 0.06 and a maximum share error of 0.03 percentage points,

compared with 1.01 and 0.55 for the AB baseline, whose parameters are not recalibrated on the expanded candidate sets.

With lower margins, every cost intercept is lower than under AB, competitiveness rises relative to AWS, and the residual intensity rises from 2.045 to 3.646. EU core presence, which is not a target, is 52.69 in the CES baseline, compared with 52.88 under AB and 40 observed.

At each CES distribution, we start from each type's best candidate and search for profitable additions, removals, and one-country replacements, continuing while profit rises. Applied to the initial AB-based candidate sets, this procedure finds a more profitable portfolio outside the set for 15 percent of type responses across the three distributions. We add the portfolios found to the common sets, recalibrate the baseline, recompute all three distributions using the iteration and stopping rules of Appendix F, and repeat. After four rounds, which add 4,274 portfolios, the search finds no more profitable portfolio for any type of any firm at the three distributions. The CES results use these extended sets; all three distributions meet the stopping criteria on both the main and validation quadratures. Unlike the AB distributions, they are not checked by searches over all feasible portfolios.

Table 7 in Section 5.1 applies the network comparisons of Section 5 under both pricing rules.³²

Profit levels and policy changes. Online Appendix Figure S.6.1 compares baseline operating profits: AB margins raise the largest providers' profits, while OVHcloud's profit level is nearly unchanged. Both the pricing rule and recalibration enter these comparisons.

Fixed baseline markups. To distinguish markup adjustment from markup levels, we also solve the AB model with its parameters unchanged and each provider's markup in each market fixed at its baseline value. The fixed operating margin is the AB margin averaged over baseline portfolio profiles with share weights. At the AB baseline distributions, these markups change operating profits and world shares by less than 0.6 percent. The baseline is not recalibrated: after network adjustment it has 168.99 expected worldwide core presences, compared with 171.35 under AB.³³

Fixing these market-specific markups lowers attenuation from about 50 to 45 percent and changes the non-EU core response from an increase of 0.36 to a decrease of 0.20. Markup adjustment therefore affects where networks adjust, while the main investment effects and price-index increases for regulated customers also arise under constant markups.

³²Responses to unchanged rivals are improved by the same add-or-drop search as under AB (Appendix F.3); on the extended sets it moves 8 percent of CES type responses, compared with 11 percent under AB. On the independent 4,096-point validation quadrature, the CES core and price contrasts change by at most 0.03.

³³Starting from the extended CES sets, the same local search extends the candidate sets for eight rounds; the last round improves one type's response by less than $10^{-6}G_f^{\text{ref}}$.

C.3 Portfolio-response comparisons

Baseline-fitted independent cutoffs. Table 6 compares the reported responses with independent country choices, holding rivals at their baseline portfolio distributions. The reported response starts from each type’s best portfolio in the common candidate sets and repeatedly accepts the most profitable single-country addition or removal until no such move improves profit (Appendix F.3).

For each provider, country and policy, we average the country’s incremental operating profit over 64 prespecified country orderings. Each increment is evaluated after adding the countries preceding it in that ordering, with rivals held at their baseline distributions. These sampled Shapley contributions average interactions across surrounding networks. We fit country-specific offsets so that the resulting independent cost cutoffs match the baseline portfolio distribution’s marginal participation probabilities. Under each policy, we hold the offsets fixed and update the averaged contributions. A country is selected when its realized location cost is below its cutoff. The fit matches baseline marginal participation, rather than the joint portfolio distribution or policy responses.

An independent cutoff differs from the inclusion threshold in Lemma A.1. That threshold conditions on the other private costs and compares the best network containing a country with the best network excluding it, reoptimizing all remaining locations. Online Appendix S.4.2 gives integration checks and additional response comparisons.

Joint changes beyond the reported responses. A network can be locally optimal even when removing a group of locations raises profit. Table C.3 reports two such responses under EU-wide eligibility, holding rivals at their baseline distributions. Every single-country addition or removal and every two-country change reduces net profit. Removing three countries together for Tencent, or five for Oracle, instead raises it. Both choices also outperform the pooled responses in Online Appendix Table S.4.2.

Table C.3: Profitable joint changes beyond local optima

Provider	Countries jointly removed	Net profit change / $10^{-4}G_f^{\text{ref}}$		
		Best single	Best pair	Joint
Oracle	UAE, Switzerland, Denmark, Philippines, South Africa	−6.45	−4.94	+1.17
Tencent	Mexico, Netherlands, Singapore	−1.92	−2.82	+7.15

Notes: Each row uses one of the original 1,536 cost types under EU-wide eligibility against unchanged baseline rivals. Changes are relative to the reported portfolio response and include realized location costs. Best single and best pair maximize net profit over all 42 one-country and 861 two-country changes, including additions, removals and replacements. Prices are recomputed on the fixed 4,096-point quadrature; independent repricing confirms the gains.

These locations complement one another in operating profit. Removing one reduces the returns to the others, making further removals more attractive. The combined location-cost saving can therefore exceed the operating-profit loss even though no individual removal pays. For Tencent, the saving is $0.26334G_f^{\text{ref}}$ and the operating-profit loss is $0.26262G_f^{\text{ref}}$. A search limited to single additions, removals or replacements cannot find it from the reported response.

The joint changes are found by enumerating six-country neighborhoods around four prespecified cost types per provider and policy. These are feasible improvements, not statements that the improved portfolios are globally optimal. Online Appendix S.4.4 gives the neighborhood selection and additional examples.

A separate search over all 42 countries on the same 64 provider-policy-cost problems closes the optimality gap in 20; the reported response agrees with the global optimum in all 20. The profitable joint changes above occur outside this completed subset. Completion depends on search difficulty, so the solved subset does not estimate an accuracy rate across cost types.

Dependence within rival networks. Rivals with the same country-level participation can create different investment incentives. A controlled rearrangement preserves each rival’s country-level core frequencies but changes which locations occur together. It changes six of 64 candidate-set choices, with the associated profit-ranking reversals reproduced in four randomizations. The National–EU-wide count contrast changes in four of 32 matched provider-types. These comparisons use diagnostic rival distributions and fixed candidate sets, not new equilibria. Online Appendix S.4.6 reports the whole-network controls and rearrangement protocol.

D. SERVICE AGGREGATION, DATA, AND CALIBRATION UNITS

We derive the unit-cost technology and then describe the data conventions. Section 4.2 describes the calibration targets and procedure; Table D.1 reports the provider-specific parameter values.

D.1 Fixed workload composition and unit cost

Fix a provider, market, policy environment, and portfolio, suppressing their indices. A unit of service requires self-contained workloads $v \in [0, 1]$ in fixed proportions. Each workload is performed at one eligible origin; different workloads may be assigned to different origins. A measure of mass one incorporates the fixed workload weights. Eligible origins and intensities t_i are common across these workloads. For $t_i > 0$, efficiency has distribution $\Pr\{Z_i(v) \leq z\} = \exp(-t_i z^{-\theta})$, with $\theta > 0$ and independence across origins. Each workload is performed at its least-cost eligible origin. Writing $\tilde{a}$ for the common efficiency scale,

$$m(v) = \min_{i:t_i>0} \frac{w_i \tau_{in}}{\tilde{a} Z_i(v)}, \quad \Pr\{m(v) > x\} = \exp(-\tilde{a}^\theta \Theta x^\theta), \quad \Theta = \sum_i t_i (w_i \tau_{in})^{-\theta}. \quad (\text{D.1})$$

Assume the average across the fixed workloads equals the distributional mean. For $\Theta > 0$, fixed requirements imply a unit cost

$$c = \int_0^1 m(v) dv = \int_0^\infty e^{-\tilde{a}^\theta \Theta x^\theta} dx = \frac{\Gamma(1 + 1/\theta)}{\tilde{a}} \Theta^{-1/\theta}. \quad (\text{D.2})$$

Substitution of $u = \tilde{a}^\theta \Theta x^\theta$ gives the last equality, finite for every $\theta > 0$. Defining $a = \tilde{a}/\Gamma(1 + 1/\theta)$ gives (2.1); the calibrated a_f is already in these units. With no eligible origin, supply is unavailable. This integrated workload heterogeneity is distinct from the private location-cost shocks.

For the core/local rule, local opportunities have intensity γ_{loc} ; a core contains these and independent additional opportunities of intensity $1 - \gamma_{\text{loc}}$. Their maximum has intensity one, preserving local supply when a core is added without double counting. This nesting is an assumption about compatible workloads, not observed product availability.

With one initial core and equal delivered input costs $w_i \tau_{in}$ across origins, adding a second core at $\theta = 3$ multiplies marginal cost by $2^{-1/3}$, a reduction of about 21 percent. Adding a local-only origin at $\gamma_{\text{loc}} = 0.5$ instead multiplies it by $1.5^{-1/3}$, a reduction of about 13 percent.

With the same eligible origins and efficiency distributions, an internal CES aggregator with elasticity $\rho > 0$, $\rho \neq 1$, gives

$$c_\rho = (\mathbb{E} m^{1-\rho})^{1/(1-\rho)} = \frac{[\Gamma(1 + (1 - \rho)/\theta)]^{1/(1-\rho)}}{\tilde{a}} \Theta^{-1/\theta}, \quad \theta > \max\{\rho - 1, 0\}.$$

The constant can again be absorbed into a , leaving the same supply exponent. The Cobb–Douglas limit at $\rho = 1$ is finite for $\theta > 0$. Integrability restricts internal substitution ρ , whereas σ governs demand across providers (Antràs et al., 2017, eqs. (5) and (9)). Setting $\rho = \sigma$ imposes $\theta > \sigma - 1$, as in Tintelnot (2017); allowing the elasticities to differ also permits $(\sigma - 1)/\theta > 1$ (Antràs et al., 2024). For example, $(\rho, \theta, \sigma) = (2, 3, 5)$ satisfies integrability with $\nu_S = 4/3$. Fixed workload composition is the $\rho = 0$ limit used in the application.

Workload classes with different eligibility sets require separate supply aggregates: fixed class quantities imply a sum of class costs, not a single aggregate with intensity equal to the fraction of supported products. An unavailable required class makes the bundle infeasible. The applications use one representative class; public shares do not identify class weights.

D.2 Expenditure, delivery, and accessibility inputs

Country expenditure is the dimensionless proxy

$$\bar{R}_n = \left[\frac{\text{GDP}_{n,2023}(I_{n,2023}/100)}{\text{GDP}_{\text{US},2023}(I_{\text{US},2023}/100)} \right]^{0.7}, \quad \bar{R}_{\text{US}} = 1.$$

GDP is nominal GDP in current U.S. dollars; I is the percentage of the population using the internet, not the number of users. The inputs use 2023 World Bank indicators, with an IMF GDP estimate and a maintained 93 percent internet-use rate for Taiwan. The exponent 0.7 compresses cross-country differences and is held fixed. The proxy is not observed cloud revenue. For EU countries the customer segments have expenditure $R_{n,\text{reg}} = 0.18\bar{R}_n$ and $R_{n,\text{private}} = 0.82\bar{R}_n$; other countries retain $\bar{R}_n$. These expenditure weights are unchanged across policy regimes.

The operating input-cost index is $w_i = 0.75 + 0.20(p_i^E/p_{\text{US}}^E) + 0.05(L_i/L_{\text{US}})$. Electricity prices are 2025 industrial prices: Eurostat band IF, excluding VAT, for the 12 model-EU countries; IEA end-use prices for industry for Canada, Chile, India, Japan, Korea, New Zealand, Serbia, Switzerland, and the United Kingdom; and the EIA industrial price for the United States. The other 20 countries are assigned the US price. Labor costs use the Penn World Table’s human-capital-adjusted compensation, with a productivity-based value for the United Arab Emirates. Here p_i^E is the electricity price and L_i the labor-cost index.

Delivered-service costs use a three-level geographic rule:

$$\tau_{in} = \begin{cases} 1, & i = n, \\ 1.15, & i \neq n, \text{ same region or within 5,000 km,} \\ 1.50, & \text{otherwise.} \end{cases} \quad (\text{D.3})$$

At $\theta = 3$, these foreign delivery wedges give an origin approximately 66 or 30 percent of an otherwise identical domestic origin’s contribution to the cost aggregator, since $1.15^{-3} \simeq 0.66$ and $1.50^{-3} \simeq 0.30$. Regions are North America, South America, Europe, Asia, the Middle East, Oceania, and Africa, and distances are measured between reference cities, one per country. All model-EU countries are in Europe, so every foreign EU origin faces the same 1.15 wedge in serving another EU country.

The initial origin–market accessibility index is

$$q_{in}^0 = -0.25 \log(1 + d_{in}/1000) + 0.201\{i = n\} + 0.151\{\tau_{in} \leq 1.15\} + 0.20(\omega_n - 1), \quad (\text{D.4})$$

where d_{in} is distance in kilometers between country locations that differ from the reference cities used in (D.3): averages of data-center facility coordinates where available and city locations otherwise. The final term is a destination-specific index from the 2025 OECD digital-policy inputs, held fixed across counterfactuals, with neutral value $\omega_n = 1$ when unavailable. Its coefficient is assigned. The proximity indicator uses the baseline delivery wedge, so a counterfactual change in τ does not mechanically change q .

For each EU destination, we reorder the eleven foreign-EU index values in (D.4) using RIPE round-trip-time rankings. The country-pair statistic is the median of minimum ping measurements between pairs of RIPE Atlas anchors (fixed measurement hosts) from the August 18, 2026 archive.

Table D.1: Provider cost scales, reference quantities, and calibrated parameters

Provider	$10^3 \ell_f^{\text{ref}}$	K_f^{ref}	Observed cores	$10^3 G_f^{\text{ref}}$	δ_f	a_f
AWS	23.8	28	28	667	-0.119	1.000
Microsoft	11.1	33	33	367	-0.040	0.817
Google	3.98	29	29	116	0.365	0.628
Oracle	1.78	27	27	48.0	0.498	0.514
Alibaba	6.04	15	17	90.5	0.675	0.692
Huawei	4.94	10	12	49.4	0.798	0.665
Tencent	2.12	11	13	23.3	0.746	0.528
OVHcloud	0.0629	18	10	1.13	1.268	0.210

Notes: δ_f and a_f are calibrated to worldwide core counts and global shares, with $a_{\text{AWS}} = 1$. The other columns give the assigned cost scales and their reference quantities; $G_f^{\text{ref}} = \ell_f^{\text{ref}} K_f^{\text{ref}}$; both are reported multiplied by 10^3 . All reference quantities and cost scaling factors remain fixed throughout calibration and policy comparisons. The factors ℓ_f^{ref} scale location costs and private-cost dispersion; G_f^{ref} also serves as a reference for reporting payoff gains. Reference counts differ from current observed core-country counts. Unrounded values are included in the replication data.

The sample retains anchor pairs observed in both directions, while computing each direction's statistic separately. Shorter round-trip times receive higher index values. Domestic indices, non-EU destinations, and non-EU origins retain q^0 . This replaces the assignment of index values within 132 directional EU country pairs; it does not estimate a latency coefficient or use latency differences elsewhere to change supply costs.

Reordering preserves the sum of $e^{\lambda_A q_{in}}$ over origins at each destination. We use the fixed all-origin frontier

$$A_n^{\text{frontier}} = \lambda_A^{-1} \log \left(1 + \sum_{i=1}^{42} e^{\lambda_A q_{in}} \right),$$

computed before applying policy eligibility. Both the ranking replacement and later policy comparisons therefore leave this reference unchanged. Otherwise an eligibility restriction could lower a provider's accessibility index and its normalization simultaneously, obscuring the service loss.

The central residual term is common across markets: $b_n = b^0 \exp(\zeta_b)$, with $b^0 = 1.778$ and calibrated log multiplier $\zeta_b = 0.140$ (both rounded). EU public and private segments inherit the same term. Its realized share is endogenous because the contributions of the eight providers change. Residual sensitivities vary this term or its exposure to the China shock without treating it as a separate strategic firm.

D.3 Provider-specific cost scaling factors

Table D.1 reports the assigned cost scaling factors $\ell_f^{\text{ref}} = G_f^{\text{ref}} / K_f^{\text{ref}}$ and their reference quantities. Its final two columns give the calibrated cost intercepts and relative competitiveness parameters from Section 4.2. The reference operating profits and core counts were computed at the initial observed networks and parameter values, then fixed before the endogenous-count calibration. They set relative location-cost scales across providers. The reference counts differ from current observed counts; for example, OVHcloud's are 18 and 10, respectively.

Changing a factor while holding the other coefficients fixed changes both systematic location costs and private-cost dispersion, and can change portfolio choices. Their values remain fixed throughout calibration and policy comparisons. Because each δ_f is calibrated, the factors affect choices through the country adjustments, distance gradient, and private-cost dispersion. Their role can equivalently be expressed as provider-specific country adjustments $\ell_f^{\text{ref}} c_i^0$, distance coefficients $\kappa \ell_f^{\text{ref}}$, and shock standard deviations $\sigma_\varepsilon \ell_f^{\text{ref}}$.

The distance D_{fi} is measured from the provider's home country, except for OVHcloud, whose home market is the model EU. Its distance is measured from the nearest model-EU country and is zero within that region. With distance measured in 10,000 kilometers, an additional 10,000 kilometers raises deterministic cost by $0.5 \ell_f^{\text{ref}}$ per core, equal to one standard deviation of that provider's private per-core shock.

The country location-cost adjustment c_i^0 equals 0.614 (rounded) in Belgium, Bahrain, Hong Kong, Israel, Morocco, the Philippines, Qatar, Serbia, Thailand, and Taiwan, the sample countries without construction-cost data, and zero in the other sample countries. The value 0.614 comes from an earlier fit of providers' location composition with core counts fixed and is held fixed in the present calibration. Belgium is the only model-EU country with a positive adjustment; the others affect the attractiveness of non-EU alternatives.

In the baseline, expected operating profit differs from G_f^{ref} by less than 1 percent for the four US providers, is 1 percent higher for Alibaba and 9 percent higher for Huawei, and is 3 percent lower for OVHcloud. Tencent's is 33 percent higher, reflecting the model's overprediction of its share and core count (Table 3).

A feasible core with a negative realized net location cost is always selected, as shown in Appendix B.4. Censoring these costs at zero retains the same optimal portfolios and lowers every feasible optimum by the same amount for a given type. Equilibrium distributions, prices, and shares are therefore unchanged, while profits net of location costs are lower.

In the EU baseline, cores with negative realized net costs make up about 34 percent of AWS's expected cores and 24 percent of Microsoft's, reflecting their negative intercepts, and 1 to 8 percent of the other providers'. Censoring lowers each provider's net profit by the same amount in every policy regime of an application. In the three EU regimes, this reduction is 13.5 to 13.9 percent of

net profit for AWS, 11.4 to 11.8 percent for Microsoft, and at most 6.4 percent for the others, the largest percentages arising under National eligibility.

D.4 Geographic composition diagnostics

We assess the geographic composition of predicted networks using forty moments: for each provider, the fractions of core presences in the model EU, Asia-Pacific, and the Americas, the mean median-normalized construction-cost index, and mean log expenditure in hosting countries. These correlated moments informed specification choices but are not targets of the reported calibration. Missing construction costs are imputed using a projection on public covariates and a median fallback.

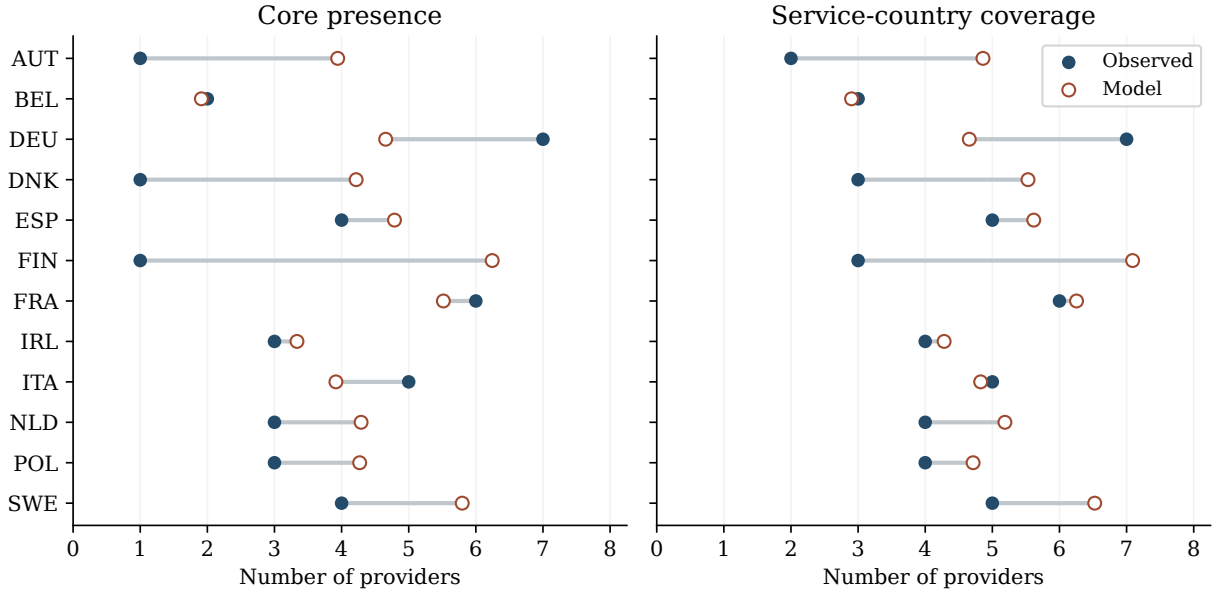


Figure D.1: EU geography: core presence and service coverage

Notes: Observed provider counts and expectations under the baseline portfolio distribution. Both panels use the same country universe and local topology. Model coverage evaluates whether the required parent cores are present for each complete portfolio before averaging.

With variable portfolio size, the moment for country feature h_{ij} is

$$m_{fj}(Q_f) = \frac{\mathbb{E}_{Q_f}[\sum_{i \in S_f} h_{ij}]}{\mathbb{E}_{Q_f}[|S_f|]}. \quad (\text{D.5})$$

The ratio of expectations weights countries by their inclusion probabilities. It differs from averaging each portfolio's fraction and retains zero-core types in the unconditional expectations. Composition is undefined if the expected core count is zero.

Table D.2: Geographic composition: observations and model

Provider	EU (%)		Asia-Pacific (%)		Americas (%)	
	Obs.	Model	Obs.	Model	Obs.	Model
AWS	21.4	32.5	42.9	26.0	14.3	17.8
Microsoft	33.3	32.0	30.3	27.7	15.2	16.6
Google	31.0	32.9	31.0	26.4	17.2	17.9
Oracle	22.2	33.1	25.9	26.1	22.2	18.4
Alibaba	11.8	27.9	52.9	45.3	17.6	8.6
Huawei	8.3	25.9	50.0	50.2	25.0	7.5
Tencent	7.7	26.6	61.5	46.7	23.1	8.7
OVHcloud	40.0	28.3	30.0	21.2	20.0	11.7

Provider	Construction index		Log expenditure	
	Obs.	Model	Obs.	Model
AWS	-0.002	-0.002	-2.185	-2.268
Microsoft	0.005	-0.009	-2.232	-2.350
Google	0.009	-0.013	-2.203	-2.321
Oracle	0.003	-0.015	-2.249	-2.313
Alibaba	0.013	0.000	-1.961	-2.332
Huawei	-0.038	-0.014	-2.420	-2.255
Tencent	0.022	-0.011	-1.889	-2.295
OVHcloud	-0.028	-0.032	-1.676	-2.477

Notes: Each cell reports the ratio in equation (D.5). Observations use core-country presences; model values use the same baseline portfolio distribution as Table 3. Construction costs enter as their ratio to the median observed cost minus one, with imputation from public covariates and a median fallback for missing observations. Expenditure is the GDP–internet proxy normalized to the US. The three regions do not exhaust the country universe. These forty correlated diagnostics informed specification assessment but were not included among the calibration targets.

Table D.2 shows that the model compresses differences across providers in the EU share of core locations: the model share lies between 26 and 33 percent across providers, against 8 to 40 percent observed. The excess EU presences come mainly from AWS, Oracle, and the three Chinese providers, while OVHcloud’s EU share is underpredicted.

E. CLOUD-NETWORK MEASUREMENT AND INSTITUTIONAL SOURCES

The inputs distinguish selectable core-country presence, installed service availability, and eligibility to serve particular customers. Online Appendix S.7 records the provider product lists and institutional sources underlying these distinctions.

E.1 Constructing core and service availability

Multiple named core regions within a country form one selectable presence. Installed local facilities can add service origins without adding selectable cores. The local-service dataset starts from 82 source entries, of which 60 pass the 42-country and product screens: 25 have a confirmed parent in the same country, nine have a confirmed parent abroad, and 26 have an unknown parent. The 34 confirmed assignments comprise 33 AWS entries and Microsoft’s Los Angeles Extended Zone. The unknown assignments comprise 25 OVHcloud entries and Alibaba’s Wuhan Local Region.

Confirmed parent support is evaluated at each complete portfolio before averaging. An unknown parent is not imputed; the local facility retains the support needed to operate. Provider documentation identifies product-support relationships rather than measured traffic routes. Online Appendix S.7.1 gives the supporting architecture and quota guides. In China, fixed partner-operated facilities provide branded services and operating profit but carry neither selectable core counts nor private core-cost draws.

E.2 Product scope and the representative workload

The dataset contains 169 core-country and 184 service-country presences across providers. Local facilities add five countries for AWS and ten for OVHcloud; several facilities in one country remain one service origin. These are availability counts, rather than capacity or workload shares.

The representative workload covers general-purpose compute supplied by both core and local facilities, with common eligibility and local-service intensity across tasks. Availability of an origin does not establish availability of every product: accelerator and storage restrictions can exclude particular workloads, whereas the absence of a managed database service need not prevent running database software on a virtual machine. Online Appendix Table S.7.1 reports the provider-specific product scope. Worldwide IaaS revenue also includes GPU-oriented supply (Gartner, 2025), whose weight the public shares do not identify. Product-specific workloads require the separate supply aggregates described in Appendix D.1.

E.3 Regulatory qualifications and operating modes

Geographic eligibility, ownership, certification, and operating arrangements are distinct restrictions. The policy comparisons vary geographic supply access while holding the latter qualifications fixed. SecNumCloud, for example, qualifies a service offer and legal entity, rather than a country's entire cloud supply. China's partner-operated regions distinguish the provider's brand from the licensed operator and directly controlled core presence. Certification to serve public customers can likewise differ from the opening of a commercial region. Online Appendix S.7.3 documents these distinctions and the relevant US provisions; Table S.7.2 summarizes the France, China, and Korea examples.

F. EU INVESTMENT MECHANISMS AND NUMERICAL CHECKS

F.1 Investment margins and international adjustment

Joint contraction or substitution. Fix rivals' portfolio distributions and the other locations T of a firm. Let Π be its expected operating profit after prices are solved. For two additional locations $a, b \notin T$, define

$$\Gamma = \Pi(T \cup \{a, b\}) - \Pi(T \cup \{a\}) - \Pi(T \cup \{b\}) + \Pi(T).$$

Thus Γ measures how the presence of a changes the incremental operating profit from b . Write $A = \Pi(T \cup \{a\}) - \Pi(T)$ and $B = \Pi(T \cup \{b\}) - \Pi(T)$. An additive operating-profit loss t from customers served only through a gives the four-choice objective, up to a constant,

$$(A - C_a - t)x_a + (B - C_b)x_b + \Gamma x_a x_b, \quad x_a, x_b \in \{0, 1\}. \quad (\text{F.1})$$

This experiment holds B and Γ fixed. A general sourcing restriction can change them as well.

Lemma F.1 (The direction of the unexposed location's response). *In (F.1), suppose costs have a joint density. As t increases, the probability of choosing b weakly decreases if $\Gamma > 0$, weakly increases if $\Gamma < 0$, and is unchanged if $\Gamma = 0$.*

Proof. Optimizing a separately with and without b makes the inclusion cutoff for b , conditional on C_a , equal to

$$\bar{C}_b(t; C_a) = B + (A - C_a - t + \Gamma)_+ - (A - C_a - t)_+, \quad z_+ = \max\{z, 0\}.$$

For $\Gamma > 0$ the positive-part difference is nondecreasing in $A - C_a - t$, so the cutoff falls with t . For $\Gamma < 0$ it rises with t . The inclusion sets are therefore nested in the stated direction for every C_a . Integrating over costs proves the claim; ties have probability zero. $\square$

With a single additive supply index, the finite-increment identity in Appendix A.7 links Γ to curvature over the interval spanned by the two additions. Convexity throughout that interval implies joint contraction, and concavity implies substitution. With both supply and access, the interaction also depends on the direction of each location's contribution. Contributions across markets add; mixed curvature requires the finite difference rather than a derivative at the starting network. The four-choice result isolates these interactions without requiring a large portfolio search.

Rival composition and investment: the construction for Table 1. Set $(\sigma, \eta, \theta) = (5, 2, 3)$, market expenditure one, no residual supply, and initial own share 0.08. One rival or seven equal rivals split the remaining share. Normalize the initial price-index sum to one and set pre-markup competitiveness to $\xi_i = s_i / (0.8 - 0.3s_i)^4$. Hold rival competitiveness and accessibility fixed while own supply rises from 1 to 1.1 or 1.2; all providers reprice at each network. Each candidate adds 0.1 supply. The resulting Γ is -2.67979×10^{-5} with one rival and 1.15632×10^{-5} with seven equal rivals.

In both cases, costs are independent $U[0, \bar{c}]$, where $\bar{c} = 0.003$, and the loss at a is $t = 0.001$. The cost boundaries remain interior over this change. Integrating the inclusion cutoff in Lemma F.1 over the cost square gives

$$\Pr_t(b) - \Pr_0(b) = -\Gamma t / \bar{c}^2.$$

Initial inclusion probabilities are 0.622774 and 0.723045; the changes are +0.297755 and -0.128480 percentage points. Piecewise integration reproduces these values, with pricing residuals below 10^{-14} . The loss can represent a separate customer market served only through a ; it does not change the common-market technology or rivals' networks. Online Appendix S.4.7 gives further examples of international adjustment.

Shared market access and domestic replication. In a symmetric two-country, core-only example, hold a country-symmetric rival distribution fixed. Let a_r be the operating profit with one core under rule r , and t_r the additional operating profit from the second, where $0 \leq t_r \leq a_r$. Costs are independent with continuous CDF F . The four net profits are 0, $a_r - C_1$, $a_r - C_2$, and $a_r + t_r - C_1 - C_2$. Neither location is selected if $\min(C_1, C_2) > a_r$; both are selected if $\max(C_1, C_2) < t_r$. Otherwise the firm selects one. Hence

$$\mathbb{E}[K_r] = 1 - [1 - F(a_r)]^2 + F(t_r)^2. \quad (\text{F.2})$$

For National regulation of expenditure share ω , let g be the operating profit from one domestic market if its entire expenditure is subject to National eligibility, including domestic competition and the exclusion of foreign origins. Thus g is defined before weighting by ω . If each core alone provides access to its domestic regulated market, separability across those markets gives $a_N = (1 - \omega)a_E + \omega g$ and $t_N = (1 - \omega)t_E + \omega g$. If $t_E < g < a_E$, National reduces the value of the first core

but raises the value of the second. Equation (F.2) then yields

$$\mathbb{E}[K_N - K_E] = \underbrace{F(t_N)^2 - F(t_E)^2}_{\text{increase in replication}} - \underbrace{([1 - F(a_N)]^2 - [1 - F(a_E)]^2)}_{\text{increase in exit}}.$$

The second bracket is the difference between the two no-core probabilities. Which margin dominates depends on the mass of cost types near each choice boundary. Online Appendix S.4.7 allows both firms to choose both countries and constructs an increase and a decrease in aggregate presence, each in every equilibrium of its example.

To separate additions from removals, we pair each cost type's evaluated pure best responses across policies. Table F.1 describes these responses; Table 4 instead aggregates the final computed portfolio distributions. EU-wide sourcing mainly adds EU cores, whereas National mainly excludes them. National produces almost no additional EU core choices relative to either unrestricted or EU-wide sourcing. Outside the EU, its small net increase relative to EU-wide combines 1.38 additions with 0.96 removals. These changes include expansions and contractions by different cost types; they are not all relocations within the same type's network.

Table F.1: Additions and removals in paired pure best responses

Policy comparison	Region	Additions	Removals	Net change
EU-wide minus unrestricted	EU	2.329	0.279	+2.050
	Non-EU	0.180	7.230	-7.049
	World	2.510	7.509	-4.999
National minus unrestricted	EU	0.009	2.622	-2.613
	Non-EU	0.060	6.687	-6.627
	World	0.069	9.309	-9.240
National minus EU-wide	EU	0.015	4.678	-4.663
	Non-EU	1.380	0.957	+0.423
	World	1.395	5.635	-4.241

Notes: Entries are expected core-country presences, summed across the eight modeled providers. The same 1,536 cost types per provider are paired across policies, using each policy's evaluated pure best responses to its final rival distributions and the fixed 4,096-point quadrature. Additions and removals are differences between policy choices, rather than observed construction or closure events. These pure responses differ slightly from the final mixed distributions used in Table 4; the largest difference in a worldwide policy contrast is 0.105 cores. Rounding can prevent displayed entries from adding exactly.

Procurement scope and complete networks. Now allow all locations to adjust, holding rivals' portfolio distribution and realized location costs fixed. Apply National eligibility to a fraction $\lambda \in [0, 1]$ of public-sector expenditure in each EU country and EU-wide eligibility to the rest. Total public-sector expenditure and unrestricted customers are unchanged. Let $\pi^r(S)$ denote expected

operating profit before location costs under rule r , evaluated against this fixed rival portfolio distribution. Each group is priced separately, with the service mapping, demand weights, and residual contribution of its respective rule. At fixed portfolios, expenditure scales operating profit without changing shares or markups under either CES or AB. Hence

$$\pi^\lambda(S) = \pi^E(S) - \lambda L(S), \quad L(S) = \pi^E(S) - \pi^N(S). \quad (\text{F.3})$$

The loss $L(S)$ includes repricing and can be negative if domestic protection raises profit. It is not a measure of cross-border sales.

Lemma F.2 (Portfolio responses to procurement scope). *For any fixed cost function $C(S)$ on the finite set of feasible portfolios, let S_λ maximize $\pi^E(S) - C(S) - \lambda L(S)$ over all feasible portfolios. If $\lambda_2 > \lambda_1$, then $L(S_{\lambda_2}) \leq L(S_{\lambda_1})$.*

Proof. Adding the optimality inequalities at the two policy values gives $(\lambda_2 - \lambda_1)[L(S_{\lambda_1}) - L(S_{\lambda_2})] \geq 0$. This holds for any maximizing choices, including ties. $\square$

The selected network becomes less exposed to the profit loss from national fragmentation. Its domestic core count need not rise: lower exposure can come from domestic replication or from contraction. With additive private costs and the smoothness conditions of equation (A.7), that identity makes the relevant transitions explicit. For a region's core count $K(S)$,

$$\frac{d\mathbb{E}K}{d\lambda} = -\frac{1}{\sigma_\varepsilon} \sum_{\{S,T\}} w_{ST}(\lambda)[K(S) - K(T)][L(S) - L(T)]. \quad (\text{F.4})$$

The nonnegative weights measure cost types near each boundary between optimal networks, which may differ in several countries. Expected core presence falls when the weighted sum of products is positive. Applying equation (A.7) to the outcome $L(S)$ gives a negative weighted sum of squared loss differences, so the expected value of $L(S)$ also weakly decreases.

A unit subsidy s to domestic cores instead adds $sK(S)$ to profit. The same optimality comparison implies that the selected domestic core count weakly increases with s , holding rivals fixed. Under the smoothness conditions above, its derivative is $\sigma_\varepsilon^{-1} \sum_{\{S,T\}} w_{ST}[K(S) - K(T)]^2 \geq 0$. Two networks with the same number of domestic cores receive the same per-core subsidy, but they can lose different amounts of profit under national sourcing. Procurement therefore changes incentives to reorganize the network even when the domestic core count is unchanged.

Table F.2 traces two illustrative cost draws in the calibration, holding rivals at their unrestricted-baseline distributions Q^O and allowing all 42 core locations to adjust. The computed AWS response drops Spain when λ passes approximately 0.483; Oracle drops the Netherlands at approximately 0.138. All other chosen locations remain unchanged.

The profit forgone from unrestricted customers is the same under both rules. A lower public-sector contribution under National is enough to make the cost saving exceed the total operating-profit loss.

Table F.2: Procurement scope and two illustrative portfolio responses

Provider	National fraction λ	Core excluded	World	EU	Non-EU	$L(S)$
<i>A. Computed best responses and policy profit losses</i>						
AWS	0–0.483	None	29	9	20	2.993
	0.483–1	Spain	28	8	20	2.684
Oracle	0–0.138	None	14	6	8	1.522
	0.138–1	Netherlands	13	5	8	1.180
<i>B. Profit difference: smaller minus larger selected network</i>						
Component					AWS	Oracle
Location-cost saving					+2.864	+3.389
Private-sector EU operating profit					–0.809	–0.956
Non-EU operating profit					–1.838	–2.104
Public-sector operating profit under EU-wide					–0.366	–0.377
Public-sector operating profit under National					–0.058	–0.035
Net profit change under EU-wide ($\lambda = 0$)					–0.149	–0.047
Net profit change under National ($\lambda = 1$)					+0.160	+0.295

Notes: λ is the National fraction within the unchanged 18-percent public-sector expenditure share. Rivals remain at Q^O . All profit entries, including $L(S)$, are percentages of the provider's G_f^{ref} . Panel B compares the smaller network with the larger one under the same rule; the two networks differ only in the core listed in Panel A. At intermediate λ , its public-sector change is the weighted average of the two public-sector rows; other components are unchanged. Breakpoints are rounded. The full-interval regret upper bounds, including pricing allowances, are below $1.7 \times 10^{-8} G_f^{\text{ref}}$ for AWS and $2.4 \times 10^{-7} G_f^{\text{ref}}$ for Oracle. The numerically unresolved intervals around the switch points have half-widths below 5.4×10^{-6} and 6.8×10^{-5} , respectively. The draws were selected from the first two providers in provider/type order with different endpoint portfolios among the ten provider–cost pairs solved under both endpoint policies; they do not estimate adjustment frequencies.

For each draw, we solve two constrained problems: one requires the core and the other excludes it. Within each constraint, the same network is optimal to the stated tolerance at both $\lambda = 0$ and $\lambda = 1$. Because every portfolio's profit is affine in λ , that network remains optimal to the same tolerance between the endpoints. Comparing the two networks locates the change in the core's inclusion decision, apart from the small numerically unresolved intervals reported in the table notes.

Rival withdrawal and incremental investment returns. In a CES market, write $y_0 < y_1$ for the firm's price-index contributions without and with an additional facility and $H > 0$ for rival and

residual contributions. Holding these two own networks fixed, their operating-profit difference is

$$D(H) = \frac{R}{\sigma} \frac{(y_1 - y_0)H}{(H + y_0)(H + y_1)}, \quad \text{sign } D'(H) = \text{sign}(y_0 y_1 - H^2).$$

Rival withdrawal raises the extra facility's return if $H^2 > y_0 y_1$. If $H^2 < y_0 y_1$, the network without it already captures enough business that withdrawal reduces the extra gain. For $H_0 < H_1$, the finite comparison has $\text{sign}[D(H_1) - D(H_0)] = \text{sign}(y_0 y_1 - H_0 H_1)$. Both identities follow by differentiating or subtracting the displayed fractions. In the full model, other own locations can also adjust; the corresponding comparison is between the optimized values with and without the core in Lemma A.1.

An inclusion threshold with the other locations reoptimized. For one prespecified Oracle cost draw, we allow all 41 other core locations to adjust under each sourcing rule and rival distribution. For each environment, we maximize profit separately over networks containing the Philippine core and networks excluding it. Adding back its location cost to the difference between these two optimized values gives its inclusion threshold. The realized Philippine cost for this draw is $53.712 \times 10^{-3} G_f^{\text{ref}}$.

Table F.3: The Philippine core's inclusion threshold after network reoptimization

Policy	Rival distribution	Inclusion threshold ($10^{-3} G_f^{\text{ref}}$)	Philippine core
EU-wide	Baseline	53.595	Excluded
EU-wide	EU-wide equilibrium	55.048	Included
National	Baseline	52.842	Excluded
National	National equilibrium	54.774	Included

Notes: All eight constrained searches cover the full 42-country choice set and attain the stated optimality tolerance. Thresholds are rounded; the lower and upper bounds differ by less than $4.7 \times 10^{-7} G_f^{\text{ref}}$ in each environment, and lie strictly on the reported side of the location cost. An independent pricing implementation reproduces all attained portfolio profits within 7.1×10^{-11} in operating-profit units.

Rival adjustment raises the Philippine core's inclusion threshold above its realized location cost under both sourcing rules, even after the rest of Oracle's network is reoptimized. The example shows a change in the investment margin under both rules, not a decomposition of the National–EU-wide aggregate difference. Online Appendix S.4.5 holds the other cores fixed to distinguish own-network composition from rival adjustment.

F.2 Portfolio convergence and global response checks

The EU calculations use 10,622–13,323 candidate portfolios per firm, 1,536 cost types, and a fixed 4,096-point rival quadrature, holding calibrated parameters fixed. Rival portfolios are integrated at each iteration, and prices are solved for every resulting portfolio profile.

Rival portfolios are integrated by mapping fixed uniform draws through each firm’s current distribution over its candidate set into complete portfolio profiles. The uniform draws stay fixed as the distributions change. Each integration point therefore represents a complete rival network profile.

At each iteration, firms update simultaneously, assigning equal weight to the current distribution and its candidate-set best-response distribution. The stopping rule requires each firm’s mean candidate-set regret to be at most $10^{-4}G_f^{\text{ref}}$ and

$$\max_f \frac{1}{2} \sum_{S_f} |\mathcal{B}_f^{\text{cs}}(Q_{-f})(S_f) - Q_f(S_f)| \leq 0.01$$

at two consecutive checks, where $\mathcal{B}_f^{\text{cs}}$ is the best response restricted to the common candidate sets. Regret includes the pricing allowance; the distributional residual is undamped and compares whole portfolios.

All three EU distributions satisfy the candidate-set stopping criteria, and the final distributions also satisfy them when payoffs are evaluated with an independent 4,096-point validation quadrature. On that quadrature, the maximum firm mean regret bounds are 5.32×10^{-6} , 5.61×10^{-6} , and $6.75 \times 10^{-6}G_f^{\text{ref}}$ for the baseline, EU-wide, and National distributions, respectively; the corresponding TV residuals are 0.00641, 0.00535, and 0.00565.

At these distributions, every full-portfolio search closes its optimality gap, and none finds a portfolio outside the candidate sets that improves on the best candidate (Table F.4). These searches use the bounds in Appendix B.5.

Four of the 36,864 type-level regret upper bounds exceed $10^{-4}G_f^{\text{ref}}$. The largest, $1.8 \times 10^{-4}G_f^{\text{ref}}$, is for an OVHcloud type under National eligibility; three of the four types also have regret lower bounds above $10^{-4}G_f^{\text{ref}}$. Their best portfolios are already candidates, so this regret arises from the strategies’ probability assignments rather than missing portfolios. These bounds apply to the simulated cost types and fixed rival quadrature; Appendix B.5 describes the distinct sources of numerical approximation error.

Table F.4: EU convergence and full-portfolio search diagnostics

<i>Panel A. Convergence on the candidate sets</i>					
Regime	Regret upper bound / G_f^{ref}			Undamped TV residual	
	Maximum firm mean	Maximum type			
Baseline	5.23×10^{-6}	1.56×10^{-4}		0.00726	
EU-wide	5.60×10^{-6}	6.70×10^{-5}		0.00771	
National	6.72×10^{-6}	1.67×10^{-4}		0.00493	

<i>Panel B. Search over all feasible country portfolios</i>					
Regime	Problems	Improving portfolios found	Searches meeting optimality-gap tolerance	Regret upper bound / G_f^{ref}	
				Maximum firm mean	Maximum type
Baseline	12,288	0	12,288	1.28×10^{-5}	1.56×10^{-4}
EU-wide	12,288	0	12,288	1.17×10^{-5}	6.74×10^{-5}
National	12,288	0	12,288	1.13×10^{-5}	1.82×10^{-4}

Notes: Both panels use 1,536 cost types per firm and the fixed 4,096-point rival quadrature used to compute the distributions. Panel A reports candidate-set regret bounds, including pricing-error allowances, and the undamped total-variation residual. Panel B searches all feasible portfolios at these distributions. Search closure uses the absolute profit-gap criterion in Appendix B.5. An improving portfolio lies outside the candidate set and outperforms its best member. Regret bounds subtract the type's expected payoff under its computed strategy from the profit upper bound and add a pricing-error allowance, then average over types within each firm or take the maximum.

F.3 Network comparisons and outcome aggregation

Let Q^O denote the baseline portfolio distribution. Fixed networks use Q^O in each policy regime. For policy r , responses to unchanged rivals use $Q^{r,u} = \bigotimes_f \hat{\mathcal{B}}_f^r(Q_{-f}^O)$. Equilibrium responses use the policy-specific distribution Q^r computed on the common candidate sets. All eight providers choose networks in the latter two comparisons, and prices are jointly solved in every portfolio profile. The hat denotes the computed response, also used in the one-firm comparisons. In the EU application, each type selects its best portfolio in the common candidate sets and then improves it by adding or dropping one country at a time. Section 4.3 describes the numerical accuracy of the equilibrium distributions. Ownership and certification effort remain fixed.

Outcome aggregation. The main EU comparison reports National minus EU-wide for the same customer markets and expenditure weights. Prices are solved for each portfolio profile before aggregation. For a customer group, let $\Delta \log \bar{P}$ denote the difference in its expenditure-weighted expected log price index; the reported percentage effect is $100[\exp(\Delta \log \bar{P}) - 1]$. Both

responses to unchanged rivals use the unrestricted baseline distribution Q^O , so this comparison does not describe a transition from the EU-wide equilibrium to National.

F.4 Supply accounting and sensitivity

EU-wide eligibility relative to the baseline. Table F.5 compares EU-wide eligibility with the unrestricted baseline using the same three network comparisons. Competitor responses reduce worldwide contraction from 7.99 to 4.89 core presences, an attenuation of about 39 percent. EU core presence rises by 1.29 against unchanged rivals and by 2.07 under equilibrium responses; contraction is concentrated outside the EU. The price-index increase for regulated customers is 28.59 percent at fixed networks, 27.93 percent against unchanged rivals, and 27.73 percent under equilibrium responses.

Table F.5: EU-wide eligibility relative to the unrestricted baseline

Outcome	Fixed networks	Responses to unchanged rivals	Equilibrium responses
<i>Core networks</i>			
World core presences	0.00	-7.99	-4.89
EU core presences	0.00	1.29	2.07
Non-EU core presences	0.00	-9.28	-6.97
<i>Price indices (%)</i>			
Regulated EU	28.59	27.93	27.73
Private-sector EU	0.00	0.58	0.27
Non-EU	0.00	0.72	0.38

Notes: EU-wide minus the unrestricted baseline under the three network comparisons in Section 5. All columns use the same baseline portfolio distribution as the reference. Responses to unchanged rivals use the common candidate sets followed by add-or-drop search, as in Table 5. Prices are jointly solved before aggregation; percentage changes use expenditure-weighted expected log price indices. Regulated EU denotes the customers subject to the restriction under EU-wide eligibility. Results use the primary 4,096-point quadrature and 1,536 cost types per firm.

Table F.6 reports eligible-provider counts and residual shares after profile-level pricing. A change in core counts need not imply the same change in service available to a particular market. At each portfolio profile, identical eligible costs, accessibility and residual contribution imply identical prices by Proposition 1. Equality of mean supply alone is insufficient, because profiles are priced before aggregation. Parent-core links enter the eligible service calculation. Table F.7 separates residual intensity from eligibility: $b_n^N = (1 - \phi)b_n^E$ makes fraction ϕ unavailable only under National. The eligibility comparison reprices fixed baseline portfolios; portfolios are not re-optimized. The observed-core comparison substitutes the observed cores while retaining the same

Table F.6: EU eligibility and residual supply across response comparisons

Outcome	Fixed networks	Responses to unchanged rivals	Equilibrium responses
Eligible modeled providers	-2.49	-2.79	-2.68
Residual share (pp)	55.97	56.95	57.03

Notes: National minus EU-wide. Eligibility counts and shares are expenditure-weighted over regulated markets after profile-level pricing. Residual supply remains locally eligible. Entries use the main 4,096-point rival quadrature; on the independent validation quadrature they change by at most 0.03.

Table F.7: EU price sensitivity at fixed networks

Assumption	Price index	Residual share (%)		
	Regulated customers (%)	EU-wide	National	
<i>A. Residual supply and baseline networks</i>				
Residual term $\times 0.5$	40.01	21.53	82.83	
Baseline residual term	27.52	34.08	90.05	
Residual term $\times 2$	17.88	49.01	94.56	
Observed cores, baseline residual	22.02	40.92	90.80	
<i>B. National eligibility of residual supply</i>				
National exposure $\phi = 0$	27.52	34.08	90.05	
National exposure $\phi = 0.25$	35.99	34.08	87.42	
National exposure $\phi = 0.5$	48.42	34.08	82.83	

Notes: Price effects compare National with EU-wide eligibility. Prices are recomputed at fixed networks, using baseline portfolios except in the observed-core row. Panel A scales the residual price-index contribution in both regimes; its final row uses observed cores with the same local-service topology and parameters. Panel B reduces the residual contribution only in National regulated markets, $b_n^N = (1 - \phi)b_n^E$. Parameters and portfolio choices are held fixed. All rows use the main 4,096-point rival quadrature.

installed local topology and parameters, measuring sensitivity to the entire baseline network configuration. On the independent 4,096-point validation quadrature, the entries of Table F.7 change by at most 0.02 percentage points.

Regional and provider incidence. With equilibrium responses, non-EU price indices rise most in nearby markets such as the United Kingdom and Switzerland, by about 0.6 percent; the increases are 0.26 percent in the United States and 0.34 percent in China. Additional non-EU presences are dispersed across countries, none gaining more than 0.05 expected core presences. Differences in Tencent's, Oracle's, and OVHcloud's responses account for about 63 percent of the world-wide attenuation. A rival's withdrawal changes the profits of both a network retaining a core and its best alternative without that core. The effect on the incentive to retain it depends on which network gains more.

Residual eligibility and provider profits. Making part of the residual contribution ineligible under National raises the price-index effect for regulated customers but changes providers' operating-profit losses little. We use the residual-eligibility changes in Panel B of Table F.7, but evaluate them at each regime's equilibrium portfolio distribution rather than common baseline portfolios. Location costs remain fixed. At $\phi = 0.25$ and $\phi = 0.5$, the price-index effect for regulated customers rises to 36.98 and 49.52 percent, while the operating-profit losses are 20.40 and 19.64 percent for OVHcloud, 4.66 and 4.50 percent for the Chinese providers, and 2.39 and 2.19 percent for the US providers. The modeled providers' share of National regulated expenditure rises from 9.80 to 16.92 percent at $\phi = 0.5$, compared with 66.83 percent under EU-wide. These comparisons hold networks fixed while all providers reprice.

F.5 Operating-profit incidence and location costs

OVHcloud's 20.81 percent operating-profit loss consists of 13.88 percentage points from regulated EU markets, 4.74 from other EU markets, and 2.19 from non-EU markets, each relative to its expected EU-wide worldwide operating profit. Regulated markets therefore account for about two-thirds of its loss. Each provider's EU operating margin changes by less than 0.07 percentage points, so lower revenue accounts for most of the operating-profit losses.

The regulated-market contribution separates into initial profit exposure and the proportional loss within those markets (Table F.8). Multiplying the two percentages and dividing by 100 gives the contribution to the worldwide loss in percentage points. OVHcloud earns a much larger fraction of its profit in regulated EU markets, while within-market loss rates are similar across provider groups. Its greater exposure therefore accounts for most of the difference in the regulated-market contribution.

Table F.8: Regulated-market profit exposure and losses

Provider group	Initial exposure (%)	Within-market loss (%)	Worldwide loss contribution (pp)
US providers	3.25	88.15	2.86
Chinese providers	2.40	86.82	2.09
OVHcloud	15.31	90.68	13.88

Notes: National relative to EU-wide eligibility under equilibrium responses. Initial exposure is regulated EU operating profit as a percentage of worldwide operating profit under EU-wide eligibility. Within-market loss is the percentage decline in regulated EU operating profit. Group values use sums of profits, before location costs. Results use the 4,096-point main rival quadrature.

Profits after location costs also fall for all eight providers; OVHcloud's loss is 20.09 percent. Lower location spending partly offsets lost operating profit; percentage changes use profit after location costs as the denominator. Online Appendix S.8 reports these comparisons (Figure S.8.1), US incidence, and one-firm responses.